

Game-Theoretically Fair Coin Toss with Arbitrary Preferences

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Abstract

Fair multi-party coin toss protocols are fundamental to many cryptographic applications. While Cleve’s celebrated result (STOC’86) showed that strong fairness is impossible against half-sized coalitions, recent works have explored a relaxed notion of fairness called Cooperative-Strategy-Proofness (CSP-fairness). CSP-fairness ensures that no coalition has an incentive to deviate from the protocol. Previous research has established the feasibility of CSP-fairness against majority-sized coalitions, but these results focused on specific preference structures where each player prefers exactly one outcome out of all possible choices. In contrast, real-world scenarios often involve players with more complicated preference structures, rather than simple binary choices.

In this work, we initiate the study of CSP-fair multi-party coin toss protocols that accommodate arbitrary preference profiles, where each player may have an unstructured utility distribution over all possible outcomes. We demonstrate that CSP-fairness is attainable against majority-sized coalitions even under arbitrary preferences. In particular, we give the following results:

- We give a reduction from achieving CSP-fairness for arbitrary preference profiles to achieving CSP-fairness for structured split-bet profiles, where each player distributes the same amount of total utility across all outcomes.
- We present two CSP-fair protocols: (1) a *size-based protocol*, which defends against majority-sized coalitions, and (2) a *bets-based protocol*, which defends against coalitions controlling a majority of the total utilities.
- Additionally, we establish an impossibility result for CSP-fair binary coin toss with split-bet preferences, showing that our protocols are nearly optimal.

Keywords: Coin-toss protocol, game-theoretic fairness.

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Contents

1	Introduction	3
1.1	Our Contributions	4
1.2	Applicability of Our Model.	8
1.3	Additional Related Work	8
2	Technical Overview	9
2.1	Reducing Arbitrary Preference To Split-Bet	9
2.2	CSP-Fairness Against Corrupted Size	10
2.3	CSP-Fairness Against Corrupted Bets	14
2.4	Impossibility	15
3	Preliminaries and Model	17
3.1	Coin Toss Protocol	17
3.2	Different formulations.	18
4	Reducibility Between Formulations	19
5	Size-Based Protocol for Split-Bet Formulation	22
5.1	size-GroupToss ^G [k] Sub-Protocol	23
5.2	Final Protocol for Coalition-Size	23
5.3	Optimizing Constraint 5.7	25
5.3.1	Optimization Allowing Non-Integer Solutions	26
5.3.2	Proof of Theorem 5.11	31
5.4	Integer Solutions	32
6	Construction by Coalition Bet	33
6.1	bet-GroupToss ^b [k] Sub-Protocol	34
6.2	Bets-Based Protocol	34
7	Concrete Examples	36
7.1	Unit-Split	36
7.2	Ranked Preference	37
8	Impossibility Results	38
A	Cryptographic Building Blocks	46
B	Upgrade To Malicious Security	48
B.1	Maliciously Secure GroupToss ^G [k] Sub-Protocol	48
B.2	Malicious Secure Final Protocol for Coalition Size	49

1 Introduction

Good friends Alice and Bob want to spend Christmas together in 2025. They are deciding between three locations: Los Angeles (Alice’s city), New York (Bob’s city), and Dallas (midpoint). However, due to the long flight, their preferences differ: Alice prefers Los Angeles over Dallas over New York, while Bob ranks them in the reverse order. To ensure fairness, they agree to select a destination via a random ternary coin toss: 0 for Los Angeles, 1 for Dallas, and 2 for New York. But how can they remotely decide a *fair* ternary coin?

The problem of fairly tossing a coin among multiple players is formulated as *coin toss protocol*. A rich body of work [AO16, BOO15, BHMO22, BHT18, Cle86, HT14, HO14, MNS16, BHLT17, DSLMM11] has focused on achieving *strong fairness* in coin toss protocols, which guarantees that an adversary controlling a set of corrupted players cannot bias the outcome towards *any* direction by a non-negligible amount. It is well known that when the adversary controls less than half of the players, strong fairness is attainable via standard multi-party computation [Yao82, MGW87, GMW19]. However, the impossibility result by Cleve [Cle86] showed that strong fairness becomes impossible if the adversary controls half of the players. Specifically, in Alice and Bob’s case, if either deviates from the protocol, they always have a strategy to bias the outcome.

Fortunately, this is not the end of the story. Despite Cleve’s impossibility result, Alice and Bob can still use a variant of Blum’s coin toss protocol [Blu83]: They each flip a private uniform ternary coin and post the commitments of the private coin to a public bulletin board (such as blockchains). Then they each open the commitments, and the outcome is the sum of the two coins modulo 3. If either fails to post a commitment or opens its commitment incorrectly, they forfeit, and the outcome defaults to the city where the other player lives. While this protocol does not achieve strong fairness since a player can always refuse to open its commitment and bias the outcome, it does achieve meaningful fairness in a *game-theoretic* sense: A rational player has no incentive to deviate from the protocol, as doing so only reduces their utility.

Game-Theoretic Fairness. Surprisingly, this natural notion of game-theoretic fairness was only recently formalized by Chung, Guo, Lin, Pass, and Shi [CGL⁺18]. They introduced *Cooperative-strategy-Proofness*, or CSP-fairness for short. Specifically, they considered N -party binary coin toss where each of the N players has a *publicly known* preference for either heads or tails. Each player gets utility 1 if the outcome matches their preference and 0 otherwise. CSP-fairness requires that any t -sized coalition cannot increase its expected utility compared to the honest behavior by deviating from the prescribed protocol. Previous works characterized (in)feasibility of CSP-fairness for binary [CGL⁺18, WAS22] and multi-sided coin toss [TSW24], where each player has a public preference of *exactly one* outcome. It is observed that while CSP-fairness is achievable against half or more-sized coalitions, the landscape is starkly different from that of strong fairness. The coalition resilience of CSP-fairness heavily relies on the actual preferences of the players: If all but one players share the same preference, CSP-fairness can be achieved against any $(N - 1)$ -sized coalition; if every outcome is preferred by the same number of players, only a half-sized coalition can be tolerated.

Our Work: Arbitrary Preference Profile. In many practical applications, players do not simply prefer one outcome; rather, they have ranked or even arbitrary utilities across outcomes. For example, imagine a community wishing to jointly decide on a random facility location from m candidates. Instead of preferring only one location, each member ranks different locations based on distance: the closer the facility, the higher the utility they get. Another example is on-chain roulette, where players stake bets on specific outcomes. Each player may put varying bets on

different outcomes and receive different payoffs depending on their stake allocation. In all these examples, players' preferences are public and non-binary. To capture these practical applications, we initiate the study of CSP-fairness for arbitrary preferences and address the following question:

Can we build CSP-fair multi-party coin toss protocols for arbitrary preference profiles against majority-sized coalitions?

1.1 Our Contributions

Our work initiates the study of game-theoretic fairness for arbitrary preferences, making both conceptual and technical contributions. On the conceptual side, we introduce two formulations of preferences that capture natural preference structures, and show that each formulation can be reduced to the other when it comes to achieving CSP-fairness against majority-sized coalitions. On the technical side, we give two multi-sided coin toss protocols: One achieves CSP-fairness against majority-sized coalitions and the other against coalitions controlling a majority of total utility. Additionally, we establish an impossibility result for binary coin toss protocols to better characterize the landscape of CSP-fairness.

To describe our main results, we briefly introduce the model. Consider N number of players jointly tossing a uniform m -sided coin. Each player i 's preference $\mathbf{p}_i$ is a vector of length m , the ℓ -th entry of which is player i 's utility if the outcome is ℓ . We also say that a player places x *bets* on an outcome ℓ if the player gets utility x when the protocol outputs ℓ . In this paper, we may use *bets* and utility interchangeably. We also assume non-negative integer utilities (see Section 1.2 for a discussion of generalizing to negative or fractional utilities). A preference profile is thus all players preferences $(\mathbf{p}_1, \dots, \mathbf{p}_N)$. A player's total bet is the sum of its bets across all outcomes.

Some players together form a coalition $\mathcal{C}$, and their joint utility is the sum of every coalition member's utility. A coalition prefers an outcome if its joint utility on that outcome exceeds its honest expected joint utility. We say that a protocol is CSP-fair for a preference profile $\mathcal{P}$ iff any t -sized coalition cannot increase their expected utility by a non-negligible amount compared to their honest expected utility.

Designing CSP-fair protocols for arbitrary preference profiles against majority-sized coalitions is inherently challenging, as no structure can be assumed on the profile. To address this, we showed that designing CSP-fair protocols for arbitrary preference profiles can be reduced to designing CSP-fair protocols for well-structured preference profiles. Formally, we define:

- **Arbitrary-bet** $\mathcal{F}_{\text{arb}}(N, m)$: In this formulation, each of the N players can put arbitrary non-negative integer bets on different outcomes $\{0, \dots, m-1\}$.
- **Split-bet** $\mathcal{F}_{\text{split}}(N, m)$: In this formulation, each player has B total bet that can be distributed across the m possible outcomes.

The preference profiles considered in previous works are a specific example of split-bet where $B = 1$. A direct generalization would be *unit-split* where each player chooses B out of m possible outcomes, and gets utility 1 for each chosen outcome and 0 otherwise. Another concrete example is *ranked preference profiles*, which are also captured by the split-bet framework with $B = m(m-1)/2$: Each player assigns $m-1$ utility on the most preferred outcome, $m-2$ to the second preference, until 0 on the least preferred outcome.

Theorem 1.1 (Reducing arbitrary preference profile to split-bet). *Fix any N and m . There is an efficient mapping $\mathcal{T} : \mathcal{F}_{\text{arb}}(N, m) \mapsto \mathcal{F}_{\text{split}}(N, m)$ such that for any arbitrary-bet preference profile $\mathcal{P} \in \mathcal{F}_{\text{arb}}(N, m)$, a CSP-fair protocol Π for $\mathcal{P}$ against t -sized coalition exists if there is a CSP-fair protocol Π' for $\mathcal{T}(\mathcal{P})$ against t -sized coalition.*

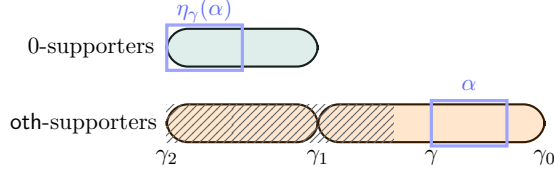


Figure 1: Illustration of a split-bet preference profile with $B = 2$. The green and orange bar each sorts the 0- and oth-supporters according to their bets on 0 in the decreasing order, respectively. Each chunk represents players with the same bets on 0. The first orange chunk bets 1 on outcome 0, and the last chunk puts no bets on 0. The γ_i 's are the margins of these chunks. The shaded area represents the set S . A coalition containing all players in the two blue rectangles does not prefer 0.

By Theorem 1.1, we only need to focus on building CSP-fair protocols for split-bet profiles. We use *cumulative bet* of an outcome ℓ to denote the sum of all players' bets on ℓ . Without loss of generality, we assume that 0 is the least preferred outcome, i.e., 0 has the lowest cumulative bet.¹ Our main result shows that, except in the special case where all outcomes receive exactly the same cumulative bet, CSP-fairness is attainable against $N/2$ -sized coalitions.

Theorem 1.2 (CSP-fairness, informal). *Assuming the existence of oblivious transfer (OT). There exists a CSP-fair m -sided N -party coin toss protocol for split-bet preference profile against t -sized malicious coalitions. Unless all outcomes are equally preferred, $t \geq N/2$.²*

To achieve the above coalition resilience, we present two protocols: the *size-based protocol*, which achieves CSP-fairness against coalitions based on their size, and the *bets-based protocol*, which achieves CSP-fairness against coalitions based on their total bet. While these two metrics are equivalent for split-bet preference profiles, each protocol outperforms the other in different scenarios depending on the preference structure. Below, we first give the general coalition resilience achieved by both protocols and compare their performance on the concrete unit-split and ranked preference profiles for a better understanding.

Size-Based Protocol. Let n_j be the number of players with j bets on outcome 0 and $\mu = \frac{B}{m}$ be the honest expected utility of each player. Then $N_0 = \sum_{i > \mu} n_i$ represents the number of players who prefer outcome 0, called the 0-supporters, and $N_{\text{oth}} = \sum_{i \leq \mu} n_i$ is the number of players who prefer other outcomes, called the oth-supporters. We sort the 0-supporters and the oth-supporters separately based on the decreasing order of their bets on 0 (break ties arbitrarily).

Let S denote the set of $0 \leq \gamma \leq N_{\text{oth}}$ such that the last $N_{\text{oth}} - \gamma$ number of oth-supporters together with all N_0 number of 0-supporters do *not* prefer 0. Let $\gamma_0 = N_{\text{oth}}$, and $\gamma_i = \gamma_{i-1} - n_{i-1}$ be the sequence of “margins” among the oth-supporters. See Figure 1 for an illustrative example.

Let $\eta_\gamma(\alpha)$ be the largest number such that the first $\eta_\gamma(\alpha)$ number of 0-supporters together with the $(\gamma + 1)$ -th to the $(\gamma + \alpha)$ -th oth-supporter does *not* prefer 0, as illustrated in Figure 1. We set $\eta_\gamma(\alpha) = \infty$ if all N_0 of the 0-supporters together with the $(\gamma + 1)$ -th to the $(\gamma + \alpha)$ -th oth-supporter do not prefer 0. Let $\text{opt}(\gamma) := \arg \min_{\alpha} |N_{\text{oth}} + \frac{N_0}{2} - \frac{\gamma}{2} - 2\alpha - \eta_\gamma(\alpha)|$.

Theorem 1.3 (Feasibility against corrupted size). *Assume oblivious transfer (OT) exists. If $\eta_0(\lfloor N_{\text{oth}}/2 \rfloor) > N_0/2 + 1$, there exists a CSP-fair m -sided coin toss protocol against any t -sized*

¹When there are multiple least preferred outcomes, break ties arbitrarily. All our protocols and fairness guarantees hold regardless of which one is chosen.

²In rare cases, $t = \lfloor N/2 \rfloor$ for odd N due to rounding effects. This happens only when the cumulative bets on all outcomes are almost the same. Similar rounding effects have also been observed in [WAS22, TSW24].

malicious coalitions where

$$N/2 \leq t = \max_{\gamma_i \in S} \left\{ \left\lfloor N_{\text{oth}} + \frac{N_0}{2} - \frac{\gamma_i}{2} - \text{opt}(\gamma_i) \right\rfloor \right\}. \quad (1)$$

As illustrated in the ternary ranked preference example in Table 6, the optimal resilience is achieved at different $\gamma_i \in S$, depending on the actual profiles. Therefore, no single γ^* universally maximizes the coalition resilience across all possible preference profiles.

In the above theorem, the condition $\eta_0(\lfloor N_{\text{oth}}/2 \rfloor) > N_0/2 + 1$ intuitively means that the first half of 0-supporters and the first half of oth-supporters together do *not* prefer 0. This generally holds except when there is a large number of “swing” oth-supporters whose bets on 0 are close to μ . In such cases, this protocol may only defend against minority-sized coalitions. Later comparisons show that when the number of swing oth-supporters is large, this protocol is no better than the bets-based protocol described below.

While we follow the traditional threshold-based model for coalition tolerance, our protocol can in fact tolerate larger-sized coalitions as long as the coalition members’ preferences satisfy certain structural conditions. The formal structural characterization is given in Constraints 5.5.

Bets-Based Protocol. An alternative approach is to measure coalition resilience based on the total bet controlled by the coalition rather than the number of players. A coalition’s total bet is the sum of all members’ total bet. A protocol is CSP-fair against coalitions of total bet β if any coalition controlling at most β total bet cannot increase its expected joint utility by deviating. We use B_ℓ to represent the cumulative bet on outcome ℓ and $B_{\text{oth}} = \sum_{\ell \neq 0} B_\ell$ represents the sum of the cumulative bet on all other outcomes. As before, without loss of generality, we assume 0 is the least preferred outcome, i.e., $B_0 \leq \frac{1}{m-1} B_{\text{oth}}$.

Theorem 1.4 (Feasibility against corrupted bets). *Assuming the existence of oblivious transfer (OT). There exists a CSP-fair m -sided coin toss protocol against any malicious coalition controlling at most β total bet, where*

$$\beta = \begin{cases} B_{\text{oth}} + \lceil \frac{1}{2} B_0 \rceil - (m-1)B_0, & \text{if } B_{\text{oth}} \geq \frac{4m-3}{2} B_0 \\ \left\lfloor \frac{m}{2m-1} B_{\text{oth}} + \frac{m}{2(2m-1)} B_0 \right\rfloor, & \text{otherwise.} \end{cases} \quad (2)$$

Unless the profile is perfectly balanced, $\beta \geq (B_0 + B_{\text{oth}})/2$.

For a split-bet preference profile where each player has B total bet, this protocol defends against any t -sized coalition where $t = \lfloor \beta/B \rfloor$.

The bets-based protocol can be used directly on arbitrary preference profiles to defend against coalitions controlling majority bets. However, if used directly, it may fail to tolerate majority-sized coalition since a few players with large bet could collectively control more than β total bet. Therefore, to achieve CSP-fairness against majority-sized coalitions for arbitrary preference profiles $\mathcal{P}$, one should first transform $\mathcal{P}$ to a split-bet profile $\mathcal{T}(\mathcal{P})$ as in Theorem 1.1 and then apply one of our two protocols on $\mathcal{T}(\mathcal{P})$.

Similar to our size-based protocol, this bets-based protocol can in fact achieve fairness against coalitions controlling more bets than the above stated threshold as long as the coalitions members’ bets satisfy certain structural conditions, as formally stated in Constraints 6.4.

Table 1: Optimal resilience for unit-bet.

Case	Size-based protocol (Theorem 1.3)	Bets-based protocol (Theorem 1.4)
$N_{\text{oth}} \geq \frac{4m-3B}{2B}N_0$	$N_{\text{oth}} + \lceil \frac{2B-m}{2B}N_0 \rceil$	$N_{\text{oth}} + \lceil \frac{2B-2m+1}{2B}N_0 \rceil$
Otherwise	$\lfloor \frac{m}{2m-B}N_{\text{oth}} + \frac{m}{2(2m-B)}N_0 \rfloor$	$\lfloor \frac{m}{2m-1}N_{\text{oth}} + \frac{2m-m/B}{2(2m-1)}N_0 \rfloor$

Table 2: Optimal resilience for ternary ranked preference.

Size-based protocol (Theorem 1.3)		Bets-based protocol (Theorem 1.4)	
Case	resilience	Case	resilience
$n_0 \geq \frac{5}{2}n_2 - \frac{1}{2}n_1$	$\lfloor n_0 + \frac{1}{2}n_1 - \frac{1}{2}n_2 \rfloor$	$n_0 \geq \frac{5}{6}n_1 + \frac{8}{3}n_2$	$\lfloor n_0 + \frac{1}{6}n_1 - \frac{2}{3}n_2 \rfloor$
$n_1 + n_2 \leq n_0 < \frac{5}{2}n_2 - \frac{1}{2}n_1$	$\lfloor \frac{2}{3}n_0 + \frac{1}{3}n_1 + \frac{1}{3}n_2 \rfloor$	Otherwise	$\lfloor \frac{3}{5}n_0 + \frac{1}{2}n_1 + \frac{2}{5}n_2 \rfloor$
Otherwise	$\lceil N/2 \rceil - 1$		

Comparison of the Two Protocols. When the number of swing **oth**-supporters is small relative to the total number of players, the size-based protocol offers stronger resilience. However, when many swing **oth**-supporters are present, the bets-based protocol provides better resilience. Below, we give the resilience of both protocols for some concrete examples for comparison.

Unit-split. Each player chooses $B < m$ out of m outcomes and gets utility 1 on each chosen outcome. In this case, we have $B_0 = N_0$ and $B_{\text{oth}} = B \cdot N - N_0$. For unit-split, the size-based protocol is always better than the bets-based protocol (Table 1), as this formulation has no swing **oth**-supporters. The coalition resilience is always at least $N/2$.

Ternary ranked preference. Each player has a rank over the three possible outcomes. Each player gets utility 2 on the most preferred outcome, utility 1 on the second preferred outcome, and 0 on the least preferred outcome. Recall that n_i is the number of players with bets i on outcome 0. Here, the n_1 players with bets 1 on outcome 0 are those swing **oth**-supporters.

From Table 2, one can see that when n_0 is much larger than $n_1 + n_2$, the size-based protocol performs better; when $n_0 \leq n_1 + n_2$, the bets-based protocol achieves better resilience.

Impossibility. To complement our protocols, we also give an impossibility result for binary split-bet preference profiles. As before, without loss of generality, we assume that 0 is the least preferred outcome.

Theorem 1.5 (Impossibility for binary split-bet). *Let N_0 and N_1 denote the number of 0-supporters and 1-supporters, respectively. Given a preference profile $\mathcal{P} \in \mathcal{F}_{\text{split}}(N, 2)$, there is no CSP-fair binary coin toss protocol against t -sized semi-malicious coalition for*

$$t \geq \max\{N_1 + \lceil N_0/2 \rceil - \theta^*, \gamma' + \lceil N_0/2 \rceil\},$$

where γ' is smallest value such that $\eta_0(\gamma') > N_0/2$, and θ^* is the maximum number such that $2\theta + \lceil (\eta_\theta(\theta) + 1)/2 \rceil \leq N_1$.

When N_1 is much larger than N_0 , this impossibility result is tight except for a gap of at most $\lceil \theta^*/2 \rceil + 2$. Specifically, our impossibility is optimal when the first $2\theta^* + 2$ number of 1-supporters have the same bets on 0. In the worst case, it is tight up to only a constant multiplicative factor.

1.2 Applicability of Our Model.

Our model aims to initiate a formal understanding of CSP-fairness for general preference structure in settings where (1) each player has a non-negative integer utility for each outcome, and (2) a coalition’s utility is the sum of its members’ utilities. We believe these assumptions capture many practical settings. Below, we elaborate on the modeling choices and their implications.

- **Extension to negative or fractional utilities:** Our non-negative integer utility assumption reflects many practical scenarios where utility represents divisible resources — such as tokens, currency, or physical items with minimal granularity. Meanwhile, our results can be extended to a broader range of utilities as long as the linear transformation $\mathcal{T}$ in Theorem 1.1 can be efficiently computed. This only requires the utilities to be encoded using a polynomial number of bits, allowing negative or fractional utilities.
- **On coalition’s joint utility:** Like prior works, we adopt the model where a coalition’s joint utility is the sum of its members’ utilities. This additive view again fits the settings where utilities correspond to divisible and transferable goods, such as monetary rewards or stake outcomes. In such cases, a coalition aims to increase its total utility to redistribute the total wealth to make every member strictly happier.

This approach inherently emphasizes the numerical scale since players with higher bets contribute more to the coalition’s joint utility. In contrast, there are contexts where players express normalized or ordinal preferences and scale invariance is desired. We emphasize that our framework can accommodate such settings through rescaling. If all players are to be treated equally, utilities can be transformed into a split-bet profile where each player distributes an equal total bet across outcomes. We leave the complete characterization of fairness under alternative notions of coalition utility that may enforce scale invariance or other desiderata as a promising future direction.

1.3 Additional Related Work

Game-theoretic fairness has also been studied in the context of leader election [CCWS21b, KMSW22, CCWS21a, KKZ24] where N players want to jointly elect a random leader. Previous work [CGL⁺18, WAS22, CCWS21b, KMSW22, KKZ24] also studied another form of game-theoretic fairness called maximin fairness, which requires that the coalition cannot harm honest users. With the recent progress on blockchains and decentralized finance, another line of work has focused on financial fairness [BK14, KMS⁺16, ADMM16, KB14, KVV16, DEF18, ATM⁺22, CMST22], where players’ collateral is used to punish misbehavior. Interestingly, our work and previous works on game-theoretic fairness achieve fairness without collateral.

The connection between game theory and multi-party computation is not new [HT04, KN08, ADGH06, OPRV09, AL11, ACH11, Kat08, DR07]. However, this line of work considers a rather different utility structure: it assumes that players prefer to get the correct outcome and learn others’ secrets without leaking their own secrets.

Organization. In Section 2, we give a technical overview. The preliminaries and formal model are introduced in Section 3. In Section 4, we establish a reduction from arbitrary preference profiles to split-bet preferences. We present our two CSP-fair coin toss protocols in Section 5 and Section 6. Section 7 gives the characterization for some concrete examples. The impossibility result is given in Section 8, and upgrade to malicious security is given in Appendix B.

2 Technical Overview

In this section, we outline the main techniques for achieving CSP-fairness against semi-malicious coalitions using a binary coin toss as an example. All of our feasibility results can be generalized to multi-sided coin tosses with techniques from [TSW24]. A semi-malicious coalition can program its randomness used in the protocol or abort early, but otherwise follows the protocol honestly. Our protocols against semi-malicious adversaries achieve information-theoretic security, and can be upgraded to malicious security using standard cryptographic techniques as seen in Appendix B.

We organize this section as follows. Section 2.1 explains how to reduce arbitrary preferences to split-bet preferences. We then illustrate the high-level idea of the two protocols and the technical challenges in Section 2.2 and Section 2.3. The impossibility is discussed in Section 2.4.

Notations. We use N_0 to denote the number of 0-supporters who prefers outcome 0 and N_1 denote the number of 1-supporters who do not prefer 0. We also use B_0 and B_1 to denote the cumulative bet on 0 and on 1, respectively. Additionally, B is used to denote the total bet each player can distribute. For a player i , we use $\mathbf{p}_i = (\text{bet}_i(0), \text{bet}_i(1))$ to denote player i 's preference where $\text{bet}_i(b)$ is player i 's bets on outcome b . A preference profile $\mathcal{P} = (\mathbf{p}_1, \dots, \mathbf{p}_N)$ is the preference of all N players. We will sort the 0-supporters and 1-supporters separately in decreasing order based on their bets on 0, breaking ties arbitrarily.

2.1 Reducing Arbitrary Preference To Split-Bet

Given an arbitrary preference profile $\mathcal{P}$, our goal is to find an efficient transformation $\mathcal{T}$ that maps $\mathcal{P}$ to a split-bet preference profile such that we can build CSP-fair protocols for $\mathcal{P}$ from any CSP-fair protocol for $\mathcal{T}(\mathcal{P})$ while maintaining the same coalition resilience.

A naive approach is to map a 0-supporter's preference to $(1, 0)$ and a 1-supporter's preference to $(0, 1)$, and then apply the protocol from [WAS22]. However, this fails to preserve coalition preferences, which is crucial for CSP-fairness. Consider a preference profile $\mathcal{P} = ((1, 0), (1, 0), (0, 4))$ of two 0-supporters and one 1-supporter as an example. If mapped naively, the resulting profile would be $\mathcal{P}' = ((1, 0), (1, 0), (0, 1))$. However, a coalition $\mathcal{C}$ containing the 1-supporter and one of the 0-supporters will prefer outcome 1 in $\mathcal{P}$ but does not prefer any outcome in $\mathcal{P}'$. Consequently, a CSP-fair protocol for $\mathcal{P}'$ could allow $\mathcal{C}$ to bias the outcome towards 1 without violating fairness, but such a strategy will surely increase the coalition's expected utility with respect to $\mathcal{P}$.

Therefore, to ensure fairness, the transformation must preserve every coalition's preference. We achieve this by enforcing *linearity*. Specifically, each player's preference profile $\mathbf{p}_i$ is mapped to

$$\mathbf{p}'_i = 2 \cdot \mathbf{p}_i + \mathbf{b}_i.$$

Here, 2 is a global scaling factor across all players and $\mathbf{b}_i$ is a length-2 shift vector where each entry equals b_i , the maximum total bet among all players minus player i 's total bet. Intuitively, linear transformations preserve coalition preferences because a coalition's joint preference is fundamentally a linear combination of all coalition members' preferences. We choose 2 as the global scaling factor to guarantee that all players' total bets after the mapping become the same, since the shift vector $\mathbf{b}_i$ increases player i 's total bet by $2 \cdot b_i$ (for general m -sided coin toss, this scaling factor will be m). After the transformation, every player's total bet becomes $2 \cdot (\text{maximum total bet among all players in } \mathcal{P})$. For example, $\mathcal{P} = ((1, 0), (1, 0), (0, 4))$ will be mapped to $\mathcal{P}' = ((5, 3), (5, 3), (0, 8))$. Unlike the naive approach, this transformation ensures that every coalition's preference remains consistent between $\mathcal{P}$ and $\mathcal{P}'$. With this preference-preserving transformation, we design CSP-fair protocol Π for $\mathcal{P}$ as follows: the N players together run a

CSP-fair protocol Π' for $\mathcal{P}' = (\mathbf{p}'_1, \dots, \mathbf{p}'_N)$ where each player i acts as if its preference is $\mathbf{p}'_i$. Since the transformation preserves coalition preferences, protocol Π achieves CSP-fairness against any t -sized coalition for $\mathcal{P}$ if Π' defends against any t -sized coalition for $\mathcal{P}'$.

2.2 CSP-Fairness Against Corrupted Size

Now we only need to focus on constructing CSP-fair protocols for split-bet profiles where each player distributes B total bet across different outcomes. A natural starting point is to generalize the protocol from [WAS22], which works for the special case of $B = 1$. They partition players into two groups: Group G_0 of the 0-supporters and G_1 of the 1-supporters. Each group runs the following sub-protocol $\text{size-GroupToss}^G[k]$ to decide a coin s_b for $b \in \{0, 1\}$, where G denotes the group running this sub-protocol and k is a parameter to be specified later. The final outcome is then determined by the value of s_0 and s_1 .

Several technical challenges arise when extending this protocol to general split-bet preference profiles:

- *Characterizing coalition preference.* In the special case where $B = 1$, a coalition's preference is b iff b -supporters dominate in the coalition. However, for general split-bet profiles, a 1-supporter may also put bets on 0 and vice versa. Thus, a coalition's preference is no longer determined solely by the number of corrupted b -supporters but also by their bet distributions.
- *Partition of G_0 and G_1 .* The trivial partition that places all b -supporters into group G_b does not always give the optimal resilience. Instead, we need to assign some of the “swing” 1-supporters to G_0 . Swing 1-supporters are players who prefer outcome 1 but still put significant bets on 0. We revise the partition to ensure that players in G_1 are aggressive 1-supporters truly opposed to outcome 0. As we will later explain, this is crucial to achieve CSP-fairness against large coalitions.
- *Optimal choice of parameters.* We need to optimize the parameter k in $\text{size-GroupToss}^G[k]$ and the number of swing 1-supporters in G_0 to achieve the best coalition resilience. This results in a rather involved integer optimization problem, as we will elaborate on later.

Below, we describe the protocol and show how to address these challenges.

size-GroupToss^G[k] sub-protocol

Sharing Phase. Each player i chooses a random coin $\text{coin}_i \xleftarrow{\$} \{0, 1\}$ and shares it using $(k+1)$ -out-of- $|G|$ Shamir secret sharing into $\{c_{i,j}\}_{j \in G}$. Send $c_{i,j}$ to player j . If any players abort, remove them from G .

Reconstruction Phase. Every player i computes the sum of the shares it received $c_i := \sum_{j \in G} c_{j,i}$ and broadcasts c_i . Players use the broadcast shares to reconstruct $s = \oplus_{i \in G} \text{coin}_i$. If the reconstruction fails, set $s = \perp$; otherwise, output s .

By the security of Shamir secret sharing (see Appendix A for formal definitions), this sub-protocol satisfies the following properties against coalition $\mathcal{C}$.

- *Binding.* The sharing phase uniquely binds to a secret s , such that the reconstruction either succeeds and outputs s , or it fails and outputs $\perp$.
- *Knowledge threshold.* If $|\mathcal{C}| \geq k + 1$, the coalition can completely control value s during the sharing phase. Otherwise, the secret s looks uniformly random to the coalition.

- *Liveness threshold.* If $|\mathcal{C}| \geq |G| - k$, the coalition can cause the reconstruction to fail and output $\perp$. Otherwise, the reconstruction phase must succeed.

CSP-fair coin toss against coalition size

Sharing phase: G_0 runs the sharing phase of $\text{size-GroupToss}^{G_0}[k_0]$. Then G_1 runs the sharing phase of $\text{size-GroupToss}^{G_1}[k_1]$.

Reconstruction phase: G_0 runs the reconstruction phase of $\text{size-GroupToss}^{G_0}[k_0]$ to reconstruct s_0 . If $s_0 = \perp$, set $s_0 = 0$. Then G_1 runs the reconstruction phase of $\text{size-GroupToss}^{G_1}[k_1]$ to reconstruct s_1 . If $s_1 = \perp$, output 0. Otherwise, output $s_0 \oplus s_1$.

Roughly speaking, this protocol achieves CSP-fairness by disincentivizing G_1 from hampering the reconstruction of s_1 . If G_1 fails to reconstruct s_1 , the protocol will output 0 as a punishment. This ensures that rational players in G_1 have no incentive to fail the reconstruction of s_1 .

As long as the following three conditions are satisfied, the above protocol achieves CSP-fairness against any coalition of size no more than $t = t_0 + t_1$ where t_b denotes the number of corrupted players in group G_b for $b \in \{0, 1\}$.

- (C1) $\mathcal{C}$ cannot control both coins. Equivalently, $t \leq k_0 + k_1 + 1$. Otherwise, $\mathcal{C}$ can force the outcome to be whatever the coalition's overall preference is.
- (C2) If $\mathcal{C}$ controls s_1 , then it cannot fail the reconstruction of s_0 . That is, $t \leq |G_0| - k_0 + k_1$. Otherwise, $\mathcal{C}$ can fail the reconstruction of s_0 and force the outcome to be s_1 it determines.
- (C3) If $\mathcal{C}$ can fail the reconstruction of s_1 , then it must NOT prefer 0.

Here, condition (C3) guarantees that any coalition is not incentivized to fail the reconstruction of s_1 . Conditioned on s_1 being successfully reconstructed, condition (C1) and (C2) together guarantee that at least one of s_0 or s_1 is uniformly random. Consequently, the final outcome must also be uniformly random. Therefore, these three conditions together ensure CSP-fairness of the protocol.

Characterizing Preference. For any fixed group partition G_0 and G_1 , we only need to choose k_0 and k_1 such that the above three conditions are satisfied to achieve CSP fairness. Observe that the first two conditions are independent of player preferences once G_0 and G_1 are fixed. However, the third condition heavily depends on player preferences and cannot be simply captured by the number of b -supporters in the coalition, as we explained in the technical challenges above.

Recall that 0- and 1-supporters are sorted by decreasing bets on 0. Let $\eta(x) < N_0$ be the maximum number such that the first $\eta(x)$ number of 0-supporters together with the first x number of 1-supporters in G_1 does not prefer 0. If all N_0 number of 0-supporters together with the first x number of 1-supporters in G_1 does not prefer 0, we set $\eta(x) = \infty$. Then (C3) can be expressed as $t \leq (|G_1| - k_1) + \eta(|G_1| - k_1)$. This is to guarantee that *any* coalition with $|G_1| - k_1$ number of 1-supporters and $\eta(|G_1| - k_1)$ number of 0-supporters — even if they are the ones with the most bets on 0 — does not prefer 0. Specifically, if $\eta(|G_1| - k_1) = \infty$, failing the reconstruction of s_1 never benefits the coalition, regardless of how many 0-supporters are corrupted.

Given G_0 and G_1 , the problem boils down to the following optimization:

Maximize t subject to

$$0 \leq k_0 \leq |G_0| - 1, 0 \leq k_1 \leq |G_1| - 1,$$

$$t \leq \min\{k_0 + k_1 + 1, |G_0| - k_0 + k_1, (|G_1| - k_1) + \eta(|G_1| - k_1)\}.$$

Partition of G_0 and G_1 . To maximize coalition resilience, we must carefully decide the partition G_0 and G_1 , since the naive approach of assigning all b -supporters to G_b is not necessarily optimal for general split-bet profiles. For example, consider a profile $\mathcal{P}^* = \{2@ (3, 0), 2@ (1, 2), 5@ (0, 3)\}$ of 9 players. This notation means that the profile contains two players with preference $(3, 0)$, two with $(1, 2)$, and five with preference $(0, 3)$. Using the naive partition $G_0 = \{2@ (3, 0)\}$ and $G_1 = \{2@ (1, 2), 5@ (0, 3)\}$, we can defend against at most $t = 4$ by choosing $k_0 = 1$, $k_1 = 3$. However, if we move the two swing 1-supporters with preference $(1, 2)$ to G_0 , now the new $G_0 = \{2@ (3, 0), 2@ (1, 2)\}$ and $G_1 = \{5@ (0, 3)\}$, which allows us to defend against $t = 5$ by picking $k_0 = 2$, $k_1 = 3$.

Why is this new partition better? Recall that the crux of the protocol is to output 0 as a punishment if the reconstruction of s_1 fails. To make this punishment effective, G_1 should consist only of aggressive 1-supporters who strongly oppose outcome 0. In the example of $\mathcal{P}^*$, the two $(1, 2)$ players are swing 1-supporters: they prefer outcome 1 but also place bets on outcome 0. If they collude with any 0-supporter, the coalition actually prefers outcome 0. Thus, to ensure that G_1 only contains aggressive 1-supporters, we move swing 1-supporters into G_0 to improve coalition resilience.

In general, we use γ to denote the number of 1-supporters in group G_0 . These players will be the γ number of 1-supporters with the most bets on 0. Then $|G_0| = N_0 + \gamma$ and $|G_1| = N_1 - \gamma$. We redefine $\eta_\gamma(x)$ to be the maximum number of 0-supporters such that the first $\eta_\gamma(x)$ number of 0-supporters together with the first x number of 1-supporters in G_1 do *not* prefer 0. As before, if all N_0 number of 0-supporters together with the first x number of 1-supporters in G_1 do not prefer 0, we set $\eta_\gamma(x) = \infty$. The problem then becomes

$$\begin{aligned} & \text{Maximize } t \text{ subject to} \\ & 0 \leq \gamma < N_1, \quad 0 \leq k_0 \leq N_0 + \gamma - 1, \quad 0 \leq k_1 \leq N_1 - \gamma - 1, \\ & t \leq \min\{k_0 + k_1 + 1, N_0 + \gamma - k_0 + k_1, (N_1 - \gamma - k_1) + \eta_\gamma(N_1 - \gamma - k_1)\}. \end{aligned} \quad (3)$$

Solving the Optimization. Let $\alpha := |G_1| - k_1$. We explain how to solve the above optimization, allowing non-integer solutions in the overview. The full proof is deferred to Section 5.3. Observe that the optimal solution of k_0 is always $|G_0|/2 = (N_0 + \gamma)/2$. Substituting this back to Equation (3), the problem reduces to finding the optimal γ and α to maximize t , subject to the constraint

$$t \leq t(\gamma, \alpha) := \min\{c_1(\gamma, \alpha), c_2(\gamma, \alpha)\},$$

where $c_1(\gamma, \alpha) = \frac{(N_0 + \gamma)}{2} + N_1 - \alpha$ and $c_2(\gamma, \alpha) = \eta_\gamma(\alpha) + \alpha$. Here, $\eta_\gamma(\cdot)$ has been extended naturally to allow both non-integer input and output.

Let $\text{opt}(\gamma)$ be the optimal choice of α given any fixed γ . From the feasible region depicted in Figure 2, $\text{opt}(\gamma)$ is either $f(\gamma)$ or $h(\gamma)$ ³ where:

$$\begin{aligned} f(\gamma) &:= \alpha \text{ such that } c_1(\gamma, \alpha) = c_2(\gamma, \alpha) \text{ if such } \alpha \text{ exists.} \\ h(\gamma) &:= \text{the minimum } \alpha \text{ such that } \eta_\gamma(\alpha) = \infty \text{ if such } \alpha \text{ exists.} \end{aligned}$$

We analyze $t(\gamma, \text{opt}(\gamma))$ for the two cases separately.

If $\text{opt}(\gamma) = f(\gamma)$: we show that $t(\gamma, f(\gamma))$ is a decreasing function in γ by analyzing $\eta_\gamma(\cdot)$.

If $\text{opt}(\gamma) = h(\gamma)$: we show that $t(\gamma, h(\gamma)) \leq \max\{t(\gamma_0, h(\gamma_0)), t(\gamma_1, h(\gamma_1))\}$, where γ_0 and γ_1 are the two margins of γ after we group players with same bets on 0 into the same chunk, as depicted in Figure 3.

³There are corner cases where $\text{opt}(\gamma) = N_1 - \gamma$, we leave it to the full proof.

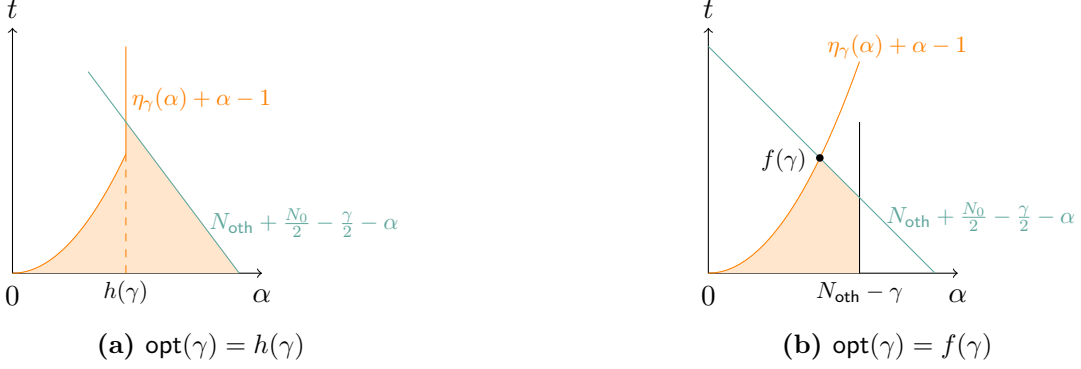


Figure 2: The orange shaded area represents the feasible region of (α, t) for any fixed γ . Here, the orange and green line represent $c_1(\gamma, \alpha)$ and $c_2(\gamma, \alpha)$, respectively.

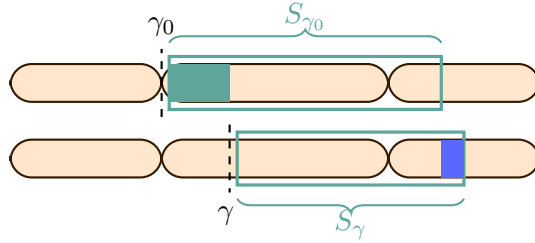


Figure 3: Illustrative explanation of Equation (4). Specifically, x and x' are the average utility of players in the solid green and blue blocks.

We use a sliding window technique to prove this claim. Let S_γ be the set of the first $h(\gamma)$ number of players in G_1 . By definition of $h(\gamma)$, players in S_γ should exactly “cancel out” all 0-supporters’ joint preference on 0: when S_γ collude with all N_0 number of 0-supporters, their average expected utility on outcome 0 is exactly $B/2$, the honest expected utility. That is, $h(\gamma)$ is the number of players needed to fail the reconstruction of s_1 such that the coalition never prefers 0.

Based on the above argument, the average utility $\bar{\mu}(\gamma)$ of players in S_γ must satisfy the following invariant for any γ :

$$(B/2 - \bar{\mu}(\gamma)) \cdot h(\gamma) = C,$$

where C is a constant that only depends on all 0-supporters’ total bet on 0. Therefore, we have

$$\begin{aligned} \frac{B}{2}(h(\gamma_0) - h(\gamma)) &= \bar{\mu}(\gamma_0) \cdot h(\gamma_0) - \bar{\mu}(\gamma) \cdot h(\gamma) \\ &= x \cdot (\gamma - \gamma_0) - x'(\gamma + h(\gamma) - \gamma_0 - h(\gamma_0)), \end{aligned} \quad (4)$$

where x and x' are the average utility of players in the green block and blue block in Figure 3, respectively. Equation (4) follows directly by observing that $\bar{\mu}(\gamma) \cdot h(\gamma)$ is the joint utility of players in S_γ . This allows us to relate changes in $h(\gamma)$ with respect to changes in γ . We can thus compare $t(\gamma, h(\gamma))$ for γ and γ_0 (respectively γ_1) since the difference between $t(\gamma, h(\gamma))$ and $t(\gamma_0, h(\gamma_0))$ is exactly $\frac{1}{2}(\gamma - \gamma_0) + (h(\gamma) - h(\gamma_0))$.

Putting the two cases together, a careful case analysis shows that the global optimal choice of γ must be one of the margins, with $\text{opt}(\gamma)$ being $f(\gamma)$ if it exists, or $h(\gamma)$ otherwise. This depends on the actual preference profiles. Our concrete example in Table 6 implies that each margin could potentially be the optimal choice. The full analysis and the integer solution are deferred to Section 5.3.

2.3 CSP-Fairness Against Corrupted Bets

Another natural approach is to consider coalition resilience in terms of the total bet controlled by a coalition. In a split-bet preference profile where each player holds exactly B total bets, defending against coalitions by corrupted size is equivalent to defending against coalitions by corrupted bets.

Recall that B_0 and B_1 denote the cumulative bet on outcome 0 and 1, respectively. Each player can be interpreted as B number of “virtual players”, each holding one bet. Specifically, player i controls $\text{bet}_i(0)$ number of virtual players with preferences $(1, 0)$ and $\text{bet}_i(1)$ with preference $(0, 1)$. For example, consider $\hat{\mathcal{P}} = ((0, 3), (2, 1), (2, 1))$ where $B = 3$. This profile can be viewed as a preference profile with 9 virtual players, four with $(1, 0)$ and five with $(0, 1)$. Thus, defending against a coalition based on the number of virtual players is equivalent to defending based on the total bet they control.

Rather than directly applying the protocol from [WAS22] to the $B \cdot N$ number of virtual players, we use weighted threshold secret sharing. This allows the protocol to work seamlessly even for arbitrary preference profiles.

In our scheme, each player participates in two sub-protocols to generate coins s_0 and s_1 . In the sub-protocol for coin s_b , players run a weighted threshold secret sharing protocol where each player’s weight equals their bets on outcome b . For example, under profile $\hat{\mathcal{P}}$ above, the first player has weight 3 while the other two players have weight 1 in the sub-protocol determining s_1 . The secret s_b can only be reconstructed from players controlling enough total weights. More formally, the access structure of the secret sharing is $\Gamma = \{S \subseteq [n] : \text{bet}_S(b) \geq k + 1\}$ where $\text{bet}_S(b)$ is the total bet of set S on outcome b . In other words, any set with $k + 1$ or more bets on outcome b can reconstruct the secret, while any set with fewer than $k + 1$ bets on outcome b learns nothing about the secret.

bet-GroupToss^b[k] sub-protocol

Let $\mathcal{O}$ be the set of players remain online. $\mathcal{O}$ is initialized as $[N]$.

Sharing Phase. Each player i chooses a random coin $\text{coin}_i \xleftarrow{\$} \{0, 1\}$. Player i shares coin_i into N shares with an access structure $\Gamma = \{S \subseteq [N] : \text{bet}_S(b) \geq k + 1\}$. Denote the set of shares as $\{c_{i,j}\}_{j \in [n]}$. Send $c_{i,j}$ to player j . If any player aborts, remove them from $\mathcal{O}$.

Reconstruction Phase. Every player i broadcast the shares $\{c_{j,i}\}_{j \in \mathcal{O}}$ it has received from players in $\mathcal{O}$. Players use $\{c_{j,i}\}_{j \in \mathcal{O}}$ to reconstruct $\{\text{coin}_i\}_{i \in \mathcal{O}}$. If any coin_i failed to be reconstructed, output $c = \perp$; otherwise, output $(\oplus_{i \in \mathcal{O}} \text{coin}_i)$.

By the property of weighted secret sharing, this sub-protocol also satisfies the binding, knowledge threshold, and liveness threshold properties with respect to the corrupted total bet, as the size-GroupToss^G[k] sub-protocol.

CSP-fair coin toss against coalition bets

Sharing phase: All players run the sharing phase of bet-GroupToss⁰[k₀] and then the sharing phase of bet-GroupToss¹[k₁].

Reconstruction phase: All players run the reconstruction phase of bet-GroupToss⁰[k₀]. Let s_0 be the reconstructed outcome. If $s_0 = \perp$, set $s_0 = 0$. Then players run the reconstruction phase of bet-GroupToss¹[k₁]. Let s_1 be the reconstructed outcome. If $s_1 = \perp$, output 0. Otherwise, output $s_0 \oplus s_1$.

Consider a coalition that controls β total bets, out of which β_b is the corrupted bets on outcome b for $b \in \{0, 1\}$. By a similar argument before, the above protocol achieves CSP fairness against any coalition controlling β total bet as long as the following three conditions are satisfied:

(C1') $\mathcal{C}$ cannot control both coins. Equivalently, $\beta \leq k_0 + k_1 + 1$.

(C2') If $\mathcal{C}$ controls s_1 , then it cannot fail the reconstruction of s_0 . That is, $\beta \leq B_0 - k_0 + k_1$.

(C3') If $\mathcal{C}$ can fail the reconstruction of s_1 , then it must NOT prefer 0. That is, if $\beta_1 \geq B_1 - k_1$, then $B_0 \leq B_1$.

The problem then becomes maximizing β subject to constraints (C1'), (C2'), and (C3'). The optimal resilience is given by

$$\beta = \begin{cases} B_1 - \lfloor \frac{B_0}{2} \rfloor, & \text{if } B_1 \geq \frac{5}{2}B_0 \\ \lfloor \frac{2}{3}B_1 + \frac{1}{3}B_0 \rfloor, & \text{otherwise.} \end{cases}$$

Equivalently, this protocol tolerates $\lfloor \frac{\beta}{B} \rfloor$ -sized coalition for split-bet profiles.

2.4 Impossibility

Our impossibility result adopts a similar approach in [CGL⁺18, WAS22, TSW24] but with more involved optimization techniques. The core idea is to transform an N -party CSP-fair protocol Π for profile $\mathcal{P}$ to a three-party protocol Π^* as follows: The N parties are partitioned into the three supernodes. Each supernode emulates the execution of the players it contains. Messages between players in the same supernode are dealt with internally, and those between players in different supernodes will become messages between supernodes. Then the protocol Π naturally transforms to a three-party protocol Π^* . The goal is to find a partition of the N players, such that if Π achieves CSP-fairness against any t -sized coalition, then the resulting Π^* satisfies the following three properties:

(C1'') *Lone-wolf condition*: $\mathcal{S}_1$ (or $\mathcal{S}_3$) cannot bias the output towards any direction.

(C2'') *Wolf-minion condition*: $\mathcal{S}_1$ (or $\mathcal{S}_3$) and $\mathcal{S}_2$ cannot bias the output towards 1.

(C3'') *$\mathbf{r}_2$ -equity condition*: for all but a negligible fraction of $\mathcal{S}_2$'s randomness $\mathbf{r}_2$, the value $|F(\mathbf{r}_2) - \frac{1}{2}|$ is negligible, where $F(\mathbf{r}_2)$ denotes the expected coin outcome in an honest execution where $\mathcal{S}_2$'s randomness is fixed to $\mathbf{r}_2$.

However, no three-party protocol satisfies the above three conditions simultaneously. Therefore, no CSP-fair protocol defend against t -sized coalition for $\mathcal{P}$.

Following this framework, we focus particularly on optimizing the partition of players into supernodes. To convey the main idea, we use $\mathcal{P}^* = \{2@ (3, 0), 2@ (1, 2), 5@ (0, 3)\}$ as an example. Recall that our feasibility from Section 2.2 gives a CSP-fair protocol for $\mathcal{P}^*$ against any 5-sized coalition. We will show that this is tight: no protocol is CSP-fair against 6-sized coalition for $\mathcal{P}^*$.

In the partition, $\mathcal{S}_1$ and $\mathcal{S}_3$ each contain $\{1@ (3, 0), 1@ (1, 2), 1@ (0, 3)\}$ and $\mathcal{S}_2$ contains the remaining three 1-supporters $\{3@ (0, 3)\}$. Assuming for the sake of contradiction that there is a CSP-fair protocol Π against 6-sized coalition, then with the above partition, the resulting three-party protocol Π^* should satisfy

1. *Lone-wolf condition*: $\mathcal{S}_1$ cannot bias the output towards 1 directly by CSP-fairness. If it can bias the outcome towards 0, then $\mathcal{S}_1$ can collude with the 0-supporter in $\mathcal{S}_3$ to form a coalition of size four that prefers 0. This would allow the coalition to bias the outcome towards 0, which contradicts CSP-fairness. The same argument holds for $\mathcal{S}_3$.

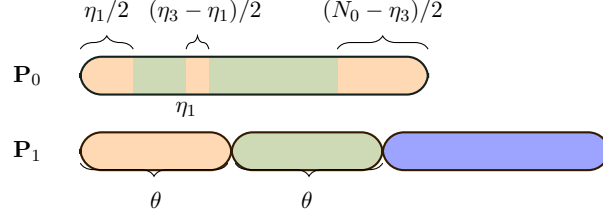


Figure 4: Illustration of partition. For simplicity, we omit the rounding in the figure. The players marked by orange are partitioned into $\mathcal{S}_1$, those marked by green are partitioned into $\mathcal{S}_3$, and those marked by blue are put into $\mathcal{S}_2$.

2. *Wolf-minion condition:* $\mathcal{S}_1$ (or $\mathcal{S}_3$) and $\mathcal{S}_2$ cannot bias the output towards 1 by CSP-fairness, since this coalition controls 6 players and prefers 1.
3. *$\mathbf{r}_2$ -equity condition:* Since $\mathcal{S}_2$ prefers 1, for all but a negligible fraction of $\mathbf{r}_2$, $F(\mathbf{r}_2)$ must not bias towards 1. Since $\mathbb{E}[F(\mathbf{r}_2)] = 1/2$ when everyone behaves honestly, it must be that $F(\mathbf{r}_2)$ cannot bias towards any direction.

But this is impossible. Thus, no protocol is CSP-fair for $\mathcal{P}^*$ against $t = 6$.

To generalize this argument for split-bet, we parametrize the partition as follows. For simplicity, we assume N_0 is even here. At a high level, $\mathcal{S}_1$ and $\mathcal{S}_3$ each contains half of 0-supporters and θ number of 1-supporters. $\mathcal{S}_2$ contains the remaining 1-supporters. However, we cannot simply put the first half of 0-supporters in $\mathcal{S}_1$ and the rest in $\mathcal{S}_3$. To get a tighter bound, we need to make the total bet on 0 in $\mathcal{S}_1$ and $\mathcal{S}_3$ as balanced as possible, as in the above example of $\mathcal{P}^*$. Therefore, we will partition as follows (See Figure 4 for an example).

Let $\mathbf{P}_0$ and $\mathbf{P}_1$ be the 0- and 1-supporters after sorting, respectively. Let $\eta_1 = \eta_0(\theta) + 1$ and $\eta_3 = \eta_\theta(\theta) + 1 \geq \eta_1$ where $\eta(\cdot)$ is defined in Section 2.2. Intuitively, η_i represents the minimum number of 0-supporters such that the 1-supporters in $\mathcal{S}_i$ together with η_i number of 0-supporters prefers 0 for $i \in \{1, 3\}$.

- $\mathcal{S}_1$ contains the first θ number of 1-supporters $\mathbf{P}_1[1, \theta]$, and half of 0-supporters:

$$\mathbf{P}_0[1, \lceil \eta_1/2 \rceil] \cup \mathbf{P}_0[\eta_1 + 1, \eta_1 + \lceil \eta_3/2 \rceil - \lceil \eta_1/2 \rceil] \cup \mathbf{P}_0[N_0/2 + (\lceil \eta_3/2 \rceil), N_0]$$

- $\mathcal{S}_3$ contains the remaining 0-supporters and the next θ number of 1-supporters $\mathbf{P}_1[\theta + 1, 2\theta]$.
- $\mathcal{S}_2$ contains the remaining 1-supporters.

With this partition, we require θ to satisfy the following. Let γ' be the minimum number such that $\eta_0(\gamma') > N_0/2$, i.e., any γ' number of 1-supporters together with the first half of 0-supporters do *not* prefer 0. As indicated by the concrete example with $\mathcal{P}^*$ above, the main constraints are posed by the lone-wolf condition and the wolf-minion condition.

1. *Non-negativity:* $0 \leq 2\theta \leq N_1$ to ensure that the partition is well-defined.
2. *Lone-wolf condition:* For $i \in \{1, 3\}$, we need to build two coalitions $\mathcal{C}$ and $\mathcal{C}'$, both containing $\mathcal{S}_i$ and both have size no more than t such that $\mathcal{C}$ prefers 0 and $\mathcal{C}'$ prefers 1. For this, we need 1.) $t \geq |\mathcal{S}_i| + \lceil \eta_i/2 \rceil$ to construct coalition $\mathcal{C}$ containing $\mathcal{S}_i$ and $\lceil \eta_i/2 \rceil$ number of 0-supporters outside $\mathcal{S}_i$ to prefer 0; and 2.) $t \geq |\mathcal{S}_i| + \gamma' - \theta$ to construct $\mathcal{C}'$ containing $\mathcal{S}_i$ and $\gamma' - \theta$ number of 1-supporters outside (if necessary) to prefer 1.

3. *Wolf-minion condition:* We need $t \geq |\mathcal{S}_i| + |\mathcal{S}_2|$ for $i \in \{1, 3\}$. In case $\mathcal{S}_i$ together with $\mathcal{S}_2$ does not prefer 1, we also require $t \geq N_0/2 + \gamma'$ to form a coalition $\mathcal{C}$ containing $\mathcal{S}_i$ and $\mathcal{S}_3$, as well as $\gamma' - (N_1 - \theta)$ number of 1-supporters outside such that $\mathcal{C}'$ that prefers 1.

Combining all these constraints, the impossibility result reduces to yet another optimization problem of minimizing t subject to the above non-negativity, lone-wolf condition, and the wolf-minion condition. The optimal t is given by

$$t = \max\{N_1 + N_0/2 - \theta^*, \gamma' + \lceil N_0/2 \rceil\},$$

where θ^* is the maximum number such that $2\theta + \lceil(\eta_\theta(\theta) + 1)/2\rceil \leq N_1$. The formal proofs can be found in Section 8.

3 Preliminaries and Model

Notations. We use $\mathbb{Z}_{\geq 0}$ for non-negative integers and $[N]$ for the set of integers $\{1, \dots, N\}$. Similarly, we use $[m]_0$ for set of integers $\{0, \dots, m-1\}$. Additionally, $[i, j]$ denotes the set of all numbers between i and j , both ends included.

We use boldface letters to denote vectors. All vectors are 1-indexed. For a vector $\mathbf{V}$ and $i, j \in [|\mathbf{V}|]$, let $\mathbf{V}[i]$ denote the i -th entry and $\mathbf{V}[i, j]$ denote the i -th to j -th entry, both sides included.

We use λ as the security parameter. Each player is modeled as probabilistic polynomial time (ppt) Turing machines and receives 1^λ as an additional input.

3.1 Coin Toss Protocol

In an N -party coin toss protocol, N players want to jointly decide a random m -sided coin. We use the term “party” and “player” interchangeably. Players can communicate with each other through pairwise private channel as well as a broadcast channel. All the communications are authenticated, i.e., each message carries the identity of the true sender. In this work, we assume that the network is synchronous, and the protocol proceeds in rounds. At the end of the protocol, the final coin outcome is a deterministic, polynomial-time computable function based on the whole broadcast history. Since players are computationally bounded, we require m and N to be at most $\text{poly}(\lambda)$ for some polynomial function $\text{poly}(\cdot)$.

Correctness. In an m -sided coin toss protocol, the outcome must be a uniformly random m -sided coin if every player behaves honestly.

Preferences and utilities. In an m -sided coin toss protocol, each player has a utility function that maps each of the m outcomes $\{0, \dots, m-1\}$ to a non-negative integer units of utility. In this paper, we use *bet* or utility interchangeably to describe each player’s utility associated with each outcome. For example, a player with bet 5 on outcome 3 will receive 5 units of utility if the protocol outputs 3. We use $\text{bet}_i(\ell)$ to denote player i ’s bet on outcome ℓ . A player i ’s *preference* $\mathbf{p}_i$ is simply the vector of its bet on all possible outcomes: $\mathbf{p}_i = (\text{bet}_i(0), \dots, \text{bet}_i(m-1))$. We also define the *total bet* of a player i as $\text{total}(\mathbf{p}_i) = \sum_{\ell=0}^{m-1} \text{bet}_i(\ell)$. As before, for each player i , we have $\text{total}(\mathbf{p}_i) \leq \text{poly}(\lambda)$ for some polynomial function $\text{poly}(\cdot)$. A *preference profile* is then a list of all of the players’ preferences $\mathcal{P} = (\mathbf{p}_1, \dots, \mathbf{p}_N)$.

Some of the players may form a coalition, also referred to as the adversary in our cryptography analysis. A coalition’s utility is the sum of all the coalition members’ utility. Specifically, for a coalition $\mathcal{C}$, its utility when the final outcome is ℓ equals $\text{bet}_{\mathcal{C}}(\ell) = \sum_{i \in \mathcal{C}} \text{bet}_i(\ell)$. The preference

of a coalition $\mathcal{C}$ is defined analogously as $\mathbf{p}_{\mathcal{C}} = (\text{bet}_{\mathcal{C}}(0), \dots, \text{bet}_{\mathcal{C}}(m-1))$. Similarly, $\text{total}(\mathbf{p}_{\mathcal{C}}) = \sum_{\ell=0}^{m-1} \text{bet}_{\mathcal{C}}(\ell)$.

We say that a coalition “prefers” an outcome ℓ , if the coalition’s utility on outcome ℓ is greater than the honest expected utility, i.e., a coalition $\mathcal{C}$ prefers outcome ℓ iff $\text{bet}_{\mathcal{C}}(\ell) > \frac{1}{m} \text{total}(\mathbf{p}_{\mathcal{C}})$. If player i prefers ℓ , we call i an ℓ -supporter.

We use *cumulative bet* of an outcome ℓ to denote the sum of all players’ bets on the outcome $\sum_{i=1}^N \text{bet}_i(\ell)$. The least preferred outcome is the one with the least cumulative bet. In this paper, we only consider *non-degenerate* preference profiles, where there is at least one ℓ -supporter for every outcome $\ell \in [m]_0$. For degenerate preference profiles, the protocol from Thyagarajan, Soni, and Wu [TSW24] achieves CSP fairness against $(N-1)$ -sized coalition.

Strategies. A strategic coalition can perform a *rushing attack*. In each round r , the coalition can wait for messages from honest players and then decide what messages to send in round r . Additionally, a *semi-malicious* coalition can perform semi-malicious strategies, which includes the following types of deviations: 1.) program the randomness used in the protocol after observing honest players’ messages; 2.) abort the protocol in some round r , after observing the honest players’ messages in round r . Once a player aborted, it does not participate in future rounds. A *malicious* coalition can perform the above two kinds of strategies, as well as sending arbitrary messages.

CSP fairness. Recall that *Cooperative-strategy-proofness* (or “CSP-fairness” for short) requires that no coalition can increase its own expected utility no matter what strategy the coalition utilizes.

Definition 3.1 (CSP-fairness [CGL⁺18]). A coin toss protocol Π satisfies *CSP-fairness* against any t -sized coalitions with respect to preference profile $\mathbf{P}$ iff for all coalitions $\mathcal{C} \subseteq [N]$ such that $|\mathcal{C}| \leq t$, for any non-uniform ppt strategy $S'_{\mathcal{C}}$, there is a negligible function $\text{negl}(\cdot)$ such that

$$U_{\mathcal{C}}(S'_{\mathcal{C}}, S^H_{-\mathcal{C}}) \leq \frac{\text{total}(\mathbf{P}_{\mathcal{C}})}{m} + \text{negl}(\lambda),$$

Where $U_{\mathcal{C}}(S'_{\mathcal{C}}, S^H_{-\mathcal{C}})$ is the expected joint utility of coalition $\mathcal{C}$ in an execution of Π when using strategy $S'_{\mathcal{C}}$ while the rest players behaves honestly.

3.2 Different formulations.

In this work, we consider the following two formulations. A *formulation* is the set of all feasible preference profiles. Here, we only consider atomic formulations where each player’s preference is restricted in the same way and independent of other players’ preferences.

Arbitrary-bet. The most general formulation allows each player to have an arbitrary non-negative integer utility on each of the m possible outcomes. Formally, each player’s preference can be described as $\mathbf{p} \in \mathbb{Z}_{\geq 0}^m$. This formulation is simply

$$\mathcal{F}_{\text{arb}}(N, m) := (\mathbb{Z}_{\geq 0}^m)^N.$$

Split-bet. This is a rather restricted formulation where each player has the same total bet of B that can be split across the m possible outcomes. In this formulation, each players’ preferences can be described as a vector $\mathbf{p} = (b_0, \dots, b_{m-1})$ such that $\sum_{\ell=0}^{m-1} b_{\ell} = B$ for the same B . Formally,

$$\mathcal{F}_{\text{split}}(N, m) := \{(\mathbf{p}_1, \dots, \mathbf{p}_N) \in (\mathbb{Z}_{\geq 0}^m)^N \mid \text{total}(\mathbf{p}_i) \text{ are the same for all } i \in [N]\}.$$

Single-bet. In this formulation, each player put bets only on one outcome, but can have different total bet. Equivalently, each player i 's preference can be described as $\mathbf{p}_i = b_i \cdot \mathbf{e}_{\ell_i}$ for some $\ell_i \in [m]_0$, where $\mathbf{e}_\ell$ is the ℓ -th standard basis vector such that only the ℓ -th entry is 1 and all other entries are 0. This formulation is formally defined as

$$\mathcal{F}_{\text{single}}(N, m) := \{(\mathbf{p}_1, \dots, \mathbf{p}_N) \in (\mathbb{Z}_{\geq 0}^m)^N \mid \text{Each } \mathbf{p}_i = b_i \cdot \mathbf{e}_{\ell_i} \text{ for some } b_i \geq 0 \text{ and } \ell_i \in [m]_0\}.$$

4 Reducibility Between Formulations

In this section, we show that the problem of achieving CSP-fairness against t -sized coalitions for a particular arbitrary-bet preference profile is reducible to the achieving CSP-fairness against t -sized coalitions for a particular split-bet preference profile and vice versa. Specifically, when $m = 2$, arbitrary-bet is equivalent to single-bet formulation.

For a preference profile $\mathcal{P}$ and a protocol Π , let $t(\mathcal{P}, \Pi)$ be the maximum value t such that when the players with preferences $\mathcal{P}$ run the protocol Π , the protocol achieves CSP fairness against any t -sized coalitions. We use $\mathbf{1}_m$ to denote the vector $(1, \dots, 1)$ of length m .

Definition 4.1 (Reducibility). Fix any N and m . Given two formulations $\mathcal{F}_1(N, m)$ and $\mathcal{F}_2(N, m)$, we say that a formulation $\mathcal{F}_1$ is *reducible* to $\mathcal{F}_2$ iff there is an efficient transformation $\mathcal{T} : \mathcal{F}_1(N, m) \mapsto \mathcal{F}_2(N, m)$, such that for any protocol Π_2 for $\mathcal{F}_2(N, m)$, we can build a protocol Π_1 for $\mathcal{F}_1(N, m)$ satisfying the following:

- On any $\mathcal{P} \in \mathcal{F}_1(N, m)$, protocol Π_1 runs Π_2 on $\mathcal{T}(\mathcal{P})$;
- The size of the coalition Π_1 can tolerate satisfies that $t(\mathcal{P}, \Pi_1) = t(\mathcal{T}(\mathcal{P}), \Pi_2)$.

The existence of a transformation that satisfies Definition 4.1 is non-trivial. As explained in Section 2.1, the transformation must preserve the relative preference of all possible coalitions. This would guarantee that if the coalition can increase its expected utility in Π_2 run on $\mathcal{T}(\mathcal{P})$, then it must be able to increase its expected utility in Π_1 run on $\mathcal{P}$.

Definition 4.2 (Preference-preserving transformation). Fix N and m . We say that a transformation $\mathcal{T} : \mathcal{F}_1(N, m) \mapsto \mathcal{F}_2(N, m)$ is *preference preserving* iff for a preference profile $\mathcal{P} = (\mathbf{p}_1, \dots, \mathbf{p}_N) \in \mathcal{F}_1(N, m)$, the transformation maps each player's preference $\mathbf{p}_i$ in to $a\mathbf{p}_i + \mathbf{b}_i$ where $a \in \mathbb{Z}_{>0}$ is the same constant scalar across all $i \in [N]$ and $\mathbf{b}_i = b_i \cdot \mathbf{1}_m \in \mathbb{Z}_{\geq 0}^m$ is a length m vector where all entries equals some $b_i \geq 0$.

Intuitively, for each player or coalition, preference-preserving transformations maintain the differences between bets on each outcome by some scalar factor.

Lemma 4.3 (Transitivity of preference-preserving transformation). *Fix N and m . If there exists a preference transformation $\mathcal{T} : \mathcal{F}_1(N, m) \mapsto \mathcal{F}_2(N, m)$ and another preference-preserving transformation $\mathcal{T}' : \mathcal{F}_2(N, m) \mapsto \mathcal{F}_3(N, m)$, then there is a preference-preserving transformation $\tilde{\mathcal{T}} = \mathcal{T}' \circ \mathcal{T}$ that maps formulation $\mathcal{F}_1(N, m)$ to $\mathcal{F}_3(N, m)$, where $\circ$ means functional composition.*

Proof. By definition, given a $\mathcal{P} = (\mathbf{p}_1, \dots, \mathbf{p}_N) \in \mathcal{F}_1$, the transformation $\mathcal{T}$ maps each player's preference $\mathbf{p}_i$ into $\mathbf{p}'_i = a\mathbf{p}_i + \mathbf{b}_i$. Similarly, $\mathcal{T}'$ will then maps each $\mathbf{p}'_i$ to $\tilde{\mathbf{p}}_i = a'\mathbf{p}'_i + \mathbf{b}'_i$. Thus, the composition $\tilde{\mathcal{T}}$ will map $\mathbf{p}_i$ into $a \cdot a'\mathbf{p}_i + (a' \cdot \mathbf{b}_i + \mathbf{b}'_i)$. By definition, this is still a preference-preserving transformation. $\square$

Lemma 4.4. *Fix N and m . A formulation $\mathcal{F}_1(N, m)$ is reducible to $\mathcal{F}_2(N, m)$ if there exists an efficient transformation $\mathcal{T} : \mathcal{F}_1(N, m) \mapsto \mathcal{F}_2(N, m)$ that is preference-preserving.*

Proof. Throughout the proof, we use $\Pi^{\mathcal{C}, S_{\mathcal{C}}}(\mathcal{P})$ to denote an execution of protocol Π among N players with preferences $\mathcal{P}$, where coalition $\mathcal{C}$ adopts strategy $S_{\mathcal{C}}$ and all other players behave honestly. We use $\Pr[\Pi^{\mathcal{C}, S}(\mathcal{P}) = \ell]$ to denote the probability that the above execution outputs ℓ . Also, let $\text{bet}_{\mathcal{C}}^{\mathcal{P}}(\ell)$ denote the bet of coalition $\mathcal{C}$ on ℓ for preference profile $\mathcal{P}$.

Given a preference-preserving transformation $\mathcal{T}$ that maps $\mathbf{p}_i$ to $a\mathbf{p}_i + \mathbf{b}_i$ where $\mathbf{b}_i = b_i \cdot \mathbf{1}_m$, we want to show that for any protocol Π_2 for $\mathcal{F}_2(N, m)$ and any $\mathcal{P} \in \mathcal{F}_1(N, m)$, and protocol Π_1 as defined in Definition 4.1, we have $t(\mathcal{P}, \Pi_1) = t(\mathcal{T}(\mathcal{P}), \Pi_2)$. For the sake of contradiction, suppose there exists some protocol Π_2 for $\mathcal{F}_2(N, m)$ and $\mathcal{P} = (\mathbf{p}_1, \dots, \mathbf{p}_n) \in \mathcal{F}_1(N, m)$ such that $t(\mathcal{P}, \Pi_1) \neq t(\mathcal{T}(\mathcal{P}), \Pi_2)$.

Case 1: $t(\mathcal{P}, \Pi_1) < t(\mathcal{T}(\mathcal{P}), \Pi_2)$. This means that there exists a coalition $\mathcal{C}$ of size $t(\mathcal{P}, \Pi_1) + 1$ and some strategy $S_{\mathcal{C}}^*$ such that $\mathcal{C}$'s expected utility increases by a non-negligible amount in an execution $\Pi_1^{\mathcal{C}, S_{\mathcal{C}}^*}(\mathcal{P})$. Equivalently, this means that, in an execution where $\mathcal{C}$ adopts $S_{\mathcal{C}}^*$ and all other players behave honestly,

$$\sum_{\ell=0}^m \left(\Pr[\Pi_1^{\mathcal{C}, S_{\mathcal{C}}^*}(\mathcal{P}) = \ell] \cdot \text{bet}_{\mathcal{C}}^{\mathcal{P}}(\ell) \right) \geq \frac{\text{total}(\mathbf{p}_{\mathcal{C}})}{m} + \frac{1}{q(\lambda)}, \quad (5)$$

where $q(\cdot)$ is some polynomial. Since Π_1 runs Π_2 on $\mathcal{T}(\mathcal{P})$, the same coalition can perform the same strategy $S_{\mathcal{C}}^*$ in protocol Π_2 on preferences $\mathcal{T}(\mathcal{P})$. The coalition's expected utility in an execution of $\Pi_2^{\mathcal{C}, S_{\mathcal{C}}^*}(\mathcal{T}(\mathcal{P}))$ is,

$$\begin{aligned} & \sum_{\ell=0}^m \left(\Pr[\Pi_2^{\mathcal{C}, S_{\mathcal{C}}^*}(\mathcal{T}(\mathcal{P})) = \ell] \cdot \text{bet}_{\mathcal{C}}^{\mathcal{T}(\mathcal{P})}(\ell) \right) \\ &= \sum_{\ell=0}^m \left(\Pr[\Pi_1^{\mathcal{C}, S_{\mathcal{C}}^*}(\mathcal{P}) = \ell] \cdot (a \cdot \text{bet}_{\mathcal{C}}^{\mathcal{P}}(\ell) + b) \right) \quad \text{where } b = \sum_{i \in \mathcal{C}} b_i \\ &= \sum_{\ell=0}^m \left(a \cdot (\Pr[\Pi_1^{\mathcal{C}, S_{\mathcal{C}}^*}(\mathcal{P}) = \ell] \cdot \text{bet}_{\mathcal{C}}^{\mathcal{P}}(\ell)) + \Pr[\Pi_1^{\mathcal{C}, S_{\mathcal{C}}^*}(\mathcal{P}) = \ell] \cdot b \right) \\ &= a \cdot \sum_{\ell=0}^m \left(\Pr[\Pi_1^{\mathcal{C}, S_{\mathcal{C}}^*}(\mathcal{P}) = \ell] \cdot \text{bet}_{\mathcal{C}}^{\mathcal{P}}(\ell) \right) + b \\ &\geq a \cdot \left(\frac{\text{total}(\mathbf{p}_{\mathcal{C}})}{m} + \frac{1}{q(\lambda)} \right) + b \quad \text{By Equation (5)} \\ &= \frac{\text{total}(\mathbf{p}'_{\mathcal{C}})}{m} + \frac{1}{q^*(\lambda)} \end{aligned}$$

where $\mathbf{p}'_{\mathcal{C}} = (\text{bet}_{\mathcal{C}}^{\mathcal{T}(\mathcal{P})}(0), \dots, \text{bet}_{\mathcal{C}}^{\mathcal{T}(\mathcal{P})}(m-1))$ and $q^*(\lambda) = \frac{q(\lambda)}{a}$. Since $q^*(\cdot)$ is also a polynomial, this contradicts CSP fairness of Π_2 against coalition of size $t(\mathcal{T}(\mathcal{P}), \Pi_2)$.

Case 2: $t(\mathcal{P}, \Pi_1) > t(\mathcal{T}(\mathcal{P}), \Pi_2)$. Similar to the argument above. Here, there must exist a coalition $\mathcal{C}$ of size $t(\mathcal{T}(\mathcal{P}), \Pi_2) + 1$ and some strategy $S_{\mathcal{C}}^*$ such that $\mathcal{C}$'s expected utility increases by a non-negligible amount in an execution $\Pi_2^{\mathcal{C}, S_{\mathcal{C}}^*}(\mathcal{T}(\mathcal{P}))$. Equivalently, this means that, in an execution where $\mathcal{C}$ adopts $S_{\mathcal{C}}^*$ and all other players behave honestly,

$$\sum_{\ell=0}^m \left(\Pr[\Pi_2^{\mathcal{C}, S_{\mathcal{C}}^*}(\mathcal{T}(\mathcal{P})) = \ell] \cdot \text{bet}_{\mathcal{C}}^{\mathcal{T}(\mathcal{P})}(\ell) \right) \geq \frac{\text{total}(\mathbf{p}'_{\mathcal{C}})}{m} + \frac{1}{q(\lambda)}, \quad (6)$$

where $q(\cdot)$ is some polynomial and $\mathbf{p}'_C = (\text{bet}_C^{\mathcal{T}(\mathcal{P})}(0), \dots, \text{bet}_C^{\mathcal{T}(\mathcal{P})}(m-1))$. Since Π_1 runs Π_2 on $\mathcal{T}(\mathcal{P})$, the same coalition can perform the same strategy S_C^* in protocol Π_1 on preferences $\mathcal{P}$. The coalition's expected utility in an execution of $\Pi_1^{C, S_C^*}(\mathcal{P})$ is,

$$\begin{aligned}
& \sum_{\ell=0}^m \left(\Pr[\Pi_1^{C, S_C^*}(\mathcal{P}) = \ell] \cdot \text{bet}_C^{\mathcal{P}}(\ell) \right) \\
&= \frac{1}{a} \cdot \sum_{\ell=0}^m \left(\Pr[\Pi_2^{C, S_C^*}(\mathcal{T}(\mathcal{P})) = \ell] \cdot \text{bet}_C^{\mathcal{T}(\mathcal{P})}(\ell) \right) - b \quad \text{where } b = \sum_{i \in C} b_i \\
&\geq \frac{1}{a} \cdot \left(\frac{\text{total}(\mathbf{p}'_C)}{m} + \frac{1}{q(\lambda)} \right) - b \quad \text{By Equation (6)} \\
&= \frac{\text{total}(\mathbf{p}_C)}{m} + \frac{1}{q^*(\lambda)}
\end{aligned}$$

where $q^*(\lambda) = a \cdot q(\lambda)$. Since $q^*(\cdot)$ is also a polynomial, this contradicts CSP fairness of Π_1 against coalition of size $t(\mathcal{P}, \Pi_1)$. $\square$

Reducibility Between Split-bet and Arbitrary-bet. With the above lemma, we only need to build preference-preserving transformations between both formulations. Below we give the formal proof that the split-bet and arbitrary-bet formulations are reducible to each other.

Theorem 4.5. *For any N and m , $\mathcal{F}_{\text{split}}(N, m)$ and $\mathcal{F}_{\text{arb}}(N, m)$ are reducible to each other.*

Proof. Showing that $\mathcal{F}_{\text{split}}(N, m)$ is reducible to $\mathcal{F}_{\text{arb}}(N, m)$ can be done using the identity transformation $\mathcal{I}(\mathcal{P}) = \mathcal{P}$ because $\mathcal{F}_{\text{split}}(N, m) \subset \mathcal{F}_{\text{arb}}(N, m)$.

By Lemma 4.4, to show $\mathcal{F}_{\text{arb}}(N, m)$ is reducible to $\mathcal{F}_{\text{split}}(N, m)$, we only need to give a transformation $\mathcal{T}_{\text{as}} : \mathcal{F}_{\text{arb}}(N, m) \mapsto \mathcal{F}_{\text{split}}(N, m)$ that is preference-preserving. Consider the transformation $\mathcal{T}_{\text{as}}$, on input $\mathcal{P}$, that maps player i 's preference $\mathbf{p}_i$ to $\mathbf{p}'_i$:

$$\mathbf{p}'_i = m \cdot \mathbf{p}_i + (B(\mathcal{P}) - \text{total}(\mathbf{p}_i)) \mathbf{1}_m, \quad (7)$$

where $B(\mathcal{P}) = \max_{j \in [n]} \{\text{total}(\mathbf{p}_j)\}$. Clearly, $\mathcal{T}_{\text{as}}$ can be efficiently computed.

Claim 4.6. *If $\mathcal{P} \in \mathcal{F}_{\text{arb}}(N, m)$, then $\mathcal{T}_{\text{as}}(\mathcal{P}) \in \mathcal{F}_{\text{split}}(N, m)$ and each player's total bet is $m \cdot B(\mathcal{P})$.*

Proof. Suppose the transformation $\mathcal{T}_{\text{as}}$ maps $\mathbf{p}_i$ is mapped to $\mathbf{p}'_i$ according to Equation (7). Then

$$\text{total}(\mathbf{p}'_i) = m \cdot \text{total}(\mathbf{p}_i) + m \cdot (B(\mathcal{P}) - \text{total}(\mathbf{p}_i)) = m \cdot B(\mathcal{P})$$

This value is the same for all players in the transformed preference profile. $\square$

Thus, by Lemma 4.4 we have shown that $\mathcal{F}_{\text{arb}}(N, m)$ is reducible to $\mathcal{F}_{\text{split}}(N, m)$. $\square$

Equivalence Between Single-bet and Constrained Arbitrary-bet We will prove the equivalence between single-bet and a specific subset of arbitrary-bet called **onearb**. Consider the formulation $\mathcal{F}_{\text{onearb}}(N, m) \subseteq \mathcal{F}_{\text{arb}}(N, m)$ where each player gets the highest utility from one specific outcome, possibly different for each player, and gets the same utility from all other outcomes. Formally,

$$\mathcal{F}_{\text{onearb}}(N, m) := \{(\mathbf{p}_1, \dots, \mathbf{p}_N) \in (\mathbb{Z}_{\geq 0}^m)^N \mid \text{Each } \mathbf{p}_i = a_i \cdot \mathbf{1}_m + b_i \cdot \mathbf{e}_{\ell_i} \text{ for } a_i, b_i \geq 0, \ell_i \in [m]_0\}.$$

For instance, the preference profile $((4, 1, 1), (2, 2, 2), (0, 0, 6)) \in \mathcal{F}_{\text{onearb}}(3, 3)$. The reason we consider this specialized **onearb** formulation is that for the special case where $m = 2$, the formulation $\mathcal{F}_{\text{onearb}}(N, 2)$ is exactly the same as the arbitrary formulation $\mathcal{F}_{\text{arb}}(N, 2)$.

Theorem 4.7. *For any N and m , $\mathcal{F}_{\text{onearb}}(N, m)$ is equivalent to $\mathcal{F}_{\text{single}}(N, m)$.*

Proof. To show that $\mathcal{F}_{\text{single}}(N, m)$ is reducible to $\mathcal{F}_{\text{onearb}}(N, m)$, we can use the identity transformation $\mathcal{I}(\mathcal{P}) = \mathcal{P}$. This works because $\mathcal{F}_{\text{single}}(N, m) \subset \mathcal{F}_{\text{onearb}}(N, m)$.

To show that $\mathcal{F}_{\text{onearb}}(N, m)$ is reducible to $\mathcal{F}_{\text{single}}(N, m)$, we define a transformation $T_{\text{os}} : \mathcal{F}_{\text{onearb}}(N, m) \mapsto \mathcal{F}_{\text{single}}(N, m)$. This transformation maps player i 's preference $\mathbf{p}_i = a_i \cdot \mathbf{1}_m + b_i \cdot \mathbf{e}_{\ell_i}$ to $\mathbf{p}'_i$ where

$$\mathbf{p}'_i = \mathbf{p}_i - (a_i)\mathbf{1}_m.$$

Clearly, T_{os} can be efficiently computed. Also, if $\mathcal{P} \in \mathcal{F}_{\text{onearb}}$, then $T_{\text{os}}(\mathcal{P}) \in \mathcal{F}_{\text{single}}$ as $\mathbf{p}'_i = b_i \cdot \mathbf{e}_{\ell_i}$ for some $\ell_i \in [m]_0$. Thus by Lemma 4.4, $\mathcal{F}_{\text{onearb}}$ is equivalent to $\mathcal{F}_{\text{single}}(N, m)$. $\square$

Corollary 4.8. *For any N and m , $\mathcal{F}_{\text{single}}(N, 2)$ is equivalent to $\mathcal{F}_{\text{split}}(N, 2)$.*

Proof. To see that $\mathcal{F}_{\text{single}}(N, 2)$ is reducible to $\mathcal{F}_{\text{split}}(N, 2)$, one can consider the transformation T_{as} defined in Equation (7). Specifically, $\mathcal{F}_{\text{single}}(N, 2) \subset \mathcal{F}_{\text{onearb}}(N, 2) = \mathcal{F}_{\text{arb}}(N, 2)$, therefore, by the same reasoning as in Theorem 4.5, the transformation T_{as} maps a preference profile in $\mathcal{F}_{\text{single}}(N, 2)$ to $\mathcal{F}_{\text{split}}(N, 2)$.

On the other hand, there is a preference-preserving transformation that maps $\mathcal{F}_{\text{split}}(N, 2)$ to $\mathcal{F}_{\text{arb}}(N, 2) = \mathcal{F}_{\text{onearb}}(N, 2)$. Since Theorem 4.7 presents a preference-preserving transformation from $\mathcal{F}_{\text{onearb}}(N, 2)$ to $\mathcal{F}_{\text{single}}$, by Lemma 4.3, we have that $\mathcal{F}_{\text{split}}(N, 2)$ is reducible to $\mathcal{F}_{\text{single}}(N, 2)$. $\square$

5 Size-Based Protocol for Split-Bet Formulation

In this section, we give a CSP-fair coin toss protocol for split-bet preference profiles against semi-malicious coalitions. By Theorem 4.5, this also yields a protocol for arbitrary-bet preference profiles. Since Blum's coin toss protocol [Blu83] achieves CSP fairness for two-party case against one corrupted player, we will focus on the case where $N \geq 3$. Without loss of generality, let 0 be the outcome with the least bets. Throughout this section, we will use a running example preference profile $\mathcal{P}^*$ for binary coin toss to help understand the protocol construction and the parameter optimization. The preference profile

$$\mathcal{P}^* = ((3, 0), (3, 0), (1, 2), (1, 2), (0, 3), (0, 3), (0, 3), (0, 3), (0, 3)) \in \mathcal{F}_{\text{split}}(9, 2).$$

We also use an abbreviation as $\{2@ (3, 0), 2@ (1, 2), 5@ (0, 3)\}$ to indicate that there are two players with preference (3, 0), two players with preference (1, 2), and the rest five players with preference (0, 3). Recall that N_0 denotes the number of 0-supporters who prefer outcome 0, and N_{oth} denote the number of the rest players who do not prefer outcome 0. For the specific example $\mathcal{P}^*$, we have $N_0 = 2$ and $N_{\text{oth}} = 7$.

As discussed in Section 2.2, we do not simply partition players based on whether they prefer 0 or not. Instead, our group G_0 contains all the 0-supporters and some swing oth-supporters. The rest aggressive oth-supporters goes to group G_{oth} . Each group G_b for $b \in \{0, \text{oth}\}$ will decide a random coin s_b using Shamir secret sharing. The final outcome is then the sum of s_0 and s_{oth} . Specifically, if the reconstruction of s_0 fails, we simply output s_{oth} ; but if the reconstruction of s_{oth} fails, we will output 0 as a punishment.

5.1 size-GroupToss^G[k] Sub-Protocol

Consider the size-GroupToss^G[k] sub-protocol defined below where G is the set of players running the protocol and k is a threshold parameter to be determined later. Specifically, the size-GroupToss^G[k] sub-protocol run among the players in G to decide a coin c while players outside G audit the protocol by observing the broadcast messages.

Protocol 5.1: size-GroupToss^G[k] sub-protocol (Semi-Malicious)

Let $\mathcal{O}$ be the set of online players, initialized as G .

Sharing Phase. Each player i chooses a random coin $\text{coin}_i \xleftarrow{\$} [m]_0$ and shares it using $(k+1)$ -out-of- $|G|$ Shamir secret sharing into $\{c_{i,j}\}_{j \in G}$ among the players in G . Send $c_{i,j}$ to player j . If any players abort, remove them from $\mathcal{O}$.

Reconstruction Phase. Every player i computes the sum of the shares it received $c_i := \sum_{j \in \mathcal{O}} c_{j,i}$ and broadcast c_i . Players use the broadcast shares to reconstruct $s = \sum_{i \in \mathcal{O}} \text{coin}_i$. If the reconstruction fails, set $s = \perp$; otherwise, output $s \bmod m$.

Lemma 5.2 (Lemma 4.2 of [TSW24]). *size-GroupToss^G[k] satisfies the following properties for any coalition $\mathcal{C}$ due to the property of Shamir secret sharing.*

- *Binding.* The sharing phase uniquely binds to a secret s , such that the reconstruction phase either succeeds and outputs s , or it fails and outputs $\perp$.
- *Knowledge threshold.* If $|\mathcal{C}| \geq k+1$, the coalition can completely control binded value s of the sharing phase. Otherwise, the binded secret s looks uniformly random to the coalition.
- *Liveness threshold.* If $|\mathcal{C}| \geq |G| - k$, the coalition can cause the reconstruction to fail and output $\perp$. Otherwise, the reconstruction phase must succeed.

5.2 Final Protocol for Coalition-Size

Protocol Parameters. Given a preference profile $\mathcal{P}$, we define some parameters that will be used in the protocol. We will explain how to choose these parameters later in Section 5.3 through optimization. Recall that we partition players into two groups G_0 and G_{oth} such that players in G_{oth} have “strong” preference against 0. We use γ to denote the number of oth-supporters that will be put into G_0 . Therefore, we have

$$|G_0| = N_0 + \gamma, \quad |G_{\text{oth}}| = N_{\text{oth}} - \gamma.$$

Given a preference profile $\mathcal{P}$, group G_0 consists of all 0-supporters and the γ number of oth-supporters who have the most bet on 0, breaking ties arbitrarily. In the example where $\mathcal{P}^* = \{2@ (3, 0), 2@ (1, 2), 5@ (0, 3)\}$, we will pick $\gamma^* = 2$, separating players into $G_0^* = \{2@ (3, 0), 2@ (1, 2)\}$ and $G_{\text{oth}}^* = \{5@ (0, 3)\}$. The coin toss protocol works as follows with the parameters k_0, k_{oth} and γ to be specified later.

Protocol 5.3: Size-based protocol (semi-malicious)

Input: The preference profile $\mathcal{P}$. **Sharing phase:**

1. G_0 runs the sharing phase of size-GroupToss^{G₀}[k₀]. G_{oth} audits.

2. If G_{oth} is non-empty, G_{oth} runs the sharing phase of $\text{size-GroupToss}^{G_{\text{oth}}}[k_{\text{oth}}]$. G_0 audits.

Reconstruction phase:

1. G_0 runs the reconstruction phase of $\text{size-GroupToss}^{G_0}[k_0]$ while G_{oth} audits. Let s_0 be the outcome of this protocol.
2. If G_{oth} is non-empty, G_{oth} runs the reconstruction phase of $\text{size-GroupToss}^{G_{\text{oth}}}[k_{\text{oth}}]$ while G_0 audits. Let s_{oth} be the outcome of this protocol. If G_{oth} is empty, let $s_{\text{oth}} = 0$.
3. If $s_{\text{oth}} = \perp$, output 0. Otherwise, if $s_0 = \perp$, set $s_0 = 0$ and output $(s_0 + s_{\text{oth}}) \bmod m$.

Coalition Resilience. For any fixed parameter k_0, k_{oth} , and γ , we can decide the coalition resilience of Protocol 5.3. Consider a coalition $\mathcal{C}$ that controls $t = t_0 + t_{\text{oth}}$ players where t_0 and t_{oth} denote the number of corrupted players in group G_0 and G_{oth} , respectively.

If $\gamma = N_{\text{oth}}$, i.e., G_{oth} is empty, the protocol degrades to a standard single-shot secret sharing and can tolerate only less than half corrupted players (and achieves strong fairness). In the following analysis, we focus on $\gamma < N_{\text{oth}}$.

For a group G_b for $b \in \{0, \text{oth}\}$ and any integer $0 \leq x \leq |G_b|$, let $\overline{M}(G_b, x)$ denote the x number of players with the most bet on 0 in G_b , breaking ties arbitrarily. For instance, when $G_0^* = \{2@ (3, 0), 2@ (1, 2)\}$, we have $\overline{M}(G_0^*, 3) = \{2@ (3, 0), 1@ (1, 2)\}$.

Definition 5.4. For any fixed $0 \leq \gamma < N_{\text{oth}}$, define $\eta_\gamma(\cdot)$ as:

$$\eta_\gamma(y) = \begin{cases} \max\{x \in [N_0] \mid \overline{M}(G_0, x) \cup \overline{M}(G_{\text{oth}}, y) \text{ not prefer } 0\} & \text{if } \overline{M}(G_0, N_0) \cup \overline{M}(G_{\text{oth}}, y) \text{ prefers } 0 \\ \infty & \text{otherwise.} \end{cases}$$

Let $\alpha = |G_{\text{oth}}| - k_{\text{oth}}$ be the number of corrupted oth-supporters needed to fail the reconstruction of s_{oth} . Intuitively, $\eta_\gamma(\cdot)$ is defined such that if a coalition controls more than $\eta_\gamma(\alpha)$ number of 0-supporters and α number of oth-players, then the coalition is able to fail the reconstruction of s_{oth} while still preferring 0. Such coalition could break the CSP-fairness. However, if the coalition only contains $\eta_\gamma(\alpha)$ number of 0-supporters and α number of oth-players, then this coalition does not prefer 0, and it is not incentivized to fail the reconstruction of s_{oth} . If $\eta_\gamma(\alpha) \geq N_0$, that implies if a coalition controls α number of oth-supporters, it shall never prefer 0.

Constraints 5.5. To achieve CSP fairness, $t = t_0 + t_{\text{oth}}$ needs to satisfy the following conditions:

- (C1) $\mathcal{C}$ cannot control both coins. Equivalently, either $t_0 \leq k_0$ or $t_{\text{oth}} \leq k_{\text{oth}}$. Otherwise, $\mathcal{C}$ can force the outcome to be whatever the coalition's overall preference is.
- (C2) If $\mathcal{C}$ controls s_{oth} , then it cannot fail the reconstruction of s_0 . That is, if $t_{\text{oth}} > k_{\text{oth}}$, then it must be that $t_0 < |G_0| - k_0$. Otherwise, $\mathcal{C}$ can control the outcome to be the s_{oth} coin it decides by failing the reconstruction of s_0 .
- (C3) If $\mathcal{C}$ can hamper the reconstruction of s_{oth} , then it must NOT prefer 0. That is, if $t_{\text{oth}} \geq |G_{\text{oth}}| - k_{\text{oth}}$, then $t_0 \leq \eta_\gamma(t_{\text{oth}})$. Otherwise, $\mathcal{C}$ can fail s_{oth} 's reconstruction to force the outcome 0, benefiting $\mathcal{C}$ compared to honest behavior.

Lemma 5.6. If (C1), (C2), (C3) are satisfied, then Protocol 5.3 achieves CSP-fairness against any coalition controlling no more than $t = t_0 + t_{\text{oth}}$ players.

Proof. By Lemma 5.2, for a coalition to fail the reconstruction of s_{oth} , we must have $t_{\text{oth}} \geq |G_{\text{oth}}| - k_{\text{oth}}$. Thus, condition (C3) implies that if the coalition fails the reconstruction of s_{oth} , the coalition's utility must be no more than its honest expected utility by the definition of $\eta_\gamma(\cdot) \geq t_0$. Therefore, any coalition is incentivized to successfully reconstruct s_{oth} .

Conditioned on s_{oth} being reconstructed, if the coalition controls the value of s_{oth} , i.e., $t_{\text{oth}} \geq k_{\text{oth}} + 1$, then by Condition (C1), $t_0 \leq k_0$. By Lemma 5.2, this means the s_o coin's value must be uniformly random to the coalition. Furthermore, (C2) stipulates that $t_0 < |G_0| - k_0$, so by Lemma 5.2, the coin s_0 is successfully reconstructed. Therefore, the protocol outcome must be uniformly random. If $t_{\text{oth}} \leq k_{\text{oth}}$, then s_{oth} must be a uniformly random coin to the coalition and successfully reconstructed. Thus, the outcome again is uniformly random, whether s_0 is successfully reconstructed or not. $\square$

Constraint 5.7: Coalition Size Parameter Constraints

Let $\alpha := |G_{\text{oth}}| - k_{\text{oth}}$.

$0 \leq \gamma < N_{\text{oth}}, 0 \leq k_0 \leq N_0 + \gamma - 1, 1 \leq \alpha \leq N_{\text{oth}} - \gamma$

(C1*) $t \leq k_0 + (N_{\text{oth}} - \gamma - \alpha) + 1$.

(C2*) $t \leq N - \alpha - k_0$.

(C3*) If $\eta_\gamma(\alpha) < N_0$ then $t \leq \eta_\gamma(\alpha) + \alpha$.

Lemma 5.8. *Protocol 5.3 with parameters k_0, k_{oth} , and γ achieves CSP fairness against any size- t coalition where t satisfies Constraint 5.7.*

Proof. We only need to show that Constraint 5.7 imply (C1), (C2), (C3). Recall that $k_{\text{oth}} = N_{\text{oth}} - \gamma - \alpha$. Constraint (C1*) implies Condition (C1) as in order to control both coins, a coalition would need at least $k_0 + k_{\text{oth}} + 2$ parties. Similarly, constraint (C2*) implies Condition (C2) as in order to control the s_{oth} coin and fail the s_0 coin, the coalition would need at least $(N_{\text{oth}} - \gamma - \alpha + 1) + (N_0 + \gamma - k_0) = N - \alpha - k_0$ parties. We now focus on proving that (C3*) implies (C3).

Observe that if $\eta_\gamma(\alpha) \geq N_0$, then when $t_{\text{oth}} \geq \alpha$, it must be that $\eta_\gamma(t_{\text{oth}}) \geq N_0 \geq t_0$. Therefore, Condition (C3) is satisfied. On the other hand, if $\eta_\gamma(\alpha) < N_0$, the constraint implies that $t \leq \eta_\gamma(\alpha) + \alpha$. Therefore, it must be that when $t_{\text{oth}} \geq \alpha$, we have $t_0 \leq t - t_{\text{oth}} \leq t - \alpha \leq \eta_\gamma(\alpha) \leq \eta_\gamma(t_{\text{oth}})$. $\square$

5.3 Optimizing Constraint 5.7

In this section, we explain how to set up the parameters k_0, k_{oth} and γ given the input preference profile $\mathcal{P}$ to maximize the coalition resistance of Protocol 5.3. For the majority of this section, we assume $\gamma \neq N_{\text{oth}}$, i.e., G_{oth} is non-empty.

Lemma 5.9. *For any fixed γ and α , setting $k_0 = \lfloor \frac{N_0 + \gamma}{2} \rfloor$ always achieves the optimal resilience of Protocol 5.3 against semi-malicious adversaries.*

Proof. Since k_0 does not appear in (C3*), for any fixed γ and α , we only need to pick k_0 that maximizes t subject to (C1*) and (C2*). Thus setting $k_0 = \lfloor \frac{N_0 + \gamma}{2} \rfloor$ gives the optimal resilience.

5.3.1 Optimization Allowing Non-Integer Solutions

Based on Lemma 5.9, we only need to find the optimal γ and α that maximizes t in Constraint 5.7. For simplicity, we will first present the proof assuming that the optimization allows non-integer solutions for α and $\eta_\gamma(\alpha)$. We will still consider only integer solutions for γ and t . Specifically, without the integer requirement, $\eta_\gamma(\alpha)$ is redefined as follows. Let $\mathbf{U}^0$ denote a vector of length N_0 where the i -th entry denotes the i -th highest bet on 0 among the 0-supporters. The vector $\mathbf{U}^{\text{oth}}$ is defined analogously: the i -th entry in $\mathbf{U}^{\text{oth}}$ is the i -th highest bet on 0 among the oth-supporters. Recall that in the split-bet formulation, every player has B total bet. Let $\mu = \frac{B}{m}$ be the expected honest utility of each player. Then every entry in $\mathbf{U}^0$ is bigger than μ while every entry in $\mathbf{U}^{\text{oth}}$ is at most μ . For the running example $\mathcal{P}^* = \{2\text{@}(3, 0), 2\text{@}(1, 2), 5\text{@}(0, 3)\}$, the two vectors $\mathbf{U}^0 = (3, 3)$ and $\mathbf{U}^{\text{oth}} = (1, 1, 0, 0, 0, 0)$.

For any $0 \leq i < j \leq N_b$, where $b \in \{0, \text{oth}\}$ and i, j need *not* to be integers, define $U_{[i,j]}^b$ as the sum of “the i -th to the j -th” largest utility on 0 in $\mathbf{U}^b$. Equivalently, $U_{[i,j]}^b$ denotes the joint utility of i -th to j -th b -supporters based on their utility on 0.

$$U_{[i,j]}^b := \left\{ \mathbf{U}^b[\lceil i \rceil] \cdot (\lceil i \rceil - i) + \mathbf{U}^b[\lfloor j \rfloor] \cdot (j - \lfloor j \rfloor) + \sum_{a=\lceil i \rceil+1}^{\lfloor j \rfloor} \mathbf{U}^b[a] \right\} \quad (8)$$

For completeness, if $\lceil i \rceil + 1 > \lfloor j \rfloor$, then we define $\sum_{a=\lceil i \rceil+1}^{\lfloor j \rfloor} \mathbf{U}^b[a] = \sum_{a=\lfloor j \rfloor+1}^{\lceil i \rceil} -\mathbf{U}^b[a]$. We also set a filler at the beginning of each vector such that $\mathbf{U}^b[0] := 0$. By the above definition, $U_{[0,x]}^0$ is exactly the utility of $\overline{M}(G_0, x)$ (allowing non-integer x) and $U_{[\gamma, \gamma+y]}^{\text{oth}}$ is exactly the utility of $\overline{M}(G_{\text{oth}}, y)$ (allowing non-integer y). For instance, for $\mathcal{P}^*$ and the corresponding $\mathbf{U}^0$ defined above, we have $U_{[0,1/3]}^0 = 1$.

Let $z := \lfloor \mu \rfloor + 1$, set $M_z := 0$ and $M_0 = N_{\text{oth}}$. We use n_j to denote the number of players who put bet j on outcome 0. For $i = z - 1, \dots, 1$, let $M_i = \sum_{j=i}^{z-1} n_j$. Roughly speaking, if we group all players with the same utility on 0 into one chunk, then M_i represents the index of each margin. See Figure 5 for an example on $\mathcal{P}^*$.

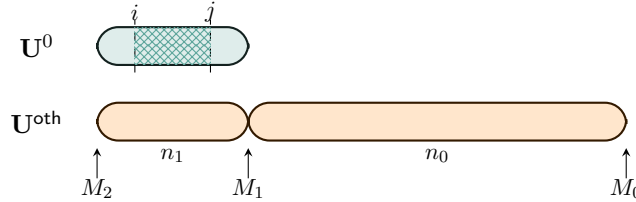


Figure 5: Illustration of variables M_i and $U_{[i,j]}^b$ for $\mathcal{P}^* = \{2\text{@}(3, 0), 2\text{@}(1, 2), 5\text{@}(0, 3)\}$. The green bar and orange bar represents $\mathbf{U}^0$ and $\mathbf{U}^{\text{oth}}$, respectively. The green hatch-shaded part corresponds to $U_{[i,j]}^b$. The variables are defines such that $N_0 = 2$ and $N_{\text{oth}} = 7$. The “margins” $M_2 = 0$, and $M_1 = n_1 = 2$, and $M_0 = n_1 + n_0 = 7$.

Given any $\gamma \leq N_{\text{oth}} - 1$ and any $1 \leq \alpha \leq N_{\text{oth}} - \gamma$, we abuse the notation and redefine $\eta_\gamma(\cdot)$ allowing non-integer solution:

$$\eta_\gamma(\alpha) := \begin{cases} x \text{ s.t. } U_{[0,x]}^0 + U_{[\gamma, \gamma+\alpha]}^{\text{oth}} = \mu \cdot (x + \alpha), & \text{if } U_{[0,N_0]}^0 + U_{[\gamma, \gamma+\alpha]}^{\text{oth}} \geq \mu \cdot (N_0 + \alpha) \\ +\infty, & \text{Otherwise} \end{cases} \quad (9)$$

Claim 5.10. For any fixed γ , $\eta_\gamma(\alpha)$ is an increasing function in α ; For any fixed α , $\eta_\gamma(\alpha)$ is an increasing function in γ .

Proof. The definition of $\eta_\gamma(\alpha)$ is the same as

$$U_{[0, \eta_\gamma(\alpha)]}^0 - \mu \cdot \eta_\gamma(\alpha) = \mu \cdot \alpha - U_{[\gamma, \gamma + \alpha]}^{\text{oth}} \quad (10)$$

For any fixed γ , when α increases, the right-hand side of Equation (10) increases. The right-hand side captures how much the $(\gamma+1)$ -th to $(\gamma+\alpha)$ -th *oth*-supporters do not like outcome 0. Therefore, the left-hand side needs to increase to make the equation holds. Therefore, $\eta_\gamma(\alpha)$ increases.

Similarly, for any fixed α , when γ increases, the right-hand side of Equation (10) increases. Thus, $\eta_\gamma(\alpha)$ increases. $\square$

Picking $k_0 = \frac{n_0 - \gamma}{2}$ by Lemma 5.9, the optimization problem becomes

$$\begin{aligned} & \text{Maximize } t \text{ subject to} \\ & 0 \leq \gamma \leq N_{\text{oth}} - 1, 1 \leq \alpha \leq N_{\text{oth}} - \gamma, \\ & t \leq t(\gamma, \alpha) := \min \{c_1(\gamma, \alpha), c_2(\gamma, \alpha)\} \\ & \text{where } c_1(\gamma, \alpha) = \frac{(N_0 - \gamma)}{2} + N_{\text{oth}} - \alpha, \quad c_2(\gamma, \alpha) = \eta_\gamma(\alpha) + \alpha. \end{aligned} \quad (11)$$

where $\eta_\gamma(\alpha)$ is defined as in Equation (9). One thing to note here is that we only consider integer solutions for t , but $t(\gamma, \alpha)$ does not need to be integers. To characterize the optimal parameters that solve the above optimization, we define the following functions.

$$\begin{aligned} \text{opt}(\gamma) &:= \arg \max_{\alpha \in [1, N_{\text{oth}} - \gamma]} t(\gamma, \alpha), \\ h(\gamma) &:= \alpha \text{ such that } \eta_\gamma(\alpha) = N_0 \text{ if such } \alpha \text{ exists,} \\ f(\gamma) &:= \alpha \text{ such that } c_1(\gamma, \alpha) = c_2(\gamma, \alpha) \text{ if such } \alpha \text{ exists.} \end{aligned}$$

Theorem 5.11 (Optimization allowing non-integer). *Let M_{z^*} be the maximum margin such that $h(M_{z^*})$ exists. Then the optimal solution of γ to Equation (11) must be one of the margins: $\gamma^* \in \{M_z, \dots, M_{z^*}, N_{\text{oth}}\}$. As a result, the optimal resilience t^* is*

$$t^* = \max \left\{ \left\{ \lfloor t(M_i, \text{opt}(M_i)) \rfloor \right\}_{i=z^*}^z, \left\lceil \frac{N}{2} \right\rceil - 1 \right\}, \quad (12)$$

where

$$\text{opt}(M_i) = \begin{cases} h(M_i), & \text{if } c_1(M_i, h(M_i)) \geq c_2(M_i, h(M_i)), \\ f(M_i), & \text{otherwise.} \end{cases}$$

To prove the theorem, we first give an illustrative explanation of the feasible region under different cases:

1. $\text{opt}(\gamma) = h(\gamma)$ iff $h(\gamma)$ exists and $c_1(\gamma, h(\gamma)) \geq c_2(\gamma, h(\gamma))$. See Figure 6a.
2. $\text{opt}(\gamma) = f(\gamma)$ iff $f(\gamma)$ exists. See Figure 6b for an example.
3. $\text{opt}(\gamma) = N_{\text{oth}} - \gamma$ iff $c_2(\gamma, N_{\text{oth}} - \gamma) < c_1(\gamma, N_{\text{oth}} - \gamma)$. See Figure 6c for an example.

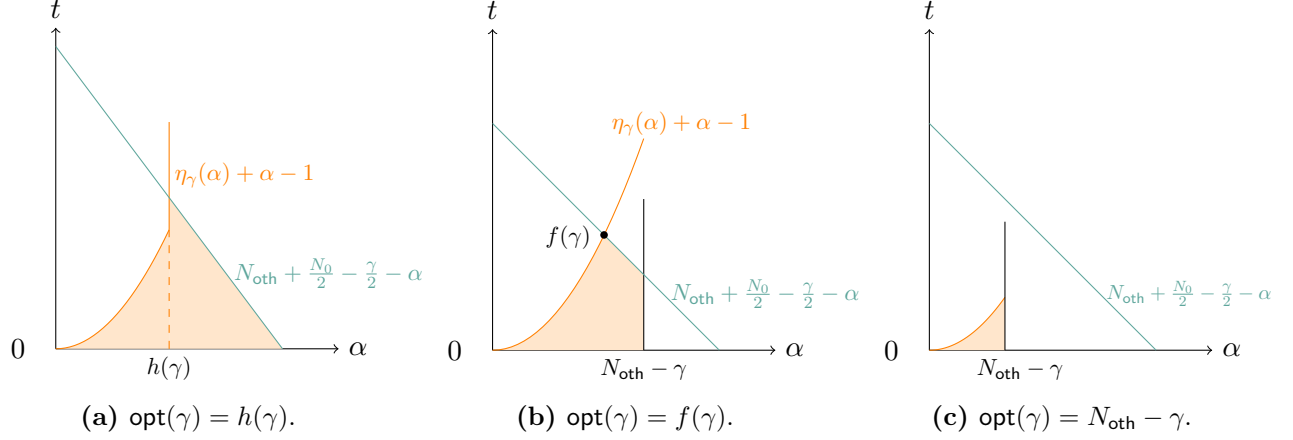


Figure 6: Feasible regions and optimal α given fixed γ . Here, the orange line represents $c_2(\gamma, \alpha)$ and the green line represents $c_1(\gamma, \alpha)$. The orange shaded area represents the feasible region of (α, t) for any fixed γ .

Lemma 5.12. Let γ_s be the maximum $\gamma \leq N_{\text{oth}} - 1$ such that $h(\gamma)$ exists. Let γ_r be the maximum γ such that $c_1(\gamma, N_{\text{oth}} - \gamma) \leq c_2(\gamma, N_{\text{oth}} - \gamma)$. Then it must be that $\gamma_r \geq \gamma_s$. Additionally,

- For any $\gamma_r < \gamma < N_{\text{oth}} - 1$, $\text{opt}(\gamma) = N_{\text{oth}} - \gamma$;
- For any $\gamma_s < \gamma \leq \gamma_r$, $\text{opt}(\gamma) = f(\gamma)$;
- For any $0 \leq \gamma \leq \gamma_s$, $\text{opt}(\gamma) = h(\gamma)$ iff $c_1(\gamma, h(\gamma)) \geq c_2(\gamma, h(\gamma))$ and $\text{opt}(\gamma) = f(\gamma)$ otherwise.

Proof. We first show that $c_1(\gamma, N_{\text{oth}} - \gamma) - c_2(\gamma, N_{\text{oth}} - \gamma)$ is an increasing function with respect to γ . Let $C := \frac{N_0}{2} + N_{\text{oth}} + 1$ be a constant. Then $c_1(\gamma, N_{\text{oth}} - \gamma) - c_2(\gamma, N_{\text{oth}} - \gamma) = C - \frac{1}{2}\gamma - 2(N_{\text{oth}} - \gamma) - \eta_\gamma(N_{\text{oth}} - \gamma)$. Define $g(\gamma) := 2(N_{\text{oth}} - \gamma) + \eta_\gamma(N_{\text{oth}} - \gamma) + \frac{1}{2}\gamma$.

Claim 5.13. For any $\gamma \leq N_{\text{oth}} - 2$, $g(\gamma + 1) \leq g(\gamma)$.

Proof. We first compare $\eta_\gamma(N_{\text{oth}} - \gamma)$ with $\eta_{\gamma+1}(N_{\text{oth}} - \gamma - 1)$. For simplicity, rewrite $\eta = \eta_\gamma(N_{\text{oth}} - \gamma)$ and $\eta' = \eta_{\gamma+1}(N_{\text{oth}} - \gamma - 1)$. By definition,

$$\begin{aligned} U_{[0, \eta]}^0 + U_{[\gamma, N_{\text{oth}}]}^{\text{oth}} &= \mu(\eta + N_{\text{oth}} - \gamma), \\ U_{[0, \eta']}^0 + U_{[\gamma+1, N_{\text{oth}}]}^{\text{oth}} &= \mu(\eta' + N_{\text{oth}} - \gamma - 1), \end{aligned}$$

Taking difference, we have

$$U_{[\eta', \eta]}^0 - \mu(\eta - \eta') = x \geq 0,$$

where x satisfies $\mathbf{U}^{\text{oth}}[\gamma + 1] = \mu - x$. Therefore, $\eta \geq \eta' + 1$, i.e., $\eta_\gamma(N_{\text{oth}} - \gamma)$ decreases as γ increases. Therefore, $g(\gamma + 1) \leq g(\gamma)$. $\square$

Therefore, for any $\gamma > \gamma_r$, we have $c_2(\gamma, N_{\text{oth}} - \gamma) < c_1(\gamma, N_{\text{oth}} - \gamma)$, which implies that the optimal solution is $\text{opt}(\gamma) = N_{\text{oth}} - \gamma$. For any $\gamma \leq \gamma_r$, we have $c_2(\gamma, N_{\text{oth}} - \gamma) \geq c_1(\gamma, N_{\text{oth}} - \gamma)$. The optimal solution is either $f(\gamma)$ or $h(\gamma)$. The lemma thus follows by the observation that $\gamma_s \leq \gamma_r$. $\square$

Case 1: $\text{opt}(\gamma) = N_{\text{oth}} - \gamma$: In this case, it must be that $\gamma > \gamma_r$. The following lemma states that the maximum coalition resilience is no better than just doing a single-shot secret sharing as picking $\gamma = N_{\text{oth}}$.

Lemma 5.14. For any $N_{\text{oth}} > \gamma > \gamma_r$, we have $t(\gamma, \text{opt}(\gamma)) < \frac{N_{\text{oth}} + N_0}{2}$.

Proof. For any $\gamma > \gamma_r$, we have $t(\gamma, \mathbf{opt}(\gamma)) \leq N_{\text{oth}} + \frac{1}{2}N_0 - \frac{1}{2}\gamma - (N_{\text{oth}} - \gamma) < \frac{N_0 + N_{\text{oth}}}{2}$. $\square$

Case 2: $\mathbf{opt}(\gamma) = f(\gamma)$: In this case, it must be that $\gamma \leq \gamma_r$ and the optimal coalition resilience is $c_2(\gamma, f(\gamma)) = c_1(\gamma, f(\gamma))$. The following lemma states that in this case, smaller γ always gives a better coalition resilience.

Lemma 5.15. *For any $\gamma_0 < \gamma_1 \leq \gamma_r$ such that $\mathbf{opt}(\gamma_0) = f(\gamma_0)$ and $\mathbf{opt}(\gamma_1) = f(\gamma_1)$, we have*

$$c_2(\gamma_0, f(\gamma_0)) > c_2(\gamma_1, f(\gamma_1)).$$

Proof. For simplicity, we set $C := \frac{N_0}{2} + N_{\text{oth}}$ as a constant. Define $\alpha_0 = f(\gamma_0)$ and $\alpha_1 = f(\gamma_1)$ be the solution to

$$2\alpha + \eta_\gamma(\alpha) + \frac{1}{2}\gamma = C \quad (13)$$

for their corresponding γ . When γ increases while α is constant, the left-hand side of Equation (13) increases. Therefore, $\alpha_1 < \alpha_0$.

By definition of $\eta_\gamma(\alpha)$,

$$\begin{aligned} U_{[0, \eta_{\gamma_0}(\alpha_0)]}^0 + U_{[\gamma_0, \gamma_0 + \alpha_0]}^{\text{oth}} &= \mu \cdot (\eta_{\gamma_0}(\alpha_0) + \alpha_0), \\ U_{[0, \eta_{\gamma_1}(\alpha_1)]}^0 + U_{[\gamma_1, \gamma_1 + \alpha_1]}^{\text{oth}} &= \mu \cdot (\eta_{\gamma_1}(\alpha_1) + \alpha_1). \end{aligned}$$

Taking differences of these two, we have

$$\left(U_{[0, \eta_{\gamma_0}(\alpha_0)]}^0 - U_{[0, \eta_{\gamma_1}(\alpha_1)]}^0 \right) - \mu(\eta_{\gamma_0}(\alpha_0) - \eta_{\gamma_1}(\alpha_1)) = \mu(\alpha_0 - \alpha_1) - \left(U_{[\gamma_0, \gamma_0 + \alpha_0]}^{\text{oth}} - U_{[\gamma_1, \gamma_1 + \alpha_1]}^{\text{oth}} \right) \geq 0,$$

where the last inequality comes from the fact that the average utility of **oth**-supporters on 0 must be no more than μ . This implies that $\eta_{\gamma_0}(\alpha_0) \geq \eta_{\gamma_1}(\alpha_1)$. Thus,

$$\begin{aligned} c_2(\gamma_0, f(\gamma_0)) - c_2(\gamma_1, f(\gamma_1)) &= c_1(\gamma_0, f(\gamma_0)) - c_1(\gamma_1, f(\gamma_1)) \\ &= \frac{1}{2}\gamma_1 + \alpha_1 - \frac{1}{2}\gamma_0 - \alpha_0 \\ &= \frac{1}{2}\gamma_1 + 2\alpha_1 - \frac{1}{2}\gamma_0 - 2\alpha_0 + (\alpha_0 - \alpha_1) \\ &> (C - \eta_{\gamma_1}(\alpha_1)) - (C - \eta_{\gamma_0}(\alpha_0)) && \text{By } \alpha_1 < \alpha_0 \\ &\geq 0 && \text{By } \eta_{\gamma_0}(\alpha_0) \geq \eta_{\gamma_1}(\alpha_1). \end{aligned}$$

The lemma thus follows. $\square$

Case 3: $\mathbf{opt}(\gamma) = h(\gamma)$: In this case, it must be that $\gamma \leq \gamma_s$ and $c_2(\gamma, h(\gamma)) \leq c_1(\gamma, h(\gamma))$. The following lemma states that in this case, picking γ to be the margin M_i always gives a better coalition resilience.

Lemma 5.16. *For any $i = z, \dots, 1$, for any $M_i \leq \gamma_0 < \gamma < \gamma_1 \leq \min\{M_{i-1}, \gamma_s\}$, it must be that*

$$\max\{c_1(\gamma_0, h(\gamma_0)), c_1(\gamma_1, h(\gamma_1))\} \geq c_1(\gamma, h(\gamma)).$$

That is, conditioned on $\mathbf{opt}(\gamma) = h(\gamma)$, the optimal solution one can get when picking γ is no better than its two margins M_i and $\min\{M_{i-1}, \gamma_s\}$.

Proof. Let $\alpha_0 = h(\gamma_0)$, $\alpha = h(\gamma)$ and $\alpha_1 = h(\gamma_1)$. By definition, we have

$$\mu(\alpha_0 + N_0) = U_{[\gamma_0, \gamma_0 + \alpha_0]}^{\text{oth}} + U_{[0, N_0]}^0, \quad (14)$$

$$\mu(\alpha + N_0) = U_{[\gamma, \gamma + \alpha]}^{\text{oth}} + U_{[0, N_0]}^0. \quad (15)$$

Taking difference between Equation (14) and Equation (15), we have $\mu(\alpha_0 - \alpha) = U_{[\gamma_0, \gamma_0 + \alpha_0]}^{\text{oth}} - U_{[\gamma, \gamma + \alpha]}^{\text{oth}}$. Observe that $U_{[\gamma_0, \gamma_0 + \alpha_0]}^{\text{oth}} - U_{[\gamma, \gamma + \alpha]}^{\text{oth}} = U_{[\gamma_0, \gamma]}^{\text{oth}} - U_{[\gamma_0 + \alpha_0, \gamma + \alpha]}^{\text{oth}}$. Since the utility in $\mathbf{U}^{\text{oth}}$ is sorted in a non-increasing order, it must be that $x_0 \geq x \geq 0$, where $\mu - x = \frac{U_{[\gamma_0, \gamma]}^{\text{oth}}}{\gamma - \gamma_0}$ is the average utility on 0 of the (γ_0) -th to the γ -th oth-supporter and $\mu - x_0 = \frac{U_{[\gamma_0 + \alpha_0, \gamma + \alpha]}^{\text{oth}}}{\gamma + \alpha - \gamma_0 - \alpha_0}$ is the average utility on 0 of the $(\gamma_0 + \alpha_0)$ -th to the $(\gamma + \alpha)$ -th oth-supporter. See Figure 7 for an illustrative example. Therefore, we have

$$\begin{aligned} \mu(\alpha_0 - \alpha) &= U_{[\gamma_0, \gamma_0 + \alpha_0]}^{\text{oth}} - U_{[\gamma, \gamma + \alpha]}^{\text{oth}} \\ &= U_{[\gamma_0, \gamma]}^{\text{oth}} - U_{[\gamma_0 + \alpha_0, \gamma + \alpha]}^{\text{oth}} \\ &= (\gamma - \gamma_0)(\mu - x) - (\gamma + \alpha - \gamma_0 - \alpha_0)(\mu - x_0). \end{aligned}$$

Rearrange, we have

$$x_0(\alpha_0 - \alpha) = (x_0 - x)(\gamma - \gamma_0). \quad (16)$$

Similarly,

$$x_1(\alpha - \alpha_1) = (x_1 - x)(\gamma_1 - \gamma), \quad (17)$$

where $\mu - x$ is the average utility of the green dashed part, which is exactly the same as the average utility of the solid green part, and $\mu - x_1$ be the average utility of the blue dashed part in Figure 7.

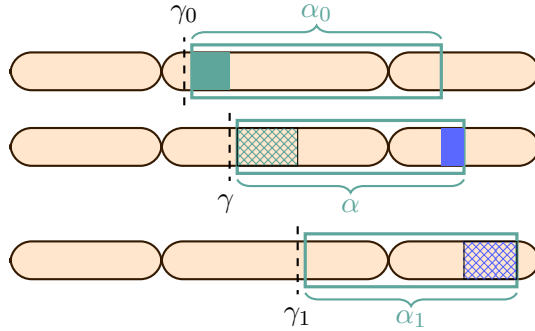


Figure 7: Illustrative explanation of the proof for Lemma 5.16. Specifically, $\mu - x$ is the average utility of players in the solid green part as well as the green hatch-shaded part. $\mu - x_0$ is the average utility of solid blue part and $\mu - x_1$ is the average utility of blue hatch-shaded part.

If $x_0 = 0$, it must be that $z - 1 = \mu$ and that $\gamma_0 + \alpha_0 \leq \gamma + \alpha \leq M_{z-1}$. In this case, we have $\alpha_0 = \alpha$ by the fact that every player's utility on 0 in the region of $[M_z, M_{z-1}]$ is the same. Therefore, $\frac{1}{2}\gamma + \alpha \geq \frac{1}{2}\gamma_0 + \alpha_0$, i.e., $c_1(\gamma_0, \alpha_0) \geq c_1(\gamma, \alpha)$.

For the rest of this proof, we will focus on the case where $x_0 > 0$. For the sake of contradiction, assume that the lemma is not true, i.e., $c_1(\gamma_0, \alpha_0) < c_1(\gamma, \alpha)$ and $c_1(\gamma_0, \alpha_0) < c_1(\gamma, \alpha)$. Equivalently, that implies

$$\begin{aligned} 0 &> c_1(\gamma_0, \alpha_0) - c_1(\gamma, \alpha) = \left(\frac{1}{2}\gamma + \alpha\right) - \left(\frac{1}{2}\gamma_0 + \alpha_0\right) = \left(\frac{1}{2} - \frac{x_0 - x}{x_0}\right)(\gamma - \gamma_0), \\ 0 &> c_1(\gamma_1, \alpha_1) - c_1(\gamma, \alpha) = \left(\frac{1}{2}\gamma + \alpha\right) - \left(\frac{1}{2}\gamma_1 + \alpha_1\right) = \left(\frac{x_1 - x}{x_1} - \frac{1}{2}\right)(\gamma_1 - \gamma). \end{aligned}$$

That is, $\frac{x_1-x}{x_1} < \frac{1}{2} < \frac{x_0-x}{x_0}$. The last step comes from solving for $(\alpha - \alpha_0)$ in Equation (16) and $(\alpha - \alpha_1)$ in Equation (17) and plugging those values into the equations respectively. However, since $\gamma_0 + \alpha_0 \leq \gamma + \alpha \leq \gamma_1 + \alpha_1$, it must be that $x_1 \geq x_0$. This leads to a contradiction since $\frac{x_1-x}{x_1} \geq \frac{x_0-x}{x_0}$ when $x_1 \geq x_0$. This completes the proof. $\square$

5.3.2 Proof of Theorem 5.11

By Lemma 5.14, picking any $\gamma \in (\gamma_r, N_{\text{oth}} - 1]$ is no better than picking $\gamma = N_{\text{oth}}$. Also, by Lemma 5.15, picking any $\gamma \in (\gamma_s, \gamma_r]$ is no better than picking the smallest γ such that $\text{opt}(\gamma) = f(\gamma)$.

Claim 5.17. $\text{opt}(\gamma_s) = f(\gamma_s)$.

Proof. To see this, note that γ_s is the maximum γ such that $h(\gamma)$ exists. That means $\gamma_s + h(\gamma_s) > N_{\text{oth}} - 1$. Otherwise if $h(\gamma_s) \leq N_{\text{oth}} - 1 - \gamma_s$, then we have

$$\eta_{\gamma_s+1}(N_{\text{oth}}-1-\gamma_s) \geq \eta_{\gamma_s}(N_{\text{oth}}-1-\gamma_s) \geq \eta_{\gamma_s}(h(\gamma_s)) \geq N_0,$$

which contradicts our choice of γ_s . Therefore,

$$\begin{aligned} c_1(\gamma_s, h(\gamma_s)) - c_2(\gamma_s, h(\gamma_s)) &= N_{\text{oth}} + \frac{N_0}{2} - \frac{1}{2}\gamma_s - N_0 - 2h(\gamma_s) \\ &< -\frac{N_0}{2} - N_{\text{oth}} - \frac{3}{2}\gamma_s + 2 \leq 0, \end{aligned}$$

where the last step follows from the assumption that $N_0 + N_{\text{oth}} \geq 3$. $\square$

Thus, by Lemma 5.15, picking any $\gamma \in (\gamma_s, \gamma_r]$ is no better than picking $\gamma = \gamma_s$. In the rest of the proof, we will show that for any $M_i = \gamma_0 \leq \gamma \leq \gamma_1 = \min\{M_{i-1}, \gamma_s\}$, the optimal coalition resilience

$$t(\gamma, \text{opt}(\gamma)) \leq \max\{t(\gamma_0, \text{opt}(\gamma_0)), t(\gamma_1, \text{opt}(\gamma_1))\}. \quad (*)$$

We will show this based on different cases:

- If $\text{opt}(\gamma) = h(\gamma)$: By Lemma 5.16, it must be that

$$\begin{aligned} t(\gamma, \text{opt}(\gamma)) &= c_1(\gamma, h(\gamma)) \leq \max\{c_1(\gamma_0, h(\gamma_0)), c_1(\gamma_1, h(\gamma_1))\} \\ &\leq \max\{t(\gamma_0, \text{opt}(\gamma_0)), t(\gamma_1, \text{opt}(\gamma_1))\}. \end{aligned}$$

Equation (*) thus holds.

- If $\text{opt}(\gamma) = f(\gamma)$: If $\text{opt}(\gamma_0) = f(\gamma_0)$, then Equation (*) holds by Lemma 5.15. If $\text{opt}(\gamma_0) = h(\gamma_0)$, then it must be that

$$\begin{aligned} t(\gamma_0, \text{opt}(\gamma_0)) &\geq c_2(\gamma_0, h(\gamma_0)) = N_0 + h(\gamma_0) \\ &\geq N_0 + h(\gamma) = c_2(\gamma, h(\gamma)) \geq c_2(\gamma, f(\gamma)) = t(\gamma, \text{opt}(\gamma)). \end{aligned}$$

Equation (*) thus holds.

By a similar reasoning, $t(\gamma_s, \text{opt}(\gamma_s))$ cannot give the maximum coalition resilience unless γ_s itself equals some M_i . Thus, Equation (12) is true. $\square$

For the example preference profile $\mathcal{P}^*$, the optimal solution to Equation (11) is picking $\gamma = 2$, $\text{opt}(\gamma) = h(\gamma) = 2$, and $\eta_\gamma(\text{opt}(\gamma)) = 2$. One can check that for $\mathcal{P}^*$, we have $\eta_0(N_{\text{oth}}/2) = \infty$. Indeed, the coalition resistance $t^* > N/2$.

5.4 Integer Solutions

Now we go back to the integer solution of Constraint 5.7. Recall that by Lemma 5.9, Constraint 5.7 boils down to the following integer optimization:

$$\begin{aligned}
& \text{Maximize } t \text{ subject to} \\
& \gamma \in [N_{\text{oth}}]_0, \alpha \in [N_{\text{oth}} - \gamma], \\
& t \leq t(\gamma, \alpha) := \min \{c_1(\gamma, \alpha), c_2(\gamma, \alpha)\} \\
& \text{where } c_1(\gamma, \alpha) = \left\lceil \frac{(N_0 - \gamma)}{2} \right\rceil + N_{\text{oth}} - \alpha, \quad c_2(\gamma, \alpha) = \eta_\gamma(\alpha) + \alpha.
\end{aligned} \tag{18}$$

Here, $\eta_\gamma(\alpha)$ reverts back to the original definition in 5.4. We abuse notation and still use $c_1(\cdot, \cdot), c_2(\cdot, \cdot)$, and $t(\cdot, \cdot)$, but now only defined for integer γ, α , and η . We also define $\text{opt}(\gamma)$ analogously for integers. We now refer to the non-integer version of these functions as $\tilde{c}_1(\cdot, \cdot), \tilde{c}_2(\cdot, \cdot)$, and $\tilde{t}(\cdot, \cdot)$.

Theorem 5.18 (Integer optimization). *Let M_{z^*} be the maximum margin such that $h(M_{z^*})$ exists. Define $\hat{t}$ as*

$$\hat{t} = \max \left\{ \{t(M_i, \text{opt}(M_i))\}_{i=z^*}^z, \left\lceil \frac{N}{2} \right\rceil - 1 \right\},$$

where

$$\text{opt}(M_i) = \begin{cases} \lfloor h(M_i) \rfloor + 1, & \text{if } \tilde{c}_1(M_i, h(M_i)) \geq \tilde{c}_2(M_i, h(M_i)) \\ \lceil f(M_i) \rceil, & \text{otherwise.} \end{cases}$$

Then the optimal resilience of Protocol 5.3 is either $\hat{t}$ or $\hat{t} + 1$.

Specifically, if the first half of 0-supporters and the first half of oth-supporters does not prefer 0. i.e., $\eta_0(\lfloor N_{\text{oth}}/2 \rfloor) \geq N_0/2 + 1$, then it must be that $\hat{t} \geq N/2$.

Proof. In this proof, we will use γ_r defined analogously for integer solutions. We use $\widetilde{\text{opt}}(\gamma)$, to denote the non-integer optimal for γ . We first eliminate consideration for when $\widetilde{\text{opt}}(\gamma) = N_{\text{oth}} - \gamma$. In this case, $t(\gamma, \text{opt}(\gamma)) \leq c_1(\gamma, \text{opt}(\gamma)) \leq \lceil \frac{N}{2} \rceil - 1$ by the same reasoning as Lemma 5.14. Here, we do no better than single-shot secret sharing. Thus we now only need to consider when $\gamma \leq \gamma_r$ by Lemma 5.12. Our strategy is to bound how far away the integer solutions can be from the non-integer solutions.

Lemma 5.19. *For any fixed $\gamma \leq \gamma_r$,*

$$\tilde{t}(\gamma, \widetilde{\text{opt}}(\gamma)) - 1 \leq t(\gamma, \text{opt}(\gamma)) < \tilde{t}(\gamma, \widetilde{\text{opt}}(\gamma)) + 1,$$

where

$$\text{opt}(\gamma) = \begin{cases} \lfloor h(M_i) \rfloor + 1, & \text{if } \tilde{c}_1(M_i, h(M_i)) \geq \tilde{c}_2(M_i, h(M_i)) \\ \lceil f(M_i) \rceil, & \text{otherwise.} \end{cases}$$

Proof. If the non-integer solutions are also integers, this trivially holds, so we can assume the non-integer solutions are indeed not integers. In the following, for any γ , let $g = \left\lceil \frac{N_0 - \gamma}{2} \right\rceil - \frac{N_0 - \gamma}{2}$. By Lemma 5.12, we can split into two cases.

Case 1: $\widetilde{\text{opt}}(\gamma) = f(\gamma)$ In this case, let $C = \widetilde{c}_1(\gamma, f(\gamma)) = \widetilde{c}_2(\gamma, f(\gamma))$. Then for any integer α , by the definition of $c_1(\cdot, \cdot)$ and $c_2(\cdot, \cdot)$, we have

$$C - (\alpha - f(\gamma)) < c_1(\gamma, \alpha) < C - (\alpha - f(\gamma)) + 1$$

and when $\alpha > f(\gamma)$:

$$C + (\alpha - f(\gamma)) - 1 < c_2(\gamma, \alpha)$$

and when $\alpha < f(\gamma)$:

$$C + (\alpha - f(\gamma)) > c_2(\gamma, \alpha)$$

Thus, when $|\alpha - f(\gamma)| \geq 2$, $t(\gamma, \alpha) = \min\{c_1(\gamma, \alpha), c_2(\gamma, \alpha)\} < \max\{t(\gamma, \lceil f(\gamma) \rceil), t(\gamma, \lfloor f(\gamma) \rfloor)\}$. Furthermore, we can say that

$$\begin{aligned} t(\gamma, \lceil f(\gamma) \rceil) &= c_1(\gamma, \lceil f(\gamma) \rceil) = C + (f(\gamma) - \lceil f(\gamma) \rceil) + g \\ &\geq C + (f(\gamma) - \lfloor f(\gamma) \rfloor) + g - 1 \\ &= c_1(\gamma, \lfloor f(\gamma) \rfloor) - 1 \geq c_2(\gamma, \lfloor f(\gamma) \rfloor). \end{aligned}$$

The last step comes from the fact that when $\alpha < f(\gamma)$, $c_2(\gamma, \alpha) < \widetilde{c}_2(\gamma, \widetilde{\text{opt}}(\gamma)) < c_1(\gamma, \alpha)$ with $c_2(\gamma, \alpha)$ and $c_1(\gamma, \alpha)$ being integers. Thus we can conclude that $\text{opt}(\gamma) = \lceil f(\gamma) \rceil$ and $\widetilde{t}(\gamma, \widetilde{\text{opt}}(\gamma)) - 1 \leq t(\gamma, \text{opt}(\gamma)) < \widetilde{t}(\gamma, \widetilde{\text{opt}}(\gamma)) + 1$ follows.

Case 2: $\widetilde{\text{opt}}(\gamma) = h(\gamma)$. For any α , by definition, $c_2(\gamma, \alpha) \leq \widetilde{c}_2(\gamma, \alpha)$ and $c_1(\gamma, \alpha) \geq \widetilde{c}_1(\gamma, \alpha)$. Thus for any $\alpha \leq h(\gamma)$, $t(\gamma, \alpha) = c_2(\gamma, \alpha)$. For any $\alpha > h(\gamma)$, $c_2(\gamma, \alpha) = \infty$, so $t(\gamma, \alpha) = c_1(\gamma, \alpha)$. This implies $\text{opt}(\gamma) \in \{\lfloor h(\gamma) \rfloor, \lfloor h(\gamma) \rfloor + 1\}$. Since $\widetilde{c}_2(\cdot, \cdot)$ increases in the second input, we know $\widetilde{c}_2(\gamma, \lfloor h(\gamma) \rfloor) \leq \lfloor \widetilde{c}_2(\gamma, h(\gamma)) \rfloor \leq \lfloor \widetilde{c}_1(\gamma, h(\gamma)) \rfloor \leq c_1(\gamma, \lfloor h(\gamma) \rfloor + 1)$. Thus $\text{opt}(\gamma) = \lfloor h(\gamma) \rfloor + 1$ and $\widetilde{t}(\gamma, \widetilde{\text{opt}}(\gamma)) - 1 \leq t(\gamma, \text{opt}(\gamma)) < \widetilde{t}(\gamma, \widetilde{\text{opt}}(\gamma)) + 1$ follows. $\square$

Using Theorem 5.11 and Lemma 5.19, we can conclude that for any $i \in \{z^*, \dots, z-1\}$, and $M_i < \gamma < M_{i+1}$,

$$\begin{aligned} t(\gamma, \text{opt}(\gamma)) &< \widetilde{t}(\gamma, \widetilde{\text{opt}}(\gamma)) + 1 && \text{Lemma 5.19} \\ &\leq \max\{\widetilde{t}(M_i, \widetilde{\text{opt}}(M_i)), \widetilde{t}(M_{i+1}, \widetilde{\text{opt}}(M_{i+1}))\} + 1 && \text{Theorem 5.11} \\ &\leq \max\{t(M_i, \text{opt}(M_i)), t(M_{i+1}, \text{opt}(M_{i+1}))\} + 2. && \text{Lemma 5.19} \end{aligned}$$

$\square$

6 Construction by Coalition Bet

Another natural construction goal is to achieve fairness against coalitions based on the total bet of the coalition. In this section, we present a protocol that achieves CSP fairness against any coalition with total bet no more than β .

Given a preference profile $\mathcal{P}$, we use B_0 and B_{oth} to respectively denote the cumulative bet on outcome 0 and on all other outcomes. Formally,

$$B_0 = \sum_{i=1}^{\ell} \text{bet}_i(0), \quad B_{\text{oth}} = \sum_{i=1}^n \sum_{\ell=1}^{m-1} \text{bet}_i(\ell).$$

We also use $\text{bet}_i(\text{oth}) = \sum_{\ell=1}^{m-1} \text{bet}_i(\ell)$ to denote the sum of player i 's bets on all non-0 outcomes.

In this protocol, we will use weighted secret sharing to defend against coalition in terms of corrupted bets. Weighted secret sharing has an access structure of the secret sharing is $\Gamma = \{S \subseteq [n] : \text{bet}_S(b) \geq k + 1\}$ where $\text{bet}_S(b) = \sum_{i \in S} \text{bet}_i(b)$ is the total bet of set S on outcome b . In other words, any set with $k + 1$ or more bets on outcome b can reconstruct the secret while any set with less than $k + 1$ bets on outcome b learns nothing about the secret. In our protocol, weighted secret sharing can be instantiated using Shamir's secret sharing in a straightforward manner: Each secret is split into B_b shares for $b \in \{0, \text{oth}\}$, and each player i gets $\text{bet}_i(b)$ number of shares.

6.1 bet-GroupToss^b[k] Sub-Protocol

Similar to Protocol 5.3, the bets-based protocol runs two sub-protocols bet-GroupToss^b[k] for $b \in \{0, \text{oth}\}$ and decide two coins c_0 and c_{oth} . Here, $k \in [B_b]$ is the threshold parameter to be chosen later. The final outcome will be determined by the value of c_0 and c_{oth} . The sub-protocol bet-GroupToss^b[k] runs a weighted secret sharing among all N players, where each player's weight is associated with its bets.

Protocol 6.1: bet-GroupToss^b[k] sub-protocol (Semi-Malicious)

Let $\mathcal{O}$ be the set of online players, initialized as $[n]$.

Sharing Phase. Each player i chooses a random coin $\text{coin}_i \xleftarrow{\$} [m]_0$. Player i shares coin_i among $[n]$ with an access structure $\Gamma = \{S \subseteq [n] : \text{bet}_S(b) \geq k + 1\}$. Denote the set of shares as $\{c_{i,j}\}_{j \in [n]}$. Send $c_{i,j}$ to player j . If any player aborts, remove them from $\mathcal{O}$.

Reconstruction Phase. Every player i broadcast the shares $\{c_{j,i}\}_{j \in \mathcal{O}}$ it has received from players in $\mathcal{O}$. Players use $\{c_{j,i}\}_{j \in \mathcal{O}}$ to reconstruct $\{\text{coin}_i\}_{i \in \mathcal{O}}$. If any coin_i failed to be reconstructed, output $c_b = \perp$; otherwise, output $(\sum_{i \in \mathcal{O}} \text{coin}_i) \bmod m$.

Lemma 6.2 (Lemma 4.2 of [TSW24]). *The above bet-GroupToss^b[k] protocol satisfies the following properties due to the property of Shamir secret sharing.*

- *Binding.* The sharing phase uniquely binds to a secret s , such that the reconstruction phase either succeeds and outputs s , or it fails and outputs $\perp$.
- *Knowledge threshold.* If a coalition $\mathcal{C}$ controls bets $\text{bet}_{\mathcal{C}}(b) \geq k + 1$, the coalition can completely control the binded value s of the sharing phase. Specifically, the coalition can reconstruct the values of coin_i from honest players and choose their value of coin_j for $j \in \mathcal{C}$ accordingly. Otherwise, the secret s looks uniformly random to the coalition.
- *Liveness threshold.* If a coalition $\mathcal{C}$ controls bet $\text{bet}_{\mathcal{C}}(b) \geq B_b - k$, the coalition can cause the reconstruction to fail and output $\perp$. Otherwise, the reconstruction phase must succeed.

6.2 Bets-Based Protocol

The full protocol runs a bet-GroupToss⁰[k₀] and a bet-GroupToss^{oth}[k_{oth}] as follows where k_0 and k_{oth} are parameters that will be determined later.

Protocol 6.3: Bets-based protocol (semi-malicious)

Sharing Phase

1. All players run the sharing phase of bet-GroupToss⁰[k₀].

2. All players run the sharing phase of $\text{bet-GroupToss}^{\text{oth}}[k_{\text{oth}}]$.

Reconstruction Phase

1. Run the reconstruction phase of $\text{bet-GroupToss}^0[k_0]$. Let s_0 be the outcome of the sub-protocol.
2. Run the reconstruction phase of $\text{bet-GroupToss}^{\text{oth}}[k_{\text{oth}}]$. Let s_{oth} be the outcome of the sub-protocol.
3. If $s_{\text{oth}} = \perp$, output 0; Otherwise, if $s_0 = \perp$, set $s_0 = 0$ and output $(s_{\text{oth}} + s_0) \bmod m$.

Constraints 6.4. Consider a coalition $\mathcal{C}$ that controls $\beta = \beta_0 + \beta_{\text{oth}}$ total bet, where β_0 and β_{oth} denote the bets on 0 and on oth. That is, $\beta_0 = \text{bet}_{\mathcal{C}}(0)$ and $\beta_{\text{oth}} = \text{bet}_{\mathcal{C}}(\text{oth})$. To achieve CSP-fairness, we must choose k_0 and k_{oth} such that the following hold.

- (C1') $\mathcal{C}$ cannot control both coins. Equivalently, either $\beta_0 \leq k_0$ or $\beta_{\text{oth}} \leq k_{\text{oth}}$. Otherwise, $\mathcal{C}$ can force the outcome to be whatever the coalition's overall preference is.
- (C2') If $\mathcal{C}$ controls s_{oth} , then it cannot fail the reconstruction of s_0 . That is, if $\beta_{\text{oth}} > k_{\text{oth}}$, then it must be that $\beta_0 < B_0 - k_0$. Otherwise, $\mathcal{C}$ can control the outcome to be the s_{oth} coin it decides by failing the reconstruction of s_0 .
- (C3') If $\mathcal{C}$ can hamper the reconstruction of s_{oth} , then it must NOT prefer 0. That is, if $\beta_{\text{oth}} \geq B_{\text{oth}} - k_{\text{oth}}$, then $\beta_0 \leq \frac{\beta}{m}$. Otherwise, $\mathcal{C}$ can fail the reconstruction of s_{oth} to force the outcome 0, benefiting $\mathcal{C}$ compared to honest behavior.

Lemma 6.5. If conditions (C1'), (C2'), (C3') are satisfied, then Protocol 6.3 achieves CSP-fairness against any coalition controlling β total bet.

Proof. Condition (C3') implies that if the coalition fails the reconstruction of s_{oth} , requiring $\beta_{\text{oth}} \geq B_{\text{oth}} - k_{\text{oth}}$ by Lemma 6.2, the coalition's utility β_0 must be no more than its honest expected utility $\frac{\beta}{m}$. Therefore, any coalition is incentivized to successfully reconstruct s_{oth} .

Conditioned on s_{oth} being reconstructed, if the coalition controls the value of s_{oth} , i.e., $\beta_{\text{oth}} \geq k_{\text{oth}} + 1$, then by Condition (C1'), $\beta_0 \leq k_0$. Thus by Lemma 6.2, the coin s_0 must be uniformly random to the coalition. Furthermore, (C2') stipulates that $\beta_0 < B_0 - k_0$, so by Lemma 6.2, the s_0 coin is successfully reconstructed. Therefore, the protocol outcome must be uniformly random. If $\beta_{\text{oth}} \leq k_{\text{oth}}$, then s_{oth} must be a uniformly random coin and successfully reconstructed. Thus, the outcome again is uniformly random, whether s_0 is successfully reconstructed or not. $\square$

Constraint 6.6: Coalition Bet Parameter Constraints

Assume $0 \leq k_0 \leq B_0$, $0 \leq k_{\text{oth}} \leq B_{\text{oth}}$

$$(C1^*) \quad \beta \leq k_0 + k_{\text{oth}} + 1$$

$$(C2^*) \quad \beta \leq k_{\text{oth}} + B_0 - k_0$$

$$(C3^*) \quad \text{if } B_{\text{oth}} - k_{\text{oth}} < (m-1)B_0, \text{ then } \beta \leq \frac{m}{m-1}(B_{\text{oth}} - k_{\text{oth}})$$

Claim 6.7. Constraint 6.6 imply (C1'), (C2'), (C3').

Table 3: Optimal resilience of Protocol 6.3 against semi-malicious adversary.

Case	k_0	k_{oth}	t
If $B_{\text{oth}} \geq \frac{4m-3}{2}B_0$	$\lfloor \frac{B_0}{2} \rfloor$	$B_{\text{oth}} - (m-1)B_0$	$\lceil \frac{B_0}{2} \rceil + B_{\text{oth}} - (m-1)B_0$
Otherwise	$\lfloor \frac{B_0}{2} \rfloor$	$\lfloor \frac{m}{2m-1}B_{\text{oth}} - \frac{m-1}{2(2m-1)}B_0 \rfloor$	$\lceil \frac{B_0}{2} \rceil + \lfloor \frac{m}{2m-1}B_{\text{oth}} - \frac{m-1}{2(2m-1)}B_0 \rfloor$

Proof. The first two are straightforward and follow the same proof structure as Lemma 5.8, so we only focus on proving the third one condition. If $B_{\text{oth}} - k_{\text{oth}} \geq (m-1)B_0$, then if a coalition can fail the reconstruction of s_{oth} , it must be the case that $\beta \geq m \cdot \beta_0$ since $\beta_0 \leq B_0 \leq \frac{1}{m-1}\beta_{\text{oth}}$. In this case, the coalition must not prefer 0. On the other hand, if $B_{\text{oth}} - k_{\text{oth}} < B_0$, then the constraint requires $\beta \leq \frac{m}{m-1}(B_{\text{oth}} - k_{\text{oth}})$. This guarantees that if $\beta_{\text{oth}} \geq B_{\text{oth}} - k_{\text{oth}}$, i.e., if the coalition can fail the reconstruction of s_{oth} , then $\beta_0 = \beta - \beta_{\text{oth}} \leq \frac{1}{m-1}\beta_{\text{oth}}$, which implies that $\beta_0 \leq \frac{\beta}{m}$. $\square$

Below, we give the optimal parameters and coalition resilience of Protocol 6.3.

7 Concrete Examples

While the above optimization describes the optimal solution, it is hard to give a closed form optimal resilience since Protocol 5.3 aims for a rather general preference profile. In this section, we give a closed form for some concrete examples and compare the coalition resilience of both protocols.

7.1 Unit-Split

In this sub-formulation of split-bet, for each player i 's preference, we have $\mathbf{p}_i \in \{0,1\}^m$. That is, each player chooses $B < m$ out of m possible outcomes, and gets utility 1 for each chosen outcome. Without loss of generality, we assume that 0 is the least preferred outcome, i.e., 0 has the cumulative bet. In this case, the expected honest utility of each player is $\mu = \frac{B}{m} < 1$. One can see that when $B = 1$, this downgrades to the special case considered in previous works where each player prefers a specific outcome. Each player either gets utility 0 on outcome 0 or utility 1 on outcome 0. Therefore, the optimal choice of γ is either $\gamma = 0$ or $\gamma = N_{\text{oth}}$. In fact, we will show that picking $\gamma = N_{\text{oth}}$ is never the optimal option for unit-split in Lemma 7.1. The optimal parameter choice is given below. ⁴

Table 4: Optimal resilience of Protocol 5.3 for the unit-bet against semi-malicious adversaries.

Case	γ	k_0	k_{oth}	t
$N_{\text{oth}} \geq \frac{4m-3B}{2B}N_0$	0	$\lfloor \frac{N_0}{2} \rfloor$	$N_{\text{oth}} - \lceil \frac{(m-B)N_0}{B} \rceil$	$N_{\text{oth}} + \lceil \frac{2B-m}{2B}N_0 \rceil$
Otherwise	0	$\lfloor \frac{N_0}{2} \rfloor$	$\lfloor \frac{m}{2m-B}N_{\text{oth}} - \frac{m-B}{2(2m-B)}N_0 \rfloor$	$\lfloor \frac{m}{2m-B}N_{\text{oth}} + \frac{m}{2(2m-B)}N_0 \rfloor$

Lemma 7.1. *For the unit-split formulation, when $N_0 < \frac{B}{m}N$, i.e., 0 is the least preferred without a tie, Protocol 5.3 achieves CSP fairness against half or more sized coalition. Specifically, the coalition resilience is given in Table 4.*

⁴For simplicity, we round the parameters as a whole when giving the expression for t . Thus the giving t might be 1 off from the optimal coalition resilience of Protocol 5.3.

Table 5: Comparison of the optimal resilience for the unit-bet formulation. Here, the size-based protocol is always at least as good as the bets-based protocol. For the specific case where $B = 1$, both protocol achieves the same resilience.

Case	Size-based protocol Protocol 5.3	bets-based protocol Protocol 6.3
$N_{\text{oth}} \geq \frac{4m-3B}{2B}N_0$	$N_{\text{oth}} + \lceil \frac{2B-m}{2B}N_0 \rceil$	$N_{\text{oth}} + \lfloor \frac{2B-2m+1}{2B} \rfloor N_0$
Otherwise	$\lfloor \frac{m}{2m-B}N_{\text{oth}} + \frac{m}{2(2m-B)}N_0 \rfloor$	$\lfloor \frac{m}{2m-1}N_{\text{oth}} + \frac{2m-m/B}{2(2m-1)}N_0 \rfloor$

Proof. When picking $\gamma = 0$, the optimal integer solution α is either $\lfloor h(0) \rfloor + 1$ or $\lceil f(0) \rceil$ as shown in Theorem 5.18. Observe that $h(0) = \frac{m-B}{B}N_0$. For any $\alpha < h(0)$, we have $\eta_0(\alpha) = \frac{B}{m-B}\alpha$. Thus, when $f(0)$ exists, it must be that $f(0) = \frac{m}{2m-B}N_{\text{oth}} - \frac{m-B}{2(2m-B)}N_0$. The optimal coalition resilience when picking $\gamma = 0$ thus follows from Theorem 5.18.

In the rest of this proof we only need to show that $t \geq \frac{N}{2}$ as long as $N_0 < \frac{B}{m}N$, i.e., when there exists at least one other outcome with more cumulative bet on 0. When picking $\alpha = \lfloor h(0) \rfloor + 1$, we have

$$t - \frac{N}{2} \geq N_{\text{oth}} + \frac{2B-m}{2B}N_0 - \frac{1}{2}N_{\text{oth}} - \frac{1}{2}N_0 = \frac{1}{2} \left(N_{\text{oth}} - \frac{m-B}{B}N_0 \right) > 0.$$

When picking $\alpha = \lceil f(0) \rceil$, we have

$$\frac{m}{2m-B}N_{\text{oth}} + \frac{m}{2(2m-B)}N_0 - \frac{N}{2} = \frac{B \cdot N_{\text{oth}} - (m-B)N_0}{2(2m-B)} > 0.$$

The final step comes from the fact that $m \cdot N_0 \leq B \cdot N = B \cdot (N_{\text{oth}} + N_0)$. $\square$

In Table 5, we give the comparison of both protocols on unit-split. Specifically, the size-based protocol is always at least as good as the bets-based protocol. For the specific case where $B = 1$, both protocol achieves the same resilience.

7.2 Ranked Preference

Next, we give the optimization for another concrete example of split-preference called the ranked preference profile. Specifically, we will give the characterization for $m = 2$ and 3 for better understanding. Instead of preferring certain outcomes, the players now have a rank over all possible outcomes and gets utility $m - i$ on the i -th preferred outcome. We use $\ell \succ \ell'$ to represent that a player ranked ℓ over ℓ' . For example, in a ternary ranked preferences, some players' rank is $2 \succ 0 \succ 1$, while some other players' rank is $1 \succ 2 \succ 0$. Each player gets utility 2 on its highest-ranked preference, 1 on the medium-ranked preference and 0 utility on its lowest-ranked preference. Note that when $m = 2$, this ranked preference boils down to the standard binary coin toss.

Formally, in an m -sided ranked preference profile, each player's preference $\mathbf{p} = (p_0, \dots, p_{m-1})$ is a permutation of $(0, 1, \dots, m-1)$. The honest expected utility of each player is therefore $\mu = \frac{m-1}{2}$. As in previous sections, without loss of generality, we assume 0 is the least preferred outcome, i.e., the cumulative bet on outcome 0 is the smallest among all possible outcomes. Specifically, we use n_i to denote the number of players who put bets i on outcome 0 for $i = 0, \dots, m-1$. Equivalently, n_i is the number of players who rank 0 at the i -th lowest position. In this case, $N_0 = \sum_{i > \mu} n_i$ is the number of players who prefer outcome 0 and $N_{\text{oth}} = \sum_{i \leq \mu} n_i$ is the number of oth-supporters. The table below presents the optimal parameter choices for $m = 2$ and $m = 3$. Since k_0 is simply $\lfloor \frac{N_0 + \gamma}{2} \rfloor$, we omit k_0 in the table.

Table 6: Coalition resilience of Protocol 5.3 for ranked preference.

# sides	Cases	γ	k_{oth}	t
$m = 2$	$n_0 \geq \frac{5}{2}n_1$	0	$n_0 - n_1$	$n_0 - \lfloor \frac{1}{2}n_1 \rfloor$
	otherwise	0	$\lfloor \frac{2}{3}n_0 - \frac{1}{6}n_1 \rfloor$	$\lfloor \frac{2}{3}n_0 + \frac{1}{3}n_1 \rfloor$
$m = 3$	$n_0 \geq \frac{5}{2}n_2 - \frac{1}{2}n_1$	n_1	$n_0 - n_2$	$\lfloor n_0 + \frac{1}{2}n_1 - \frac{1}{2}n_2 \rfloor$
	$n_1 + n_2 \leq n_0 < \frac{5}{2}n_2 - \frac{1}{2}n_1$	0	$\lfloor \frac{1}{3}n_0 + \frac{2}{3}n_1 + \frac{1}{6}n_2 \rfloor$	$\lfloor \frac{2}{3}n_0 + \frac{1}{3}n_1 + \frac{1}{3}n_2 \rfloor$
	otherwise	$n_1 + n_0$	N/A	$\lceil \frac{1}{2}n_0 + \frac{1}{2}n_1 + \frac{1}{2}n_2 \rceil - 1$

Claim 7.2. *For the ranked preference profile with $m = 2$ and $m = 3$, the above table gives the optimal parameter choice for Protocol 5.3.*

Proof. When $m = 2$, the problem boils down to the standard binary coin toss. The parameter choices thus follow from [WAS22]. For ternary ranked preference, by Theorem 5.18, γ is either 0 or n_1 .

We first show that $t(0, h(0)) < t(n_1, h(n_1))$. To see this, observe that $\mathbf{U}^0[i]$ for $i = 1, \dots, n_1$ is exactly 1. Therefore, $h(0) = n_1 + h(n_1)$. Hence, $t(0, h(0)) < t(n_\mu, h(n_\mu))$.

Moreover, if both $f(0)$ and $f(n_1)$ exists, we have $t(0, f(0)) = t(n_1, f(n_1))$. Again, by the fact that $\mathbf{U}^0[i]$ for $i = 1, \dots, n_1$ is exactly 1, it must be that $f(0) > n_1$. Therefore, $\eta_0(\alpha) = \alpha - n_1$ for $\alpha > n_1$ and $\eta_{n_1}(\alpha) = \alpha$. Substitute back into $c_2(\gamma, \alpha)$, we get $\frac{1}{2}n_1 + f(n_1) = f(0)$. This implies that $t(0, f(0)) = t(n_1, f(n_1))$ when $f(0)$ and $f(n_1)$ exist.

Therefore, the optimal solution is either $t(0, f(0)) = t(n_1, f(n_1))$ or $t(n_1, h(n_1))$. The claim thus follows from the fact that $f(0) = \frac{1}{3}n_0 + \frac{2}{3}n_1 + \frac{1}{6}n_2$ and $h(n_1) = n_0 - n_2$. $\square$

Below, we give the resilience of both protocols for ternary ranked preference, since binary ranked preference is equivalent to unit-split with $B = 1$. In this case, $B_0 = 2n_2 + n_1$ and $B_{\text{oth}} = n_2 + 2n_1 + 3n_0$. From Table 2, one can see that when n_0 is much larger than $n_1 + n_2$, the size-based protocol performs better; when $n_0 \leq n_1 + n_2$, the bets-based protocol achieves better resilience.

Table 7: Comparison of the optimal resilience for the ternary ranked preference. When n_0 is much larger than $n_1 + n_2$, the size-based protocol performs better; when $n_0 \leq n_1 + n_2$, the bets-based protocol achieves better resilience.

Size-based protocol Protocol 5.3		Bets-based protocol Protocol 6.3	
Case	Resilience	Case	Resilience
$n_0 \geq \frac{5}{2}n_2 - \frac{1}{2}n_1$	$\lfloor n_0 + \frac{1}{2}n_1 - \frac{1}{2}n_2 \rfloor$	$n_0 \geq \frac{5}{6}n_1 + \frac{8}{3}n_2$	$\lfloor n_0 + \frac{1}{6}n_1 - \frac{2}{3}n_2 \rfloor$
$n_1 + n_2 \leq n_0 < \frac{5}{2}n_2 - \frac{1}{2}n_1$	$\lfloor \frac{2}{3}n_0 + \frac{1}{3}n_1 + \frac{1}{3}n_2 \rfloor$	Otherwise	$\lfloor \frac{3}{5}n_0 + \frac{1}{2}n_1 + \frac{2}{5}n_2 \rfloor$
Otherwise	$\lceil N/2 \rceil - 1$		

8 Impossibility Results

In this section, we prove an impossibility result for binary coin toss to illustrate a landscape for game-theoretic fairness with general split preference. The impossibility result follows a similar technique as in [CGL⁺18, WAS22]. As before, we assume that 0 is the least preferred outcome.

The key to the impossibility result is to show that if a binary coin toss protocol Π' for a preference profile $\mathcal{P}$ achieving CSP-fairness against any t -sized semi-malicious coalition, then we can construct a three-party protocol Π among three supernodes $\mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3$ that satisfy the following conditions.

- (IC1) *Lone-wolf condition*: a semi-malicious, non-uniform p.p.t. adversary controlling $\mathcal{S}_1$ (or $\mathcal{S}_3$) cannot bias the output towards either outcome.
- (IC2) *Wolf-minion condition*: a semi-malicious, non-uniform p.p.t. adversary controlling $\mathcal{S}_1$ (or $\mathcal{S}_3$) and $\mathcal{S}_2$ cannot bias the output towards 1.
- (IC3) *$\mathbf{r}_2$ -equity condition*: for all but a negligible fraction of $\mathcal{S}_2$'s randomness $\mathbf{r}_2$, the value $|F(\mathbf{r}_2) - \frac{1}{2}|$ is negligible, where $F(\mathbf{r}_2)$ denotes the expected coin outcome in an honest execution where $\mathcal{S}_2$'s randomness is fixed to $\mathbf{r}_2$.

However, no three-party protocol achieves the above three conditions simultaneously.

Lemma 8.1 (Theorem 2.5 of [WAS22]). *No binary coin toss protocol Π among three players $\mathcal{S}_1, \mathcal{S}_2$ and $\mathcal{S}_3$ satisfying the above (IC1), (IC2), and (IC3) simultaneously.*

In the rest of this section, we will show how to partition the players into three supernodes $\mathcal{S}_1, \mathcal{S}_2$, and $\mathcal{S}_3$ such that all the above three conditions hold. Without loss of generality, given a protocol Π' , given a specific partition for the three supernodes, the three-party protocol Π runs as follows: $\mathcal{S}_1$ sends a message in the first round. In the following rounds, every supernode sends messages. In the last round, only $\mathcal{S}_3$ sends messages. The three supernodes then each simulates the execution of all players it contains. Any message between players within the same supernode are dealt with internally, and any message from a player in one supernode $\mathcal{S}_i$ to a player in another supernode $\mathcal{S}_j$ where $j \neq i$ is treated as a message from $\mathcal{S}_i$ to $\mathcal{S}_j$.

We are only left with the specific partition of the players. How to partition the players heavily relies on the preference profile $\mathcal{P}$. Below, we give the optimal partition for different formulations.

Let $\mathbf{P}_b$ for $b \in \{0, 1\}$ be the vector of identities of b -supporters, sorted based on non-increasing order of bet on 0. Let N_b denote the number of b -supporters. For any integers $0 \leq i \leq j \leq |\mathbf{P}_b|$, let $\mathbf{P}_b[i, j]$ denote the set of the i -th to the j -th b -supporter. If $i > j$, we simply treat $\mathbf{P}_b[i, j]$ as an empty set.

Roughly speaking, $\mathcal{S}_1$ and $\mathcal{S}_3$ each contains half of the 0-supporters and θ number of the 1-supporters. Remaining players all goes to $\mathcal{S}_2$. However, to make the total bet on $\mathcal{S}_1$ and $\mathcal{S}_3$ more “balanced”, we will not simply put the first $N_0/2$ number of 0-supporters into $\mathcal{S}_1$ and the rest to $\mathcal{S}_3$. Instead, we use a more fine-grained partition. Let $\bar{\eta} = \eta_\theta(\theta) + 1$ and $\underline{\eta} = \eta_0(\theta) + 1$ where $\eta(\cdot)$ is defined in Definition 5.4. Intuitively, the θ number of 1-supporters in $\mathcal{S}_3$ together with $\bar{\eta}$ number of 0-supporters prefer outcome 0, and the same holds for $\mathcal{S}_1$ with $\underline{\eta}$.

- $\mathcal{S}_1$ contains θ number of 1-supporters $\mathbf{P}_1[1, \theta]$, and $\lfloor \frac{N_0}{2} \rfloor$ number of 0-supporters:

$$\mathbf{P}_0[1, \lceil \underline{\eta}/2 \rceil] \cup \mathbf{P}_0[\underline{\eta} + 1, \underline{\eta} + \lceil \bar{\eta}/2 \rceil - \lceil \underline{\eta}/2 \rceil] \cup \mathbf{P}_0 \left[\left\lceil \frac{N_0}{2} \right\rceil + (\lceil \bar{\eta}/2 \rceil), N_0 \right]$$

- $\mathcal{S}_3$ contains the remaining 0-supporters and θ number of 1-supporters $\mathbf{P}_1[\theta + 1, 2\theta]$.
- $\mathcal{S}_2$ contains the remaining 1-supporters.

By the above partition, it is possible that $\mathcal{S}_2$ is empty if $\theta = N_1/2$. However, in such case, the impossibility directly follows from the lone-wolf condition. We want to choose θ such that if the protocol Π' achieves CSP-fairness against any t -sized coalition, then the resulting three-party protocol satisfies all three conditions. Define

$$\gamma' = \min \left\{ \gamma \mid \eta_0(\gamma) > \left\lceil \frac{N_0}{2} \right\rceil \right\}, \gamma'' = \begin{cases} \min\{\gamma \mid \eta_\gamma(\gamma) \geq N_0\}, & \text{if } \eta_{N_1/2}(N_1/2) \geq N_0 \\ \infty, & \text{otherwise.} \end{cases}$$

Specifically, t must satisfy the following constraints.

Table 8: Constraints for t based on N_0 , N_1 , and θ .

Non-negative	Lone-wolf	Wolf-minion	$\mathbf{r}_2$ -equity
$0 \leq 2\theta \leq N_1$	$t \geq \theta + \lceil \bar{\eta}/2 \rceil + \lceil N_0/2 \rceil$	$t \geq \gamma' + \lceil N_0/2 \rceil$	$t \geq N_1 - 2\theta$
$\bar{\eta} \leq N_0$	$t \geq \gamma' + \lceil N_0/2 \rceil$	$t \geq N_1 + \lceil N_0/2 \rceil - \theta$	

In the proof below, for a set of players G and an integer $x \leq |G|$, let $\overline{M}(G, x)$ denote the set of x players that have the most bet on 0 in G and let $\underline{M}(G, x)$ denote the x players that have the least bet on 0 in G . If $x < 0$ in either of the aforementioned functions, treat it as if $x = 0$.

Lemma 8.2. *Let Π' be a protocol that is CSP-fair against any non-uniform p.p.t., semi-malicious coalition of size t . If θ and t satisfy the non-negative and lone-wolf constraints in Table 8, then the three-party protocol Π satisfies the lone-wolf condition (IC1).*

Proof. For the sake of contradiction, assume that $\mathcal{S}_1$ can bias the outcome towards 0 by a non-negligible amount. Consider a coalition $\mathcal{C}$ that consists of $\mathcal{S}_1$ and $\overline{M}(\mathcal{S}_3, \lceil \eta/2 \rceil)$. By the above defined partition, such a coalition exists since $\underline{\eta} \leq \bar{\eta} \leq N_0$. See Figure 8a for an illustrative expression of $\mathcal{C}$. Because $\mathcal{S}_3$ contains $\lceil N_0/2 \rceil \geq \lceil \eta/2 \rceil$ number of 0-supporters, all players in $\overline{M}(\mathcal{S}_3, \lceil \eta/2 \rceil)$ are 0-supporters. Coalition $\mathcal{C}$ thus contains the θ number of 1-supporters from $\mathcal{S}_1$ and at least $\underline{\eta} = \eta_0(\theta)$ number of 0-supporters with the most bets on 0, this coalition prefers 0 by definition. This coalition can have $\mathcal{S}_1$ adopt the strategy that bias towards 0 while having players in $\mathcal{S}_3$ behave honestly. This strategy strictly increases their expected utility. This contradicts CSP-fairness of Π' since $t \geq \lceil \frac{N_0}{2} \rceil + \theta + \lceil \bar{\eta}/2 \rceil \geq |\mathcal{C}|$.

If $\mathcal{S}_1$ can bias the outcome towards 1, then consider a coalition $\mathcal{C}'$ containing $\mathcal{S}_1$ and another $\max\{\gamma' - \theta, 0\}$ number of oth-supporters outside $\mathcal{S}_1$. By definition of γ' , this coalition $\mathcal{C}'$ prefers outcome 1 (see Figure 8b). Such coalition can have $\mathcal{S}_1$ adopt the strategy that bias the outcome towards 1 while the remaining players behaving honestly. This contradicts CSP-fairness of Π' against t -sized coalition since $t \geq \gamma' + \lceil \frac{N_0}{2} \rceil \geq |\mathcal{C}'|$. The same argument holds for $\mathcal{S}_3$. $\square$



(a) Coalition $\mathcal{C}$ consists of all hatch-shaded parts. Such coalition prefers 0.

(b) A coalition $\mathcal{C}'$ of all hatch-shaded parts. Such coalition prefers 1.

Figure 8: The players marked by orange (either solid or shaded) are in $\mathcal{S}_1$, those marked by green (either solid or shaded) are in $\mathcal{S}_3$, and remaining players marked by blue are in $\mathcal{S}_2$.

Lemma 8.3. *Let Π' be a protocol that is CSP-fair against any non-uniform p.p.t., semi-malicious coalition of size t . If θ and t satisfy the non-negative and wolf-minion constraints in Table 8, then the three-party protocol Π satisfies the wolf-minion condition (IC2).*

Proof. For the sake of contradiction, suppose the coalition $\mathcal{C} = \mathcal{S}_1 \cup \mathcal{S}_2$ can bias the outcome towards 1 by a non-negligible amount. If the number of 1-supporters in $\mathcal{C}$ exceeds γ' , i.e., $N_1 - \theta \geq \gamma'$, then coalition prefers 1. This contradicts the CSP-fairness of Π' since $t \geq N_1 - \theta + \lceil N_0/2 \rceil \geq |\mathcal{C}|$.

If $N_1 - \theta < \gamma'$, then consider a coalition $\mathcal{C}'$ containing $\mathcal{S}_1 \cup \mathcal{S}_2$ and $\gamma' - (N_1 - \theta)$ number of 1-supporters from $\mathcal{S}_3$. Such coalition must exist since $\gamma' \leq N_1$. Again by definition, coalition $\mathcal{C}'$ prefers 1. This contradicts the CSP-fairness of Π' since $t \geq \gamma' + \lceil N_0/2 \rceil \geq |\mathcal{C}'|$. The same argument holds for $\mathcal{C}' = \mathcal{S}_3 \cup \mathcal{S}_2$. $\square$

Lemma 8.4. *Let Π' be a protocol that is CSP-fair against any nonuniform p.p.t., semi-malicious coalition of size t . If θ and t satisfy the non-negative and $\mathbf{r}_2$ -equity constraints in Table 8, then Π satisfies the $\mathbf{r}_2$ -equity condition (IC3).*

Proof. Note that $\mathcal{S}_2$ only contains 1-supporters. Thus $\mathcal{S}_2$ must prefer 1. Suppose for the sake of contradiction that $\mathbf{r}_2$ -equity condition is not true, that is, for a non-negligible fraction of $\mathcal{S}_2$'s randomness $\mathbf{r}_2$, the difference $|F(\mathbf{r}_2) - \frac{1}{2}| = 1/p(\lambda)$ for some polynomial $p(\cdot)$. We separate the rest of proof into two cases.

Case 1. Suppose that for a non-negligible fraction of $\mathcal{S}_2$'s randomness $\mathbf{r}_2$, the value $F(\mathbf{r}_2) - \frac{1}{2} \geq 1/p(\lambda)$. Let $\mathbf{r}_2^*$ be one such instance of $\mathbf{r}_2$ where $F(\mathbf{r}_2^*) - \frac{1}{2} \geq \text{negl}(\lambda)$. An adversary $\mathcal{A}^*$ that controls $\mathcal{S}_2$ and receives $\mathbf{r}_2^*$ as an advice can program its randomness to $\mathbf{r}_2^*$ but otherwise behave honestly. This increases $\mathcal{S}_2$'s expected utility by a non-negligible amount, which contradicts CSP-fairness of Π' since $t \geq N_1 - 2\theta = |\mathcal{S}_2|$.

Case 2. Suppose, however, that for a non-negligible fraction of $\mathcal{S}_2$'s randomness $\mathbf{r}_2$, $\frac{1}{2} - F(\mathbf{r}_2) \geq 1/p(\lambda)$. By correctness, $\mathbb{E}_{\mathbf{r}_2}[F(\mathbf{r}_2)] = \frac{1}{2}$. Thus, there must exist a non-negligible fraction of $\mathbf{r}_2$ such that $F(\mathbf{r}_2) - \frac{1}{2} \geq 1/q(\lambda)$ for some polynomial $q(\cdot)$. By the same reasoning as case 1, there is a contradiction. $\square$

Combining Lemma 8.2, Lemma 8.3 and Lemma 8.4, we conclude that given N_0, N_1 , no CSP-fair coin toss protocol defend against t -sized coalition for any t satisfying the constraints in Table 8. The remaining task is to find the minimum t subject to all the constraints.

If we clean up all the constraints in Table 8, we end up with the following optimization problem:

$$\begin{aligned} & \text{Minimize } t, \text{ subject to} \\ & 0 \leq \theta \leq \min\{N_1/2, \gamma'' - 1\}, \\ & t \geq g_1(\theta) := N_1 + \lceil N_0/2 \rceil - \theta, \\ & t \geq g_2(\theta) := \theta + \lceil \bar{\eta}/2 \rceil + \lceil N_0/2 \rceil, \\ & t \geq \gamma' + \lceil N_0/2 \rceil. \end{aligned}$$

Theorem 8.5. *Fix any N . Given a preference profile $\mathcal{P} \in \mathcal{F}_{\text{split}}(N, 2)$, there is no CSP-fair binary coin toss protocol against t -sized semi-malicious coalition for*

$$t \geq \max\{N_1 + \lceil N_0/2 \rceil - \theta^*, \gamma' + \lceil N_0/2 \rceil\},$$

where

$$\theta^* = \max\{\theta \mid \theta \leq \gamma'' - 1 \text{ and } 2\theta + \lceil (\eta_\theta(\theta) + 1)/2 \rceil \leq N_1\}. \quad (19)$$

This set is non-empty as $\theta = 0$ satisfy the conditions.

Proof. Since $\gamma' + \lceil N_0/2 \rceil$ is a constant that does not depend on θ , we only need to choose θ that minimizes $\max\{g_1(\theta), g_2(\theta)\}$. Observe that $g_2(N_1/2) \geq g_1(N_1/2)$. Therefore, picking θ to be $N_1/2$ cannot be the optimal solution unless $g_2(N_1/2) = g_1(N_1/2)$. The optimal solution to θ is therefore the maximum θ such that $\theta \leq \gamma'' - 1$ and that $g_2(\theta) \leq g_1(\theta)$. Equivalently, the optimal θ^* is the one that satisfies Equation (19). The theorem thus follows by combining Lemma 8.2, Lemma 8.3, Lemma 8.3, and Lemma 8.1. $\square$

Corollary 8.6. *For any binary-split preference profile, Protocol 5.3 is almost optimal except for a gap no more than*

$$\begin{cases} \gamma' - \lfloor N_1/2 \rfloor + 1, & \text{if } \gamma' > N_1/2 \\ \lceil (\theta^* + 1)/2 \rceil + 1, & \text{if } \gamma' \leq N_1/2 \text{ and } \eta_{\theta^*+1}(\theta^* + 1) + \lceil \frac{3}{2}(\theta^* + 1) \rceil > N_1 + \lceil N_0/2 \rceil \\ \lceil N_0/2 \rceil + 2\theta^* - N_1 + 4, & \text{otherwise.} \end{cases}$$

Proof. If $N_1 - \theta^* < \gamma'$, then it must be that $\gamma' > \frac{N_1}{2}$. Since Protocol 5.3 defend against at least $(\lceil N/2 \rceil - 1)$ -sized coalition when $\gamma' \geq \frac{N_1}{2}$, the gap is at most $\gamma' - \lfloor N_1/2 \rfloor + 1$.

In the rest of this proof, we will focus on the case where $N_1 - \theta^* \geq \gamma'$. Consider setting $\gamma = \alpha = \theta^* + 1$ in Protocol 5.3, where θ^* is defined in Equation (19). Then Protocol 5.3 achieves coalition resilience

$$t^* = \min \left\{ N_1 + \left\lceil \frac{N_0 - (\theta^* + 1)}{2} \right\rceil - (\theta^* + 1), \eta_{\theta^*+1}(\theta^* + 1) + (\theta^* + 1) \right\}.$$

We consider the following cases:

Case 1: If $t^* = N_1 + \left\lceil \frac{N_0 - (\theta^* + 1)}{2} \right\rceil - (\theta^* + 1)$, the gap is at most $\lceil (\theta^* + 1)/2 \rceil + 1$.

Case 2: Otherwise, it must be that $\eta_{\theta^*+1}(\theta^* + 1) \leq N_0$. Therefore, $\theta^* < \gamma'' - 1$. By definition of θ^* , we have

$$2(\theta^* + 1) + \lceil (\eta_{\theta^*+1}(\theta^* + 1) + 1)/2 \rceil > N_1.$$

This implies that

$$\eta_{\theta^*+1}(\theta^* + 1) + 2 \geq 2\lceil (\eta_{\theta^*+1}(\theta^* + 1) + 1)/2 \rceil > 2(N_1 - 2(\theta^* + 1)),$$

or equivalently, $\eta_{\theta^*+1}(\theta^* + 1) \geq 2(N_1 - 2(\theta^* + 1)) - 1$. Substitute this into the difference between the coalition resilience and the impossibility, we have

$$\begin{aligned} & N_1 + \lceil N_0/2 \rceil - \theta^* - (\eta_{\theta^*+1}(\theta^* + 1) + (\theta^* + 1)) \\ & \leq N_1 + \lceil N_0/2 \rceil - 2\theta^* - 1 - (2(N_1 - 2(\theta^* + 1)) - 1) \\ & = \lceil N_0/2 \rceil + 2\theta^* - N_1 + 4. \end{aligned}$$

$\square$

References

- [ACH11] Gilad Asharov, Ran Canetti, and Carmit Hazay. Towards a game theoretic view of secure computation. In *Eurocrypt*, 2011. 1.3
- [ADGH06] Ittai Abraham, Danny Dolev, Rica Gonen, and Joseph Halpern. Distributed computing meets game theory: Robust mechanisms for rational secret sharing and multiparty computation. In *PODC*, 2006. 1.3
- [ADMM16] Marcin Andrychowicz, Stefan Dziembowski, Daniel Malinowski, and undefinedukasz Mazurek. Secure multiparty computations on bitcoin. *Commun. ACM*, 59(4):76–84, March 2016. 1.3
- [AL11] Gilad Asharov and Yehuda Lindell. Utility dependence in correct and fair rational secret sharing. *Journal of Cryptology*, 24(1), 2011. 1.3
- [AO16] Bar Alon and Eran Omri. Almost-optimally fair multiparty coin-tossing with nearly three-quarters malicious. In *TCC*, 2016. 1
- [ATM⁺22] Lukas Aumayr, Sri AravindaKrishnan Thyagarajan, Giulio Malavolta, Pedro Moreno-Sanchez, and Matteo Maffei. Sleepy channels: Bi-directional payment channels without watchtowers. In *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security, CCS '22*, page 179–192, New York, NY, USA, 2022. Association for Computing Machinery. 1.3
- [BHLT17] Niv Buchbinder, Iftach Haitner, Nissan Levi, and Eliad Tsfadia. Fair coin flipping: Tighter analysis and the many-party case. In *Proceedings of the Twenty-Eighth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 2580–2600. SIAM, 2017. 1
- [BHMO22] Amos Beimel, Iftach Haitner, Nikolaos Makriyannis, and Eran Omri. Tighter bounds on multiparty coin flipping via augmented weak martingales and differentially private sampling. *SIAM Journal on Computing*, 51(4):1126–1171, 2022. 1
- [BHT18] Itay Berman, Iftach Haitner, and Aris Tentes. Coin flipping of any constant bias implies one-way functions. *Journal of the ACM (JACM)*, 65(3):1–95, 2018. 1
- [BK14] Iddo Bentov and Ranjit Kumaresan. How to use bitcoin to design fair protocols. In *CRYPTO*, pages 421–439, 2014. 1.3
- [Blu83] Manuel Blum. Coin flipping by telephone a protocol for solving impossible problems. *SIGACT News*, 1983. 1, 5
- [BOO15] Amos Beimel, Eran Omri, and Ilan Orlov. Protocols for multiparty coin toss with a dishonest majority. *Journal of Cryptology*, 28(3):551–600, 2015. 1
- [CCWS21a] Kai-Min Chung, T.-H. Hubert Chan, Ting Wen, and Elaine Shi. Game-theoretic fairness meets multi-party protocols: The case of leader election. In *CRYPTO*, 2021. <https://eprint.iacr.org/2020/1591>. 1.3
- [CCWS21b] Kai-Min Chung, T.-H. Hubert Chan, Ting Wen, and Elaine Shi. Game-theoretic fairness meets multi-party protocols: The case of leader election. In Tal Malkin and Chris Peikert, editors, *Advances in Cryptology – CRYPTO 2021*, pages 3–32, Cham, 2021. Springer International Publishing. 1.3

- [CGL⁺18] Kai-Min Chung, Yue Guo, Wei-Kai Lin, Rafael Pass, and Elaine Shi. Game theoretic notions of fairness in multi-party coin toss. In *TCC*, 2018. [1](#), [1.3](#), [2.4](#), [3.1](#), [8](#)
- [Cle86] Richard Cleve. Limits on the security of coin flips when half the processors are faulty. In *Proceedings of the eighteenth annual ACM symposium on Theory of computing*, pages 364–369, 1986. [1](#)
- [CMST22] Hao Chung, Elisaweta Masserova, Elaine Shi, and Sri AravindaKrishnan Thyagarajan. Rapidash: Foundations of side-contract-resilient fair exchange. Cryptology ePrint Archive, Paper 2022/1063, 2022. <https://eprint.iacr.org/2022/1063>. [1.3](#)
- [DEF18] Stefan Dziembowski, Lisa Ekey, and Sebastian Faust. Fairswap: How to fairly exchange digital goods. In *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security, CCS '18*, page 967–984, New York, NY, USA, 2018. Association for Computing Machinery. [1.3](#)
- [DR07] Yevgeniy Dodis and Tal Rabin. Cryptography and game theory. In *AGT*, 2007. [1.3](#)
- [DSLMM11] Dana Dachman-Soled, Yehuda Lindell, Mohammad Mahmoody, and Tal Malkin. On the black-box complexity of optimally-fair coin tossing. In *Theory of Cryptography Conference*, pages 450–467. Springer, 2011. [1](#)
- [GMW19] Oded Goldreich, Silvio Micali, and Avi Wigderson. How to play any mental game, or a completeness theorem for protocols with honest majority. In *STOC*. 2019. [1](#)
- [Gol04] Oded Goldreich. *Foundations of Cryptography, Volume 2*. Cambridge university press Cambridge, 2004. [A](#), [A.2](#)
- [HO14] Iftach Haitner and Eran Omri. Coin flipping with constant bias implies one-way functions. *SIAM Journal on Computing*, 43(2):389–409, 2014. [1](#)
- [HT04] Joseph Halpern and Vanessa Teague. Rational secret sharing and multiparty computation. In *STOC*, 2004. [1.3](#)
- [HT14] Iftach Haitner and Eliad Tsfadia. An almost-optimally fair three-party coin-flipping protocol. In *Proceedings of the forty-sixth annual ACM symposium on Theory of computing*, pages 408–416, 2014. [1](#)
- [IOZ14] Yuval Ishai, Rafail Ostrovsky, and Vassilis Zikas. Secure multi-party computation with identifiable abort. In *Advances in Cryptology–CRYPTO 2014: 34th Annual Cryptology Conference, Santa Barbara, CA, USA, August 17–21, 2014, Proceedings, Part II 34*, pages 369–386. Springer, 2014. [A](#), [A.2](#)
- [Kat08] Jonathan Katz. Bridging game theory and cryptography: Recent results and future directions. In *TCC*, 2008. [1.3](#)
- [KB14] Ranjit Kumaresan and Iddo Bentov. How to use bitcoin to incentivize correct computations. In *Proceedings of the 2014 ACM SIGSAC Conference on Computer and Communications Security, Scottsdale, AZ, USA, November 3–7, 2014*, pages 30–41. ACM, 2014. [1.3](#)
- [KKZ24] Ohad Klein, Ilan Komargodski, and Chenzhi Zhu. On the (im) possibility of game-theoretically fair leader election protocols. In *Theory of Cryptography Conference*, pages 383–412. Springer, 2024. [1.3](#)

- [KMS⁺16] Ahmed E. Kosba, Andrew Miller, Elaine Shi, Zikai Wen, and Charalampos Papamanthou. Hawk: The blockchain model of cryptography and privacy-preserving smart contracts. In *IEEE Symposium on Security and Privacy, SP 2016, San Jose, CA, USA, May 22-26, 2016*, pages 839–858. IEEE Computer Society, 2016. 1.3
- [KMSW22] Ilan Komargodski, Shin’ichiro Matsuo, Elaine Shi, and Ke Wu. log*-round game-theoretically-fair leader election. In Yevgeniy Dodis and Thomas Shrimpton, editors, *Advances in Cryptology – CRYPTO 2022*, pages 409–438, Cham, 2022. Springer Nature Switzerland. 1.3
- [KN08] Gillat Kol and Moni Naor. Cryptography and game theory: Designing protocols for exchanging information. In *TCC*, 2008. 1.3
- [KVV16] Ranjit Kumaresan, Vinod Vaikuntanathan, and Prashant Nalini Vasudevan. Improvements to secure computation with penalties. In *ACM CCS*, 2016. 1.3
- [MGW87] Silvio Micali, Oded Goldreich, and Avi Wigderson. How to play any mental game. In *Proceedings of the Nineteenth ACM Symp. on Theory of Computing, STOC*, pages 218–229. ACM New York, NY, USA, 1987. 1, A, A.2
- [MNS16] Tal Moran, Moni Naor, and Gil Segev. An optimally fair coin toss. *Journal of Cryptology*, 29(3):491–513, 2016. 1
- [OPRV09] Shien Jin Ong, David C. Parkes, Alon Rosen, and Salil P. Vadhan. Fairness with an honest minority and a rational majority. In *TCC*, 2009. 1.3
- [TSW24] Sri AravindaKrishnan Thyagarajan, Pratik Soni, and Ke Wu. Game-theoretically fair distributed sampling. In *Annual International Cryptology Conference*, pages 207–239. Springer, 2024. 1, 2, 2, 2.4, 3.1, 5.2, 6.2, B, B.3
- [WAS22] Ke Wu, Gilad Asharov, and Elaine Shi. A complete characterization of game-theoretically fair, multi-party coin toss. In Orr Dunkelman and Stefan Dziembowski, editors, *Advances in Cryptology – EUROCRYPT 2022*, pages 120–149, Cham, 2022. Springer International Publishing. 1, 2, 1.3, 2.1, 2.2, 2.3, 2.4, 7.2, 8, 8.1
- [Yao82] Andrew C Yao. Protocols for secure computations. In *FOCS*, 1982. 1

Supplementary Materials

A Cryptographic Building Blocks

A function $\text{negl} : \mathbb{N} \rightarrow \mathbb{R}$ is negligible in n if for every polynomial $\text{poly}(\cdot)$, there exists $n_0 \in \mathbb{N}$ s.t. for every $n \geq n_0$ it holds that $|\text{negl}(n)| \leq \text{poly}(n)$. We say that two distribution ensembles $\{X_n\}$ and $\{Y_n\}$ are computationally indistinguishable, denoted as $\{X_n\} \equiv_c \{Y_n\}$ if for any ppt adversary $\mathcal{A}$, for any n , we have $|\Pr[x \leftarrow X_n : \mathcal{A}(x) = 1] - \Pr[y \leftarrow Y_n : \mathcal{A}(y) = 1]| \leq \text{negl}(n)$.

Secret Sharing.

Definition A.1 (Access Structure). Let $[N]$ be the set of players. A collection $\Gamma \subseteq 2^{[N]}$ is monotone if $B \in \Gamma$ and $B \subset C$, then $C \in \Gamma$.

An *access structure* is a monotone collection $\Gamma \subseteq 2^{[N]}$ of non-empty subsets of $[N]$. Sets in Γ are called authorized sets.

A secret sharing with access structure Γ consists of two algorithms **Share** and **Recon**.

1. The sharing algorithm **Share**($\cdot$) takes a secret s as input and output N number of shares $(s_1, \dots, s_N)$: The sharing algorithm takes as input a secret s , and outputs N shares of the secret.
2. The reconstruction takes as input a set I of secret shares $\{s_i\}_{i \in I}$ and output the secret if $I \in \Gamma$ or $\perp$ if $I \notin \Gamma$.

The secret sharing scheme with access structure Γ satisfies the following properties:

1. **Correctness**: For any secret s and any sets $I \in \Gamma$, the reconstruction algorithm returns the correct share given $\{s_i\}_{i \in I}$. Specifically:

$$\Pr[\text{Recon}(I, \{s_i\}_{i \in I}) = s : (s_1, \dots, s_n) \leftarrow \text{Share}(s)] = 1.$$

2. **Security**: Any set $I \notin \Gamma$ learns no information about the secret. Formally, for any secrets s and s' , for any sets $I \subset [N]$ where $I \notin \Gamma$, it must be that

$$\{\{s_i\}_{i \in I} : (s_1, \dots, s_n) \leftarrow \text{Share}(s)\} \equiv_c \{\{s'_i\}_{i \in I} : (s'_1, \dots, s'_n) \leftarrow \text{Share}(s')\},$$

where $\equiv_c$ means computational indistinguishability.

Specifically, a k -out-of- N Shamir secret sharing uses access structures $\Gamma = \{I \subset [N], |I| \geq k+1\}$, and weighted threshold secret sharing uses access structures $\Gamma = \{I \subset [N], \sum_{i \in I} w_i \geq k+1\}$, where w_i is player i 's assigned weight.

Shamir's secret sharing uses polynomials on finite fields. To share a secret s , the sharing algorithm randomly chooses a degree $k-1$ polynomial $f(\cdot)$ over some finite field $\mathbb{F}$ such that $f(s) = 0$. Each s_i is then set to $f(i)$. The reconstruction algorithm simply recovers the polynomial $f(\cdot)$ using interpolation. Shamir's secret sharing additionally satisfies the following *linearity* property: For any secrets s and s' , for any sets $I \subset [N]$ such that $|I| \geq k$, the following holds:

$$\Pr[\text{Recon}(I, \{s_i + s'_i\}_{i \in I}) = s + s' : (s_1, \dots, s_n) \leftarrow \text{Share}(s), (s'_1, \dots, s'_n) \leftarrow \text{Share}(s')] = 1.$$

Equivalently one can reconstruct $s + s'$ from the sum of the shares of s and s' .

Weighted threshold secret sharing can be instantiated directly using Shamir's secret sharing if all w_i are integers. Specifically, each secret is split into $W = \sum_{i \in [N]} w_i$ shares and each player i gets w_i number of them.

Identifiable abort. Multi-party computation with identifiable abort [MGW87, Gol04, IOZ14] achieves the following guarantees: either the computation succeed, or if the computation fails, honest players receive a set of identities of misbehaved players who led to the failure. Here we give the ideal-world execution assuming the existence of some trusted authority specifically for inputless function $\mathcal{F}$. We refer the readers to [IOZ14] for the details on the instantiation of this ideal-world execution and how to achieve identifiable abort for functions with inputs.

Let $\mathcal{A}$ be an adversary that controls a set of parties $\mathcal{C} \subset [n]$. Recall that each player receives the security parameter λ as an input, and the adversary may additionally receive some auxiliary inputs. Given a function $\mathcal{F}$, the ideal execution works as follows:

The trusted authority first samples some randomness r and compute each player's output $y_1, \dots, y_N$ based on the function $\mathcal{F}$. It then sends the corresponding outputs for corrupted players $\{y_i\}_{i \in \mathcal{C}}$ to the adversary $\mathcal{A}$. The adversary $\mathcal{A}$ then decides whether to fail this execution. If not, it sends `ok` to the trusted authority, after which the trusted authority will send the corresponding outputs $\{y_j\}_{j \notin \mathcal{C}}$ to honest players. Otherwise, if $\mathcal{A}$ wants to fail the current execution, it sends some (fail, i) for some $i \in \mathcal{C}$ to the trusted authority. The trusted authority will then send (fail, i) to all honest players. At the end of the execution, honest players output whatever they received from the trusted authority, and the adversary outputs an arbitrary function based on its view.

Lemma A.2 ([MGW87, Gol04, IOZ14]). *Assuming the existence of oblivious transfers. For any N -party inputless functionality $\mathcal{F}$, there is a protocol Π that securely computes $\mathcal{F}$ with identifiable abort.*

Signature scheme A signature scheme Sig consists of three algorithms:

- Key generation algorithm $(\text{sk}, \text{vk}) \leftarrow \text{KeyGen}(1^\lambda)$: takes λ as input and outputs a pair of keys sk for sign messages and vk for verification.
- Signing algorithm $\sigma \leftarrow \text{Sign}(\text{sk}, \text{msg})$: takes in the signing key key and a message msg , and outputs a signature σ .
- Verification algorithm $b \leftarrow \text{Ver}(\text{vk}, \text{msg}, \sigma)$: takes the verification key vk , the message msg and the signature σ and outputs a single bit. It output bit 1 for accept and 0 for reject.

A signature scheme must satisfy correctness and unforgeability.

- **Correctness:** The verification must accept any correctly generated signatures. Formally, for any messages msg ,

$$\Pr[\text{Ver}(\text{vk}, \text{msg}, \sigma) = 1 : (\text{sk}, \text{vk}) \leftarrow \text{KeyGen}(1^\lambda), \sigma \leftarrow \text{Sign}(\text{sk}, \text{msg})] = 1.$$

- **Unforgeability:** Adversary cannot forge signature of honest players. Specifically, for any non-uniform p.p.t. adversary $\mathcal{A}$, there exists a negligible function $\text{negl}(\cdot)$ such that for any λ , probability $\Pr[\text{Expt}^{\mathcal{A}}(1^\lambda) = 1] \leq \text{negl}(\lambda)$, where $\text{Expt}^{\mathcal{A}}(1^\lambda)$ describes the following experiments:

- $(\text{sk}, \text{vk}) \leftarrow \text{KeyGen}(1^\lambda)$;
- $(\text{msg}^*, \sigma^*) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{Sign}}(\cdot)}(1^\lambda, \text{vk})$, where $\mathcal{O}_{\text{Sign}}(\cdot)$ is a signing oracle that returns $\sigma \leftarrow \text{Sign}(\text{sk}, \text{msg})$ on query msg .
- Output 1 iff $\text{Ver}(\text{vk}, \text{msg}^*, \sigma^*) = 1$ and msg^* not queried before.

Commitment scheme A commitment scheme contains two phases:

- **Commit phase:** For a message msg , the committer samples some randomness r and sends $\text{com} = \text{Commit}(\text{msg}, r)$ to the receiver.
- **Open phase:** The committer sends the opening (msg, r) to the receiver. The receiver accepts if the opening is correct and rejects otherwise.

A commitment scheme must satisfy the following properties.

- *Binding:* For any $\text{msg} \neq \text{msg}'$, for any r and r' , $\text{Commit}(\text{msg}, r) \neq \text{Commit}(\text{msg}', r')$.
- *Hiding:* For any msg and msg' , we have $\{\text{Commit}(\text{msg}, r) : r \xleftarrow{\$} U\} \equiv_c \{\text{Commit}(\text{msg}', r'), r' \xleftarrow{\$} U\}$.

B Upgrade To Malicious Security

We follow the same framework used in [TSW24] to upgrade Protocol 5.3 to malicious security. Protocol 6.3 can be upgraded using the exactly same technique. All the building blocks needed to construct the malicious protocols are introduced in Appendix A.

B.1 Maliciously Secure $\text{GroupToss}^G[k]$ Sub-Protocol

In the malicious version, players will run MPC with identifiable abort Appendix A to securely compute the following ideal functionality to generate secret shares, commitment of the secret, and signatures of the shares.

Functionality B.1: $\mathfrak{F}_{\text{sharegen}}^{b, \mathcal{O}}[k]$ Ideal Functionality

1. Sample $(\text{sk}, \text{vk}) \leftarrow \text{Sig.KeyGen}(1^\lambda)$ where $\text{Sig} := (\text{KeyGen}, \text{Sign}, \text{Ver})$ is a signature scheme.
2. Sample $s \xleftarrow{\$} [m]_0$ and randomness $\mathbf{r} \xleftarrow{\$} \{0, 1\}^\lambda$. Let $\text{com} := \text{Commit}(s, \mathbf{r})$.
3. Share $s, \mathbf{r}$, and com using $(k+1)$ -out-of- $|\mathcal{O}|$ secret sharing and get $\{s_i, r_i, \text{com}_i\}_{i \in \mathcal{O}}$. Sign all the shares $\sigma_i := \text{Sig.Sign}(\text{sk}, \text{com}_i)$ and $\sigma'_i := \text{Sig.Sign}(\text{sk}, (s_i, r_i))$ for any $i \in \mathcal{O}$.
4. Each player i receives $(\text{vk}, s_i, r_i, \text{com}_i, \sigma_i, \sigma'_i)$.

The sub-protocol then runs as follows.

Protocol B.2: $\text{size-GroupToss}^G[k]$ sub-protocol (Malicious)

Let $\mathcal{O}$ be the set of online players, initialized as G .

Sharing Phase. Repeat the following until success:

Players in $\mathcal{O}$ run MPC with identifiable abort to securely compute the ideal functionality $\mathfrak{F}_{\text{sharegen}}^{b, \mathcal{O}}[k]$, described below. If the protocol aborts, every honest player obtains the identity of some corrupt player j^* . Remove j^* from $\mathcal{O}$. At the end of this stage, each player $i \in \mathcal{O}$ has obtained the tuple $(\text{vk}, s_i, r_i, \text{com}_i, \sigma_i, \sigma'_i)$ from $\mathfrak{F}_{\text{sharegen}}^{b, \mathcal{O}}[k]$.

Vote Phase. Each player broadcasts its verification key vk . Let vk' be the verification key that gathered most votes. If vk' received no more than k votes, the voting phase fails.

Otherwise, let $\mathcal{O}_{\text{vk}'}$ be the set of players in $\mathcal{O}$ that received vk' from $\mathfrak{F}_{\text{sharegen}}^{b, \mathcal{O}}[k]$. Each player $i \in \mathcal{O}_{\text{vk}'}$ broadcasts (com_i, σ_i) .

Let $\mathfrak{S}$ denote the set of all broadcast (com_j, σ_j) pairs where σ_j is a valid signature of com_j under vk' . If $|\mathfrak{S}| \geq k + 1$ and all shares com_j reconstruct uniquely to the value com , then each player records the reconstructed com . Otherwise, the vote phase fails.

Reconstruction Phase. If the vote phase failed, output $s = \perp$. Otherwise, each player $i \in \mathcal{O}_{\text{vk}'}$ broadcasts the tuple (s_i, r_i, σ'_i) . Every player use those (s_j, r_j) with a valid signature σ'_j to reconstruct the secret $(s, \mathbf{r})$. If the reconstruction succeed and $(s, \mathbf{r})$ is a valid opening of com , output s . Otherwise, output $s = \perp$.

Lemma B.3 (Lemma 7.4 of [TSW24]). *Protocol B.2 satisfies the following properties for any coalition $\mathcal{C} \subseteq G$ due to the property of Shamir secret sharing.*

- *Binding.* If the voting phase succeed, then at the end of the voting phase, a unique secret s is binded such that the reconstruction phase either succeeds and outputs s , or it fails and outputs $\perp$.
- *Knowledge threshold.* If $|\mathcal{C}| \geq k+1$, the coalition controls the value of s binded to at the end of the voting phase. Otherwise, $\mathcal{C}$'s view at the end of the voting phase is computationally independent of the coin value s . That is, if the voting phase does not fail, for any non-uniform p.p.t $\mathcal{C}$ with $|\mathcal{C}| \leq k$, there exists a simulator $\mathcal{S}(1^\lambda)$ such that,

$$(s, \text{view}_{\mathcal{C}}) \equiv_c (\text{Uniform}, \mathcal{S}(1^\lambda)) \quad (20)$$

where $\text{view}_{\mathcal{C}}$ is the coalition's view at the end of the voting phase and Uniform denotes a values uniformly randomly sampled from $[m]_0$.

- *Liveness threshold.* If $|\mathcal{C}| \geq \min\{|G| - k, \lceil |G|/2 \rceil\}$, the coalition can cause the reconstruction to output $\perp$. Otherwise, the reconstruction phase must succeed.

B.2 Malicious Secure Final Protocol for Coalition Size

Protocol B.4: Final Protocol for Coalition Size (Malicious)

In this protocol, $\text{size-GroupToss}^{G_0}[k_0]$ refers to the malicious secure version of the protocol.

Input: The preference profile $\mathcal{P}$.

Sharing phase:

(S-0) G_0 runs the sharing phase of $\text{size-GroupToss}^{G_0}[k_0]$. G_{oth} audits.

(S-oth) If G_{oth} is non-empty, G_{oth} runs the sharing phase of $\text{size-GroupToss}^{G_{\text{oth}}}[k_{\text{oth}}]$. G_0 audits.

Voting phase:

(V-oth) If G_{oth} is non-empty, G_{oth} runs the voting phase of $\text{size-GroupToss}^{G_{\text{oth}}}[k_{\text{oth}}]$. G_0 audits.

(V-0) G_0 runs the voting phase of $\text{size-GroupToss}^{G_0}[k_0]$. G_{oth} audits.

Reconstruction phase:

(R-0) G_0 runs the reconstruction phase of $\text{size-GroupToss}^{G_0}[k_0]$ while G_{oth} audits. Let s_0 be the outcome of this protocol.

(R-oth) If G_{oth} is non-empty, G_{oth} runs the reconstruction phase of $\text{size-GroupToss}^{G_{\text{oth}}}[k_{\text{oth}}]$ while G_0 audits. Let s_{oth} be the outcome of this protocol. If G_{oth} is empty, let $s_{\text{oth}} = 0$.

(Output) If $s_{\text{oth}} = \perp$, output 0. Otherwise, if $s_0 = \perp$, set $s_0 = 0$ and output $(s_0 + s_{\text{oth}}) \bmod m$.

Note that in the above protocol the order of the two groups in the vote and reconstruction phases is flipped to ensure fairness. Consider a coalition $\mathcal{C}$ that controls $t = t_0 + t_{\text{oth}}$ players where t_0 and t_{oth} denote the number of corrupted players in group G_0 and G_{oth} , respectively. For Protocol B.4 to achieve CSP fairness, t needs to satisfy the following conditions:

- (MC1) $\mathcal{C}$ cannot control both coins. Equivalently, either $t_0 \leq k_0$ or $t_{\text{oth}} \leq k_{\text{oth}}$. Otherwise, $\mathcal{C}$ can force the outcome to be whatever the coalition's overall preference is.
- (MC2) If $\mathcal{C}$ controls s_{oth} , then it cannot fail the reconstruction of s_0 . That is, if $t_{\text{oth}} > k_{\text{oth}}$, then it must be that $t_0 < \min\{|G_0| - k_0, \lceil |G_0|/2 \rceil\}$. Otherwise, $\mathcal{C}$ can control the outcome to be the s_{oth} coin it decides by failing the reconstruction of s_0 .
- (MC3) If $\mathcal{C}$ can hamper the reconstruction of s_{oth} , then it must NOT prefer 0. That is, if $t_{\text{oth}} \geq \min\{|G_{\text{oth}}| - k_{\text{oth}}, \lceil |G_{\text{oth}}|/2 \rceil\}$, then $t_0 \leq \eta_\gamma(t_{\text{oth}})$. Otherwise, $\mathcal{C}$ can force the outcome to be 0 by failing the reconstruction of s_{oth} and gain benefit compared to honest behaviors.

Lemma B.5. *If conditions (MC1), (MC2), (MC3) are satisfied, then Protocol 5.3 achieves CSP-fairness against any coalition controlling no more than t players.*

Proof. Condition (MC3) implies that if the coalition fails the reconstruction of s_{oth} , the coalition's utility must be no more than its honest expected utility from the definition of $\eta_\gamma(\cdot)$. Therefore, we only need to show that conditioned on s_{oth} being successfully reconstructed, the outcome $(s_0 + s_{\text{oth}}) \bmod m$ is computationally indistinguishable from a uniformly random coin.

Case 1: $t_{\text{oth}} \leq k_{\text{oth}}$. Consider the following hybrid experiments. Here, $\mathcal{A}$ is the non-uniform p.p.t adversary that controls $\mathcal{C}$.

- (Real): A real execution of $\text{GroupToss}^{G_0}[k]$ from Step (S-0) to Step (R-0) in the $\mathfrak{F}_{\text{sharegen}}^{b, \mathcal{O}}[k]$ -hybrid world. At this moment, $(s_0 + s_{\text{oth}}) \bmod m$ is well-defined.
- (Hybrid): The simulator S^* interacts with $\mathcal{A}$ as follows:
 1. In Step (S-0), S^* emulates the honest players and interacts with $\mathcal{A}$. Let **state** denote the adversary's current state after this.
 2. In Step (S-oth) and Step (V-oth), the simulator samples $s_{\text{oth}} \xleftarrow{\$} [m]_0$, runs $\mathcal{S}(1^\lambda)$ defined in Equation (20) based on **state**, and resets $\mathcal{A}$'s state based on the output of $\mathcal{S}(1^\lambda)$.
 3. In Steps (V-0) and (R-0), S^* continues to interact with $\mathcal{A}$. At this point, $(s_0 + s_{\text{oth}}) \bmod m$ is well defined.

By construction, $(s_0 + s_{\text{oth}}) \bmod m$ in Hybrid is a uniformly random m -coin. The two experiments Real and Hybrid are indistinguishable by the knowledge threshold property of $\text{GroupToss}^{G_{\text{oth}}}[k_{\text{oth}}]$. Therefore, $(s_0 + s_{\text{oth}}) \bmod m$ in Real is computationally indistinguishable from a uniformly random m -coin.

Case 2: $t_{\text{oth}} > k_{\text{oth}}$. By (MC1) and (MC2), $t \leq \min\{k_{\text{oth}}, |G_0| - k_{\text{oth}} - 1, \lceil |G_0|/2 \rceil\}$. Thus, s_0 must successfully reconstruct, meaning we can use a similar technique to Case 1. Note that by the end of (V-oth), both coins s_0 and s_{oth} are determined. Consider the following sequence of hybrids, where $\mathcal{A}$ denotes a non-uniform p.p.t adversary that controls $\mathcal{C}$.

- (Real): A real execution of $\text{GroupToss}^{G_0}[k]$ from Step (S-0) to Step (V-0) in the $\mathfrak{F}_{\text{sharegen}}^{b,\mathcal{O}}[k]$ -hybrid world. At this moment, $(s_0 + s_{\text{oth}}) \bmod m$ is well-defined.
- (Hybrid): The simulator S^* interacts with $\mathcal{A}$ as follows: first the simulator samples $s_{\text{oth}} \xleftarrow{\$}[m]_0$. Then, in Step (S-0) through Step (V-0), S^* runs $\mathcal{S}(1^\lambda)$ defined in Equation (20). At this moment, $(s_0 + s_{\text{oth}}) \bmod m$ is well-defined.

By construction, $(s_0 + s_{\text{oth}}) \bmod m$ in Hybrid is a uniformly random m -coin. By the knowledge threshold property of $\text{GroupToss}^{G_0}[k_0]$, the two hybrids, Real and Hybrid are indistinguishable as $t_0 \leq k_0$. Therefore, $(s_0 + s_{\text{oth}}) \bmod m$ in Real is computationally indistinguishable from a uniformly random m -coin. $\square$

Lemma B.6. *Assuming the existence of OT, then if $\eta_0(\lfloor N_{\text{oth}}/2 \rfloor) > \frac{1}{2}N_0 + 1$, the optimal parameter choices from Theorem 5.18 satisfies (MC1), (MC2), and (MC3).*

Proof. Theorem 5.18 chooses $k_0 = \lceil |G_0|/2 \rceil$ and γ^* to be one of the margins such that $h(\gamma^*)$ exists, and the optimal α^* to be $\text{opt}(\gamma^*)$. These parameters satisfy (MC1), (MC2). In the rest of this proof, we focus on proving that they satisfy the third constraint (MC3). Since $\eta_0(\lfloor N_{\text{oth}}/2 \rfloor) > \frac{1}{2}N_0 + 1$, by the same proof as in Theorem 5.18, we have that t^* must be strictly larger than $N/2$ except for the case where $\gamma^* = 0, \alpha^* = \lfloor N_{\text{oth}}/2 \rfloor$ and $t^* = \lceil N/2 \rceil$. We consider these two cases separately.

- If $\gamma^* = 0, k_0 = \lceil N_0/2 \rceil, \alpha^* = \lfloor N_{\text{oth}}/2 \rfloor$, and $t = \lceil N/2 \rceil$: In this case, (MC3) is trivially satisfied.
- If $t(\gamma^*, \alpha^*) > N/2$. In this case, we show that α^* must be no more than $\lceil \frac{G_{\text{oth}}}{2} \rceil$ and (MC3) is therefore satisfied. Seeking contradiction, assume that $\alpha > \lceil \frac{G_{\text{oth}}}{2} \rceil$. This means $\alpha \geq \lfloor \frac{G_{\text{oth}}}{2} + 1 \rfloor$. Now consider the value of $t(\gamma, \alpha)$.

$$\begin{aligned}
t(\gamma, \alpha) &\leq \left\lceil \frac{N_0 - \gamma}{2} \right\rceil + N_{\text{oth}} - \alpha \leq \left\lceil \frac{N_0 - \gamma}{2} \right\rceil + N_{\text{oth}} - \left\lfloor \frac{G_{\text{oth}}}{2} + 1 \right\rfloor \\
&\leq \frac{N_0 - \gamma}{2} + N_{\text{oth}} - \frac{G_{\text{oth}}}{2} \\
&= \frac{N_0 - \gamma}{2} + N_{\text{oth}} - \frac{N_{\text{oth}} - \gamma}{2} \\
&= \frac{N_0}{2} + \frac{N_{\text{oth}}}{2}.
\end{aligned}$$

This contradicts $t(\gamma, \alpha) > \frac{N}{2}$. $\square$

Theorem B.7. *When $\eta_0(\lfloor N_{\text{oth}}/2 \rfloor) > \frac{1}{2}N_0 + 1$ then Protocol B.4 achieves the same coalition resilience as the semi-malicious Protocol 5.3 and $t \geq N/2$.*

Proof. Combining Lemma B.6 and Theorem 5.18. $\square$